

第八章 正弦稳态电路分析

知识结构

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8.3 阻抗和导纳

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概念

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8.1 相量及其基本性质

8.1.1 基本概念

- 对线性、非时变电路而言，知道了任一正弦量的稳态响应，也就知道了电路对其他任意信号的响应。
- 直接求解法 复杂；相量分析法（实数域→复数域）简单。

有关复数的**复习**：

$$\operatorname{Re}[x + jy] = x \quad \operatorname{Im}[x + jy] = y$$

$$x + jy = A(\cos \alpha + j \sin \alpha) = Ae^{j\alpha} = A \angle \alpha$$

$$A_1 \angle \alpha_1 \times A_2 \angle \alpha_2 = A_1 A_2 \angle (\alpha_1 + \alpha_2)$$

$$A_1 \angle \alpha_1 / (A_2 \angle \alpha_2) = (A_1 / A_2) \angle (\alpha_1 - \alpha_2)$$

相量——正弦量的一种数学表示

对一给定的正弦量（电压或电流）： $u(t) = U_m \cos(\omega t + \varphi)$

利用欧拉公式

正弦量三要素—— U_m 、 ω 、 φ

$$U_m \cos(\omega t + \varphi) \Leftrightarrow U_m e^{j(\omega t + \varphi)} = (U_m e^{j\varphi}) e^{j\omega t}$$

$$\dot{U}_m = U_m e^{j\varphi} = U_m \angle \varphi \quad \text{振幅相量, 简称相量}$$

$$u(t) = \operatorname{Re}[U_m e^{j(\omega t + \varphi)}] = \operatorname{Re}(\dot{U}_m e^{j\omega t}) = \operatorname{Re}[\dot{U}_m \angle(\omega t)]$$

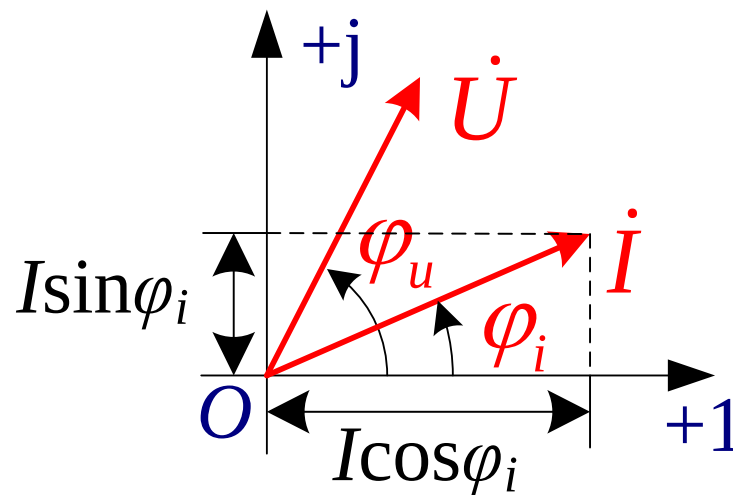
$$U_m \cos(\omega t + \varphi) \quad \Leftrightarrow \quad \dot{U}_m$$

相量变换

- 知道了表示正弦量的相量，就确定了振幅和相位，但不能确定频率[需事先给定]。
- 用正弦函数表达正弦量及其相量

$$i(t) = I_m \sin(\omega t + \varphi) \quad i(t) = \text{Im}[\dot{I}_m \angle(\omega t)]$$

- 相量法主要用来计算强制函数为正弦量时的线性、时不变微分方程的特解[正弦稳态解]。
- 相量在复平面上的表示——相量图



例：已知相量，求正弦量 ($f=50\text{Hz}$)

$$\dot{U}_{1\text{m}} = 50\angle -30^\circ \text{V}, \dot{U}_{2\text{m}} = 100\angle 150^\circ \text{V}$$

对任意正弦量，有

$$A\angle\varphi \Leftrightarrow A\cos(\omega t + \varphi) \quad \text{必须给定}\omega$$

$$\dot{U}_{1\text{m}} = 50\angle -30^\circ \text{V} \Rightarrow u_1(t) = 50\cos(314t - 30^\circ)\text{V}$$

$$\dot{U}_{2\text{m}} = 100\angle 150^\circ \text{V} \Rightarrow u_2(t) = 100\cos(314t + 150^\circ)\text{V}$$

练习：求下列正弦电流的相量，选择正确答案。

(1) $i_1(t) = 5\cos(314t + 60^\circ)\text{A}$

(2) $i_2(t) = -10\sin(314t + 60^\circ)\text{A}$

(3) $i_3(t) = -4\cos(314t + 60^\circ)\text{A}$

A. $\dot{I}_{1\text{m}} = 5\angle 60^\circ \text{A}$ $\dot{I}_{2\text{m}} = -10\angle 150^\circ \text{A}$ $\dot{I}_{3\text{m}} = 4\angle -120^\circ \text{A}$

B. $\dot{I}_{1\text{m}} = 5\angle 60^\circ \text{A}$ $\dot{I}_{2\text{m}} = -10\angle 60^\circ \text{A}$ $\dot{I}_{3\text{m}} = 4\angle -120^\circ \text{A}$

C. $\dot{I}_{1\text{m}} = 5\angle 60^\circ \text{A}$ $\dot{I}_{2\text{m}} = 10\angle 150^\circ \text{A}$ $\dot{I}_{3\text{m}} = 4\angle 120^\circ \text{A}$

D. $\dot{I}_{1\text{m}} = 5\angle 60^\circ \text{A}$ $\dot{I}_{2\text{m}} = 10\angle 150^\circ \text{A}$ $\dot{I}_{3\text{m}} = 4\angle -120^\circ \text{A}$

E. $\dot{I}_{1\text{m}} = 5\angle 60^\circ \text{A}$ $\dot{I}_{2\text{m}} = 10\angle -150^\circ \text{A}$ $\dot{I}_{3\text{m}} = 4\angle -120^\circ \text{A}$

8.1.2 有效值 平均值 有效值相量

周期电流、电压的瞬时值是随时间变化的，瞬时值并不能刻画周期量的整体大小。此时，可引入有效值、平均值的概念。

定义：对一周期量 $x(t)$ ，有效值、平均值规定为

$$X = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt} \quad X_a = \frac{1}{T} \int_0^T |x(t)| dt$$

有效值的物理含义：

$$\begin{cases} I = \sqrt{\frac{1}{T} \int_0^T i^2 dt} \\ U = \sqrt{\frac{1}{T} \int_0^T u^2 dt} \end{cases} \Rightarrow \begin{cases} I^2 RT = R \int_0^T i^2 dt \\ \frac{U^2 T}{R} = \frac{\int_0^T u^2 dt}{R} \end{cases}$$

在一个周期
内消耗在电
阻 R 上的能量

就作功的能力而言， I 与 $i(t)$ 、 U 与 $u(t)$ 是等效的。

正弦量的有效值与有效值相量

$$u(t) = U_m \cos(\omega t + \varphi)$$

$$U = \sqrt{\frac{1}{T} \int_0^T [U_m \cos(\omega t + \varphi)]^2 dt} = \sqrt{\frac{1}{T} \int_0^T \frac{U_m^2}{2} [\cos(2\omega t + 2\varphi) + 1] dt} = \frac{1}{\sqrt{2}} U_m$$

$$I = \frac{1}{\sqrt{2}} I_m = 0.707 I_m \quad \text{正弦波的有效值为其振幅的 } 1/\sqrt{2} \text{。}$$

有效值可以代替振幅作为正弦量的一个特征值。交流电表的读数值一般是指有效值。

用有效值来表示正弦量：**有效值相量**

$$u(t) = U_m \cos(\omega t + \varphi) = \sqrt{2}U \cos(\omega t + \varphi) \leftrightarrow \dot{U} = U \angle \varphi$$

8.1.3 相量的基本性质

(1) 线性性质

如果 $f_1(t) \leftrightarrow \dot{A}_1$, $f_2(t) \leftrightarrow \dot{A}_2$, 则 $\alpha_1 f_1(t) + \alpha_2 f_2(t) \leftrightarrow \alpha_1 \dot{A}_1 + \alpha_2 \dot{A}_2$
其中 α_1, α_2 为实数。

证明:

$$\begin{aligned}\alpha_1 f_1(t) + \alpha_2 f_2(t) &= \alpha_1 \operatorname{Re}(\dot{A}_1 e^{j\omega t}) + \alpha_2 \operatorname{Re}(\dot{A}_2 e^{j\omega t}) \\ &= \operatorname{Re}(\alpha_1 \dot{A}_1 e^{j\omega t}) + \operatorname{Re}(\alpha_2 \dot{A}_2 e^{j\omega t}) \\ &= \operatorname{Re}[(\alpha_1 \dot{A}_1 + \alpha_2 \dot{A}_2) e^{j\omega t}]\end{aligned}$$

例：若 $f(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t)$, $\dot{A} \Leftrightarrow f(t)$, 证明 $\dot{A} = K_1 - jK_2$ 。

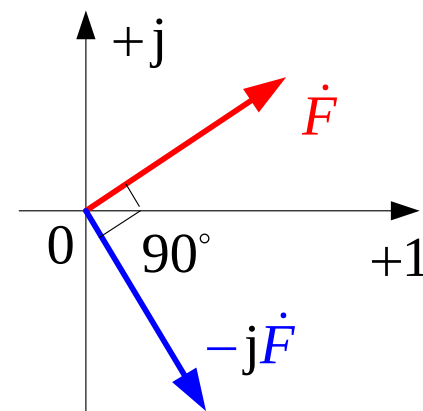
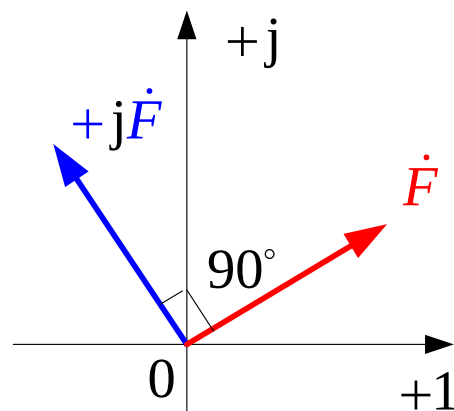
解一（三角方法）：

$$f(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t) = \sqrt{K_1^2 + K_2^2} \cos(\omega t - \tan^{-1} \frac{K_2}{K_1})$$

$$\Rightarrow \dot{A} = \sqrt{K_1^2 + K_2^2} \angle(-\tan^{-1} \frac{K_2}{K_1}) = K_1 - jK_2$$

解二（相量法）：

$$\dot{A} = K_1 \angle 0^\circ + K_2 \angle -90^\circ = K_1 - jK_2$$



j 的作用：旋转 90°

(2) 微分性质

如果 $f(t) \leftrightarrow \dot{A}_m$, 则 $\frac{df(t)}{dt} \leftrightarrow j\omega \dot{A}_m$ 。

证明:
$$\begin{aligned}\frac{d}{dt} \operatorname{Re}(\dot{A}_m e^{j\omega t}) &= \frac{d}{dt} [A_m \cos(\omega t + \varphi)] = -\omega A_m \sin(\omega t + \varphi) \\ &= \operatorname{Re}[j\omega A_m e^{j(\omega t + \varphi)}] = \operatorname{Re}[j\omega \dot{A}_m e^{j\omega t}]\end{aligned}$$

推广: n 阶微分性质?

例: 若 $f(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t)$, $\dot{A} \Leftrightarrow f(t)$, 证明 $\dot{A} = K_1 - jK_2$ 。

$$f(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t) = K_1 \cos(\omega t) - \frac{1}{\omega} \frac{d}{dt} [K_2 \cos(\omega t)]$$

$$\Rightarrow \dot{A} = K_1 \angle 0^\circ - \frac{1}{\omega} j\omega K_2 \angle 0^\circ = K_1 - jK_2$$

(3) 积分性质:

如果 $f(t) \leftrightarrow \dot{A}_m$, 则 $\int f(t)dt \leftrightarrow \frac{\dot{A}_m}{j\omega}$ 。

推广: n 阶积分性质?

例: 若 $f(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t)$, $\dot{A} \Leftrightarrow f(t)$, 证明 $\dot{A} = K_1 - jK_2$ 。

$$f(t) = K_1 \cos(\omega t) + K_2 \sin(\omega t) = K_1 \cos(\omega t) + \omega K_2 \int \cos(\omega t) dt$$

$$\Rightarrow \dot{A} = K_1 \angle 0^\circ + \frac{1}{j\omega} \omega K_2 \angle 0^\circ = K_1 - jK_2$$

练习：求下列正弦量的相量，选择正确答案。

$$f(t) = 5\cos(\omega t) - 5\sin(\omega t) + 10\sqrt{2}\cos(\omega t - 135^\circ)$$

- A. $f(t) = 5\sqrt{2}\cos(\omega t + 45^\circ)$
- B. $f(t) = 5\sqrt{2}\cos(\omega t - 135^\circ)$
- C. $f(t) = 0$
- D. $f(t) = 4\cos(\omega t + 45^\circ)$
- E. $f(t) = 5\cos(\omega t - 135^\circ)$

相量变换在求解常系数微分方程在正弦激励下**特解**中的应用。

$$a_0 \frac{d^n x}{dt^n} + a_1 \frac{d^{n-1} x}{dt^{n-1}} + \cdots + a_{n-1} \frac{dx}{dt} + a_n x = A_m \cos(\omega t + \varphi)$$

引入相量 $\dot{A}_m = A_m e^{j\varphi}$, $\dot{X}_m = X_m e^{j\phi}$

$$\Rightarrow [a_0 (j\omega)^n + a_1 (j\omega)^{n-1} + \cdots + a_n] \dot{X}_m = \dot{A}_m$$

$$\Rightarrow \dot{X}_m = \frac{\dot{A}_m}{[a_0 (j\omega)^n + a_1 (j\omega)^{n-1} + \cdots + a_n]}$$

分母不能为零；等于零时应如何处理？

$$\Rightarrow X_m = \frac{A_m}{[(a_n - a_{n-2}\omega^2 + \cdots)^2 + (a_{n-1}\omega - a_{n-3}\omega^3 + \cdots)^2]^{1/2}}$$

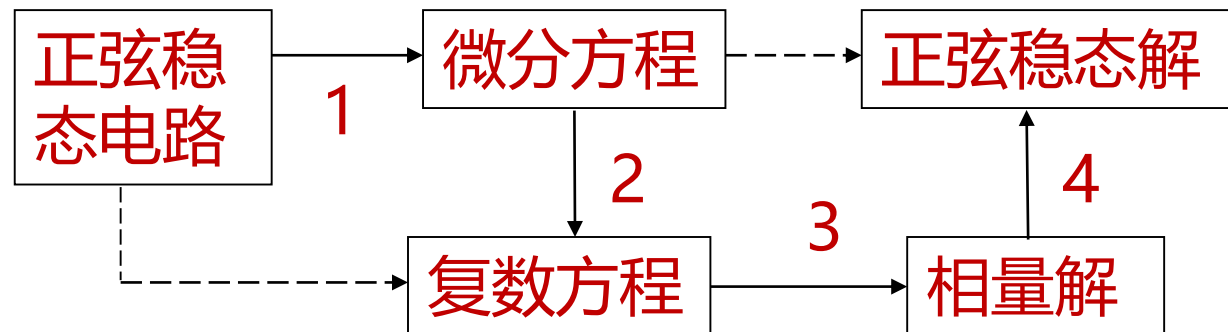
$$\phi = \varphi - \tan^{-1} \frac{a_{n-1}\omega - a_{n-3}\omega^3 + \cdots}{a_n - a_{n-2}\omega^2 + \cdots}$$

$x(t) = X_m \cos(\omega t + \phi)$

正弦稳态电路的主定理：

任意个角频率相同的正弦量以及任意个这种正弦量的任意阶导数的代数和，仍为一个同频率的正弦量。

正弦稳态电路的相量法求解过程：



8.2 两类约束的相量形式

8.2.1 基尔霍夫定律的相量形式

对单一频率正弦激励下的线性非时变电路正弦稳态电路，其电压、电流必为同频率的正弦波。由KCL得：

$$\sum_{k=1}^K i_k = \sum_{k=1}^K \operatorname{Re}(\dot{I}_{km} e^{j\omega t}) = \operatorname{Re}(e^{j\omega t} \sum_{k=1}^K \dot{I}_{km}) = 0 \Rightarrow \boxed{\sum_{k=1}^K \dot{I}_{km} = 0}$$

$$\boxed{\sum_{k=1}^K \dot{U}_{km} = 0} \quad \text{KVL}$$

8.2.2 基本电路元件VCR的相量形式

在关联参考方向下，线性时不变电阻、电容及电感元件的VCR方别为：

$$u = Ri, \quad i = C \frac{du}{dt}, \quad u = L \frac{di}{dt}$$

利用相量的性质，得到

$$\begin{cases} \dot{U}_m = R\dot{I}_m \\ \dot{I}_m = j\omega C\dot{U}_m \\ \dot{U}_m = j\omega L\dot{I}_m \end{cases}$$

$$\begin{cases} \dot{U}_m = R\dot{I}_m \\ \dot{U}_m = \frac{1}{j\omega C}\dot{I}_m \\ \dot{U}_m = j\omega L\dot{I}_m \end{cases}$$

● 相量欧姆定律

VCR

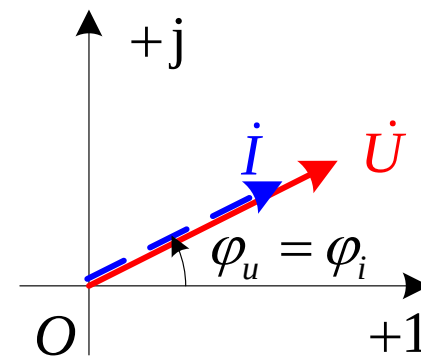
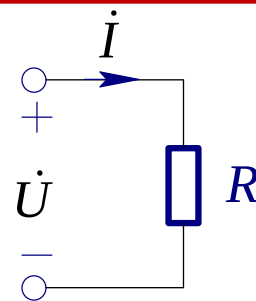
相量模型

相量图

电阻

$$u = Ri \Leftrightarrow \dot{U}_m = R\dot{I}_m$$

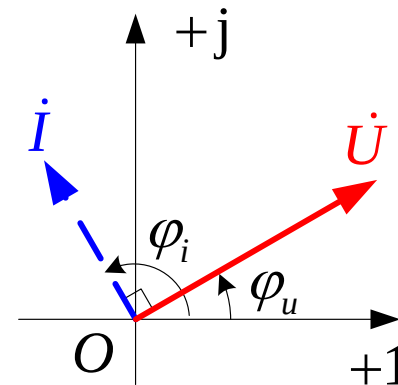
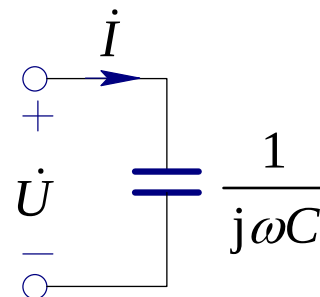
$$\begin{cases} U_m = RI_m \\ \varphi_u = \varphi_i \end{cases}$$



电容

$$i = C \frac{du}{dt} \Leftrightarrow \dot{I}_m = j\omega C \dot{U}_m$$

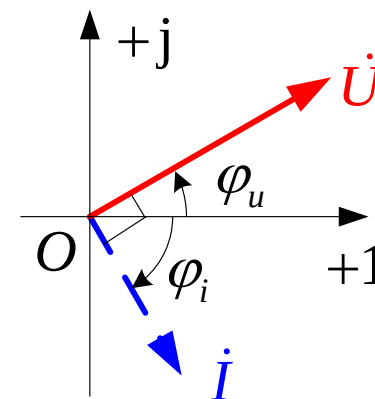
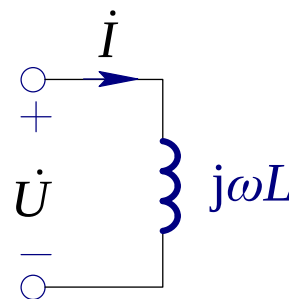
$$\begin{cases} I_m = \omega C U_m \\ \varphi_i = \varphi_u + 90^\circ \end{cases}$$



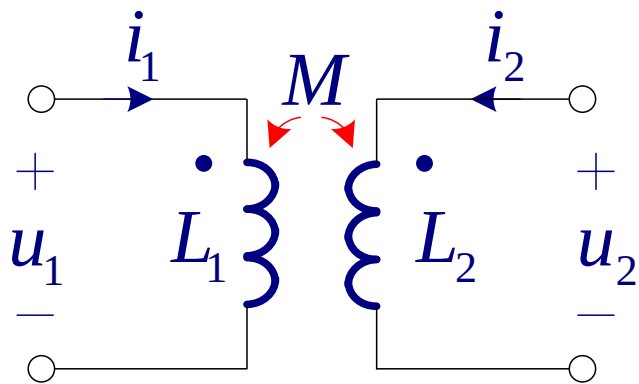
电感

$$u = L \frac{di}{dt} \Leftrightarrow \dot{U}_m = j\omega L \dot{I}_m$$

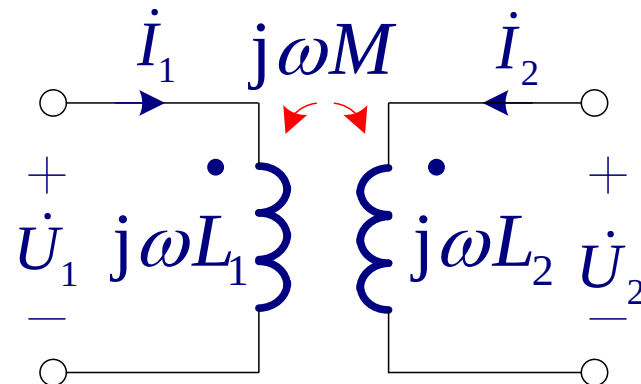
$$\begin{cases} U_m = \omega L I_m \\ \varphi_u = \varphi_i + 90^\circ \end{cases}$$



8.2.3 耦合电感元件VCR的相量形式



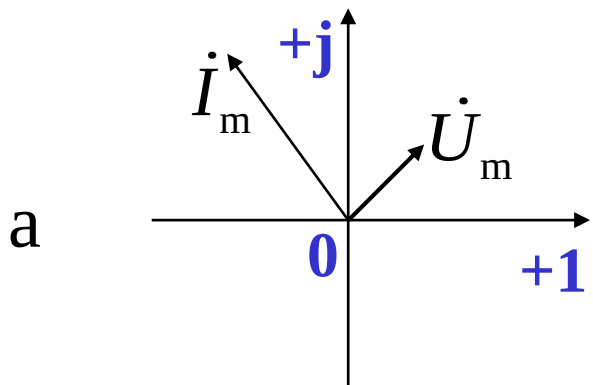
$$\begin{bmatrix} u_1 \\ u_2 \end{bmatrix} = \begin{bmatrix} L_1 & M \\ M & L_2 \end{bmatrix} \begin{bmatrix} \frac{di_1}{dt} \\ \frac{di_2}{dt} \end{bmatrix}$$



$$\begin{bmatrix} \dot{U}_1 \\ \dot{U}_2 \end{bmatrix} = \begin{bmatrix} j\omega L_1 & j\omega M \\ j\omega M & j\omega L_2 \end{bmatrix} \begin{bmatrix} \dot{I}_1 \\ \dot{I}_2 \end{bmatrix}$$

练习：下述正确的是

- A. 已知1H电感元件的端电压 $u(t) = \cos(t) \text{ V}$,
流经元件的电流为 $i(t) = \cos(t + 90^\circ) \text{ A}$ 。
- B. 图a是电感元件的相量图。
- C. 对电感元件, 有: $\dot{U}_m = j\omega L \dot{I}_m$ 。
- D. 对电容元件, 有: $\dot{U}_m = j\omega C \dot{I}_m$ 。
- E. 相量乘以复数仍是一个相量。



8.3 阻抗和导纳

8.3.1 阻抗和导纳的定义

元件在正弦稳态时的电压相量与电流相量之比，称为该元件的**阻抗**(impedance)，记为 Z ，阻抗的倒数定义为**导纳**(admittance)，记为 Y

$$\frac{\dot{U}_m}{\dot{I}_m} = Z \quad Y = \frac{1}{Z} = \frac{\dot{I}_m}{\dot{U}_m}$$

● 相量欧姆定律

$$\dot{I}_m = Y \dot{U}_m$$

三种基本元件的相量关系可统一为

$$\left. \begin{aligned} \dot{U}_m &= R \dot{I}_m \\ \dot{U}_m &= \frac{1}{j\omega C} \dot{I}_m \\ \dot{U}_m &= j\omega L \dot{I}_m \end{aligned} \right\} \Rightarrow \boxed{\dot{U}_m = Z \dot{I}_m} \Rightarrow \begin{cases} Z_R = R \\ Z_C = -j \frac{1}{\omega C} \\ Z_L = j\omega L \end{cases}$$

$$Y_R = \frac{1}{R} = G$$

$$Y_C = j\omega C$$

$$Y_L = \frac{1}{j\omega L} = -j \frac{1}{\omega L}$$

讨论：将阻抗中的虚数部分称为电抗，记为 $X = \text{Im}[Z]$ 。将导纳中的虚数部分称为电纳，记为 $B = \text{Im}[Y]$ 。

$$X_C = \text{Im}[Z_C] = -\frac{1}{\omega C}, \quad X_L = \omega L, \quad B_C = \text{Im}[Y_C] = \omega C, \quad B_L = -\frac{1}{\omega L}$$

电容的电抗-容抗
感抗
容纳
感纳

不含独立电源单口网络的阻抗和导纳

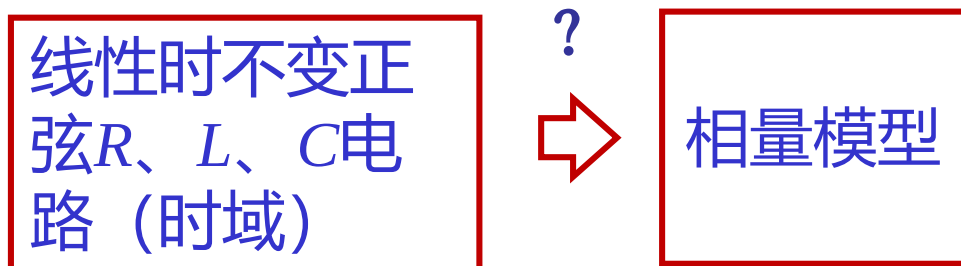
$$Z = \frac{\dot{U}}{\dot{I}} = \frac{U \angle \varphi_u}{I \angle \varphi_i} = \frac{U}{I} \angle (\varphi_u - \varphi_i) \quad \overset{|Z|=\frac{U}{I}, \varphi_Z=\varphi_u-\varphi_i}{=} \quad |Z| \angle \varphi_Z$$

$$\varphi_Z(\omega) \overset{Z(j\omega)=R(\omega)+jX(\omega)}{=} \tan^{-1} [X(\omega)/R(\omega)] \quad |\varphi_Z| \leq 180^\circ$$

$$|Y(j\omega)| = \frac{1}{|Z(j\omega)|}, \quad \varphi_Y(\omega) = -\varphi_Z(\omega)$$

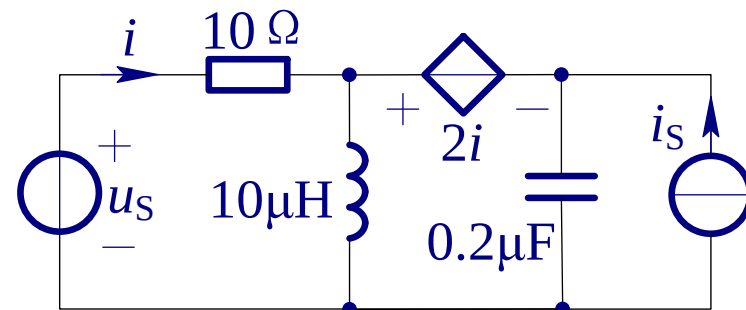
8.3.2 相量模型

对正弦稳态电路，它所受到的约束仍然为两类：拓扑约束和元件约束。



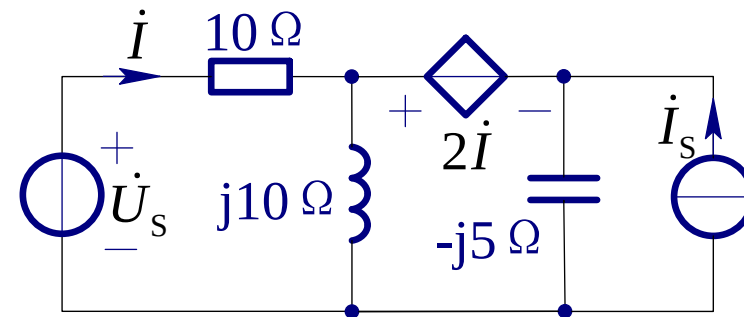
用相量法分析正弦稳态电路的基本步骤：

- (1) 作相量模型
- (2) 应用一般电路分析方法进行分析
- (3) 结果转换为时域形式 (正弦时间函数)



$$u_s = 10 \cos(10^6 t + 60^\circ) \text{ V}$$

$$i_s = 5 \sin(10^6 t + 60^\circ) \text{ A}$$



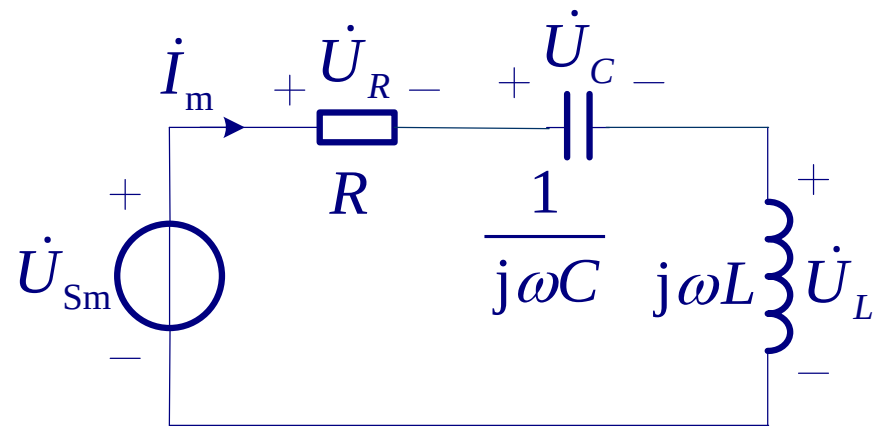
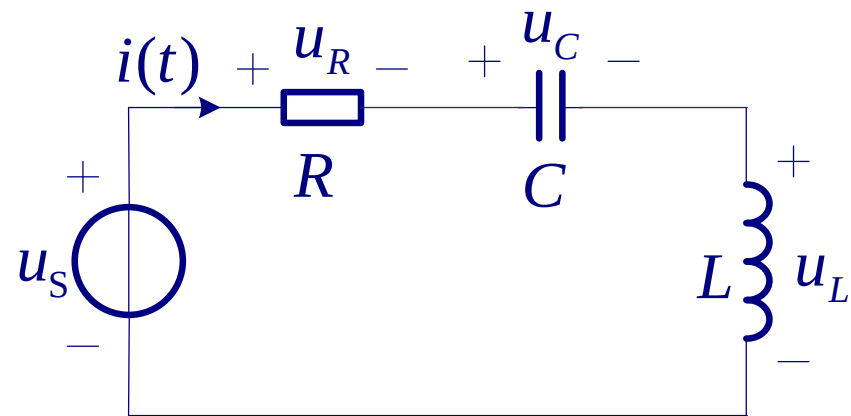
例：已知 $u_S(t) = U_{Sm} \cos(\omega t + \psi_u)$ ，求稳态电流 $i(t)$ 。

$$Z = Z_R + Z_L + Z_C = R + j\omega L + \frac{1}{j\omega C}$$

$$= \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2} \angle \tan^{-1}[(\omega L - \frac{1}{\omega C}) / R]$$

$$\dot{I}_m = \frac{\dot{U}_{Sm}}{Z} = \frac{U_{Sm}}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \angle \left(\psi_u - \tan^{-1}[(\omega L - \frac{1}{\omega C}) / R] \right)$$

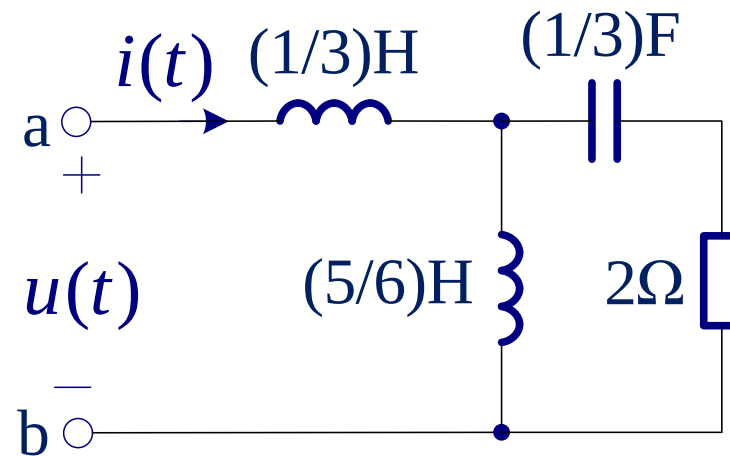
$$i(t) = \frac{U_{Sm}}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} \cos \left[\omega t + \psi_u - \tan^{-1}[(\omega L - \frac{1}{\omega C}) / R] \right]$$



例：如图，求输入阻抗函数，如 $i(t) = \cos(3t + 45^\circ)$ ，求稳态电压 $u(t)$ ， $\omega = 6\text{rad/s}$ 呢？

$$Z_{ab}(j\omega) = \left[\frac{j\omega}{3} + \left(2 + \frac{3}{j\omega} \right) \parallel \frac{j5\omega}{6} \right]$$

$$= \left[\frac{j\omega}{3} + \frac{j5\omega(j2\omega + 3)}{18 + j12\omega - 5\omega^2} \right] \Omega$$



$$Z_{ab}(j3) = 2\sqrt{2} \angle 45^\circ \Omega$$

$$u(t) = 2\sqrt{2} \cos(3t + 45^\circ + 45^\circ) \text{V}$$

$$Z_{ab}(j6) = 3.13 \angle 48.9^\circ \Omega$$

$$u(t) = 3.13 \cos(6t + 45^\circ + 48.9^\circ) \text{V}$$

8.4 正弦稳态电路的分析

8.4.1 简单正弦稳态电路的分析

例：求各稳态电流。

$$Z = 1.5 + (j1) // (1 - 2j) = \left(1.5 + \frac{(j1)(1 - 2j)}{(j1) + (1 - 2j)} \right)$$

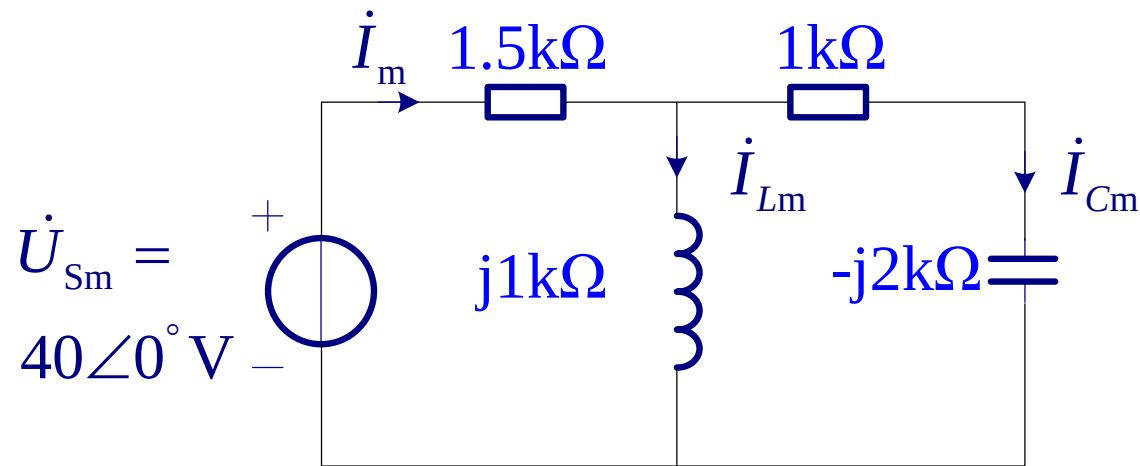
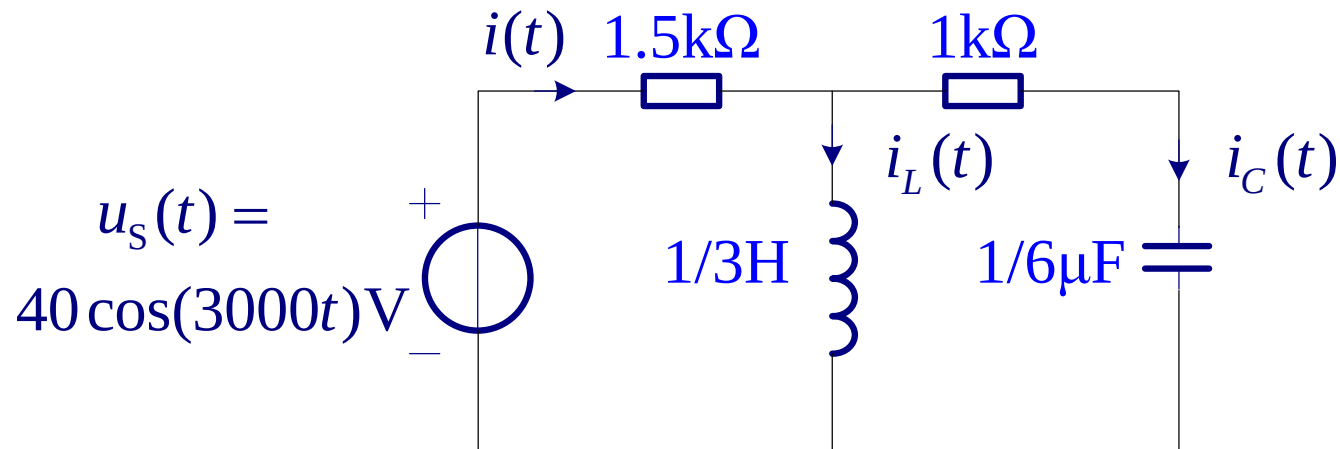
$$= 2.5 \angle 36.9^\circ \text{ k}\Omega$$

$$\dot{I}_m = \frac{\dot{U}_{Sm}}{Z} = \frac{40 \angle 0^\circ}{2.5 \angle 36.9^\circ} = 16 \angle -36.9^\circ \text{ mA}$$

$$\dot{I}_{Cm} = \dot{I}_m \frac{j1}{1 + j1 - 2j} = \dots = 11.3 \angle 98.1^\circ \text{ mA}$$

$$\dot{I}_{Lm} = \dot{I}_m \frac{1 - j2}{1 + j1 - 2j} = \dots = 25.3 \angle -55.3^\circ \text{ mA}$$

$$i(t) = 16 \cos(3000t - 36.9^\circ) \text{ mA}; \quad i_C(t) = 11.3 \cos(3000t + 98.1^\circ) \text{ mA}; \quad i_L(t) = 25.3 \cos(3000t - 55.3^\circ) \text{ mA}$$



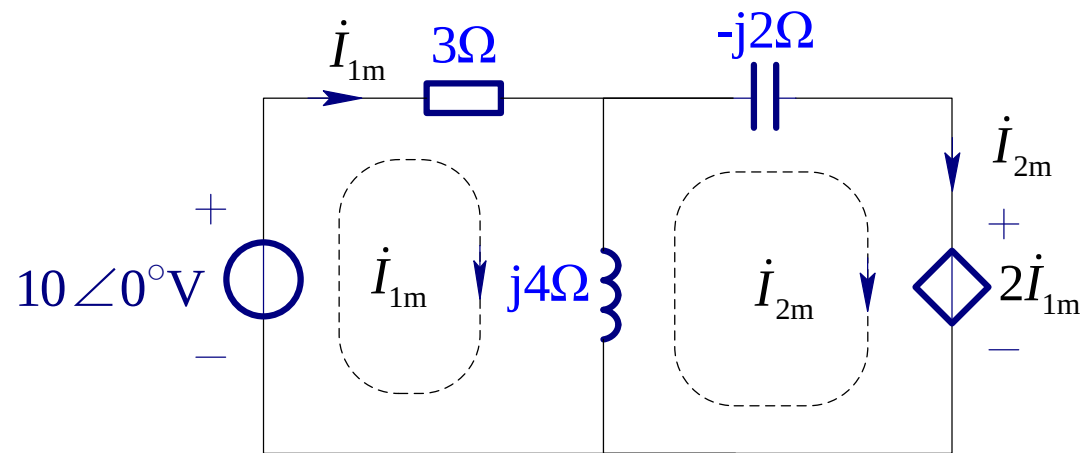
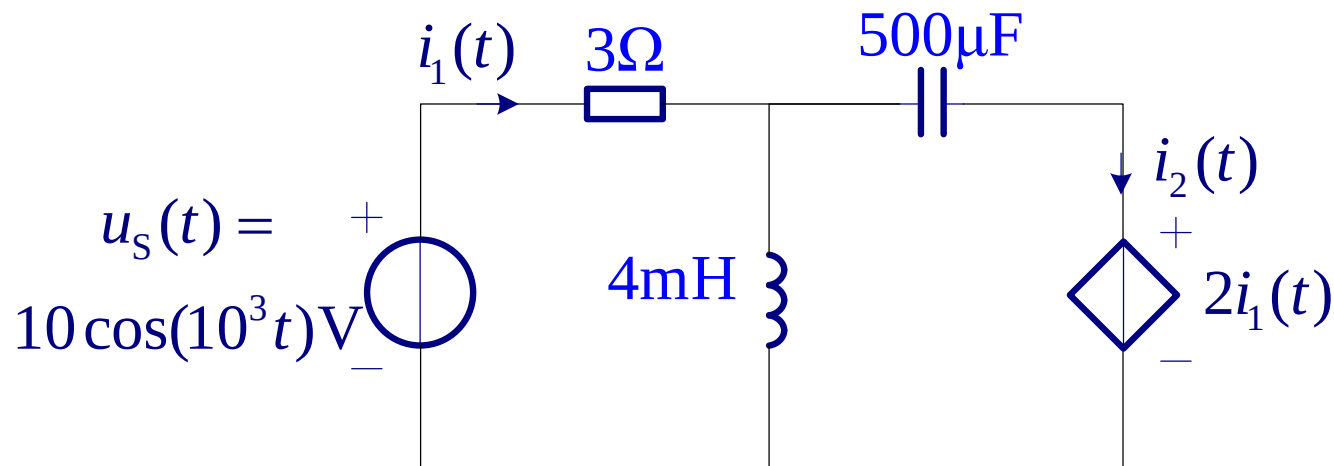
例：求图示电路中的电流。

用回路（网孔）法。

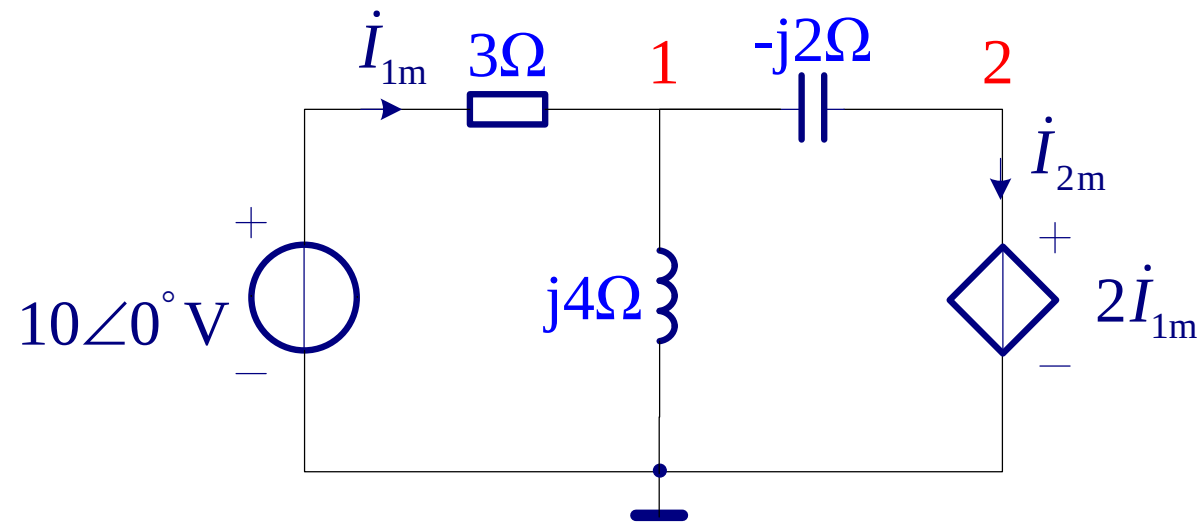
$$\begin{cases} (3 + 4j)\dot{I}_{1m} - j4\dot{I}_{2m} = 10\angle 0^\circ \\ -4j\dot{I}_{1m} + (j4 - j2)\dot{I}_{2m} = -2\dot{I}_{1m} \end{cases}$$

$$\begin{cases} \dot{I}_{1m} = 1.0769 + 0.6154j = 1.24\angle 29.7^\circ \text{ A} \\ \dot{I}_{2m} = 1.5385 + 2.3077j = 2.77\angle 56.3^\circ \text{ A} \end{cases}$$

$$\begin{cases} i_1(t) = 1.24\cos(10^3t + 29.7^\circ) \text{ A} \\ i_2(t) = 2.77\cos(10^3t + 56.3^\circ) \text{ A} \end{cases}$$



接上例，用节点法求解：



$$\begin{cases} (\frac{1}{3} + \frac{1}{j4} - \frac{1}{j2})\dot{U}_1 + \frac{1}{j2}\dot{U}_2 = \frac{10\angle 0^\circ}{3} \\ \dot{U}_2 = 2\dot{I}_{1m} = \frac{2}{3}(10\angle 0^\circ - \dot{U}_1) \end{cases}$$

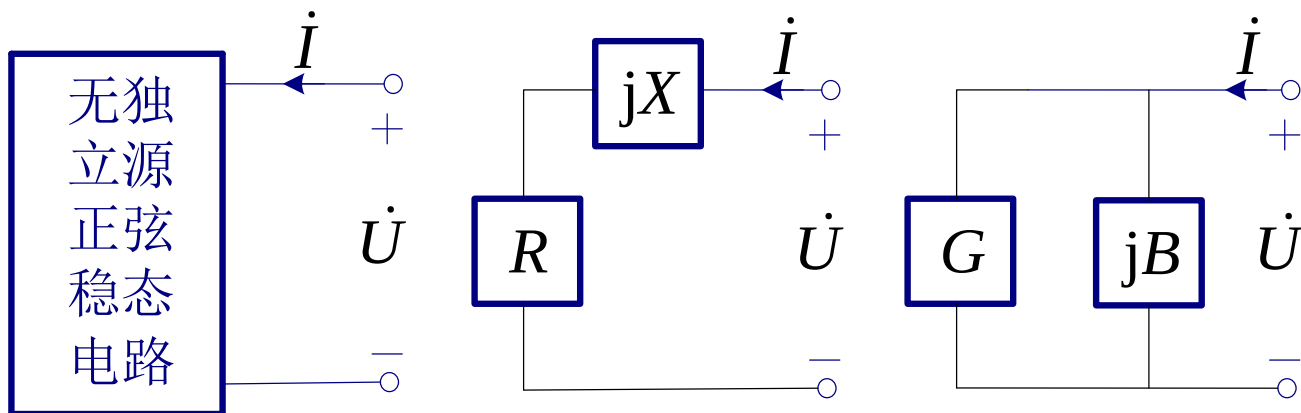
$$\Rightarrow \begin{cases} \dot{U}_1 = 6.7692 - j1.8462 \\ \dot{U}_2 = 2.1538 + j1.2308 \end{cases}$$

$$\begin{cases} \dot{I}_{1m} = 1.0769 + 0.6154j = 1.24\angle 29.7^\circ \text{ A} \\ \dot{I}_{2m} = 1.5385 + 2.3077j = 2.77\angle 56.3^\circ \text{ A} \end{cases}$$

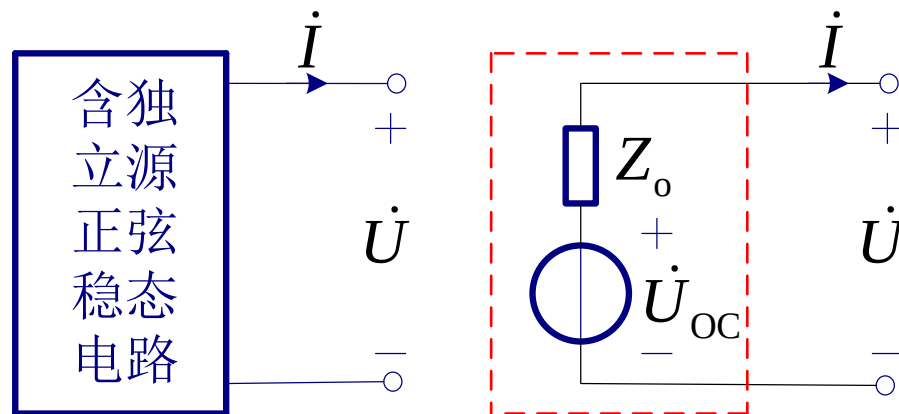
$$\Rightarrow \begin{cases} \dot{I}_{1m} = (10\angle 0^\circ - \dot{U}_1) / 3 = 1.0769 + 0.6154j = 1.24\angle 29.7^\circ \text{ A} \\ \dot{I}_{2m} = (\dot{U}_1 - \dot{U}_2) / (-j2) = 1.5385 + 2.3077j = 2.77\angle 56.3^\circ \text{ A} \end{cases}$$

8.4.2 相量模型的等效

第一篇中介绍的等效概念及方法完全适合相量模型。



$$\begin{aligned} \dot{U} &= Z\dot{I}, \quad Z = R + jX \\ \dot{I} &= Y\dot{U}, \quad Y = G + jB \end{aligned} \quad Z = \frac{1}{Y} \Rightarrow \begin{cases} G = \frac{R}{R^2 + X^2} \\ B = -\frac{X}{R^2 + X^2} \end{cases} \Rightarrow \begin{cases} R = \frac{G}{G^2 + B^2}, \quad X = -\frac{B}{G^2 + B^2} \end{cases}$$

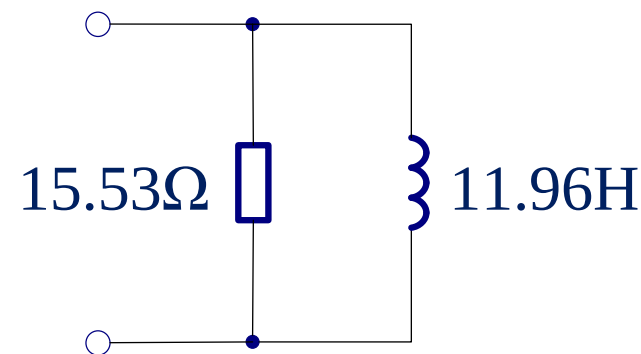
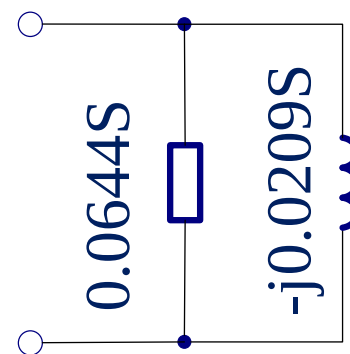
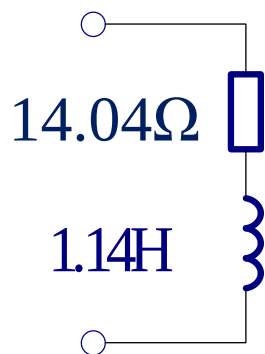
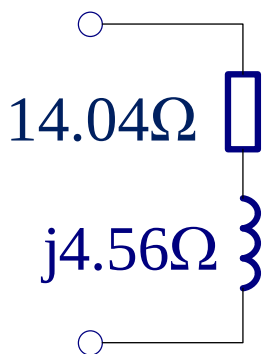
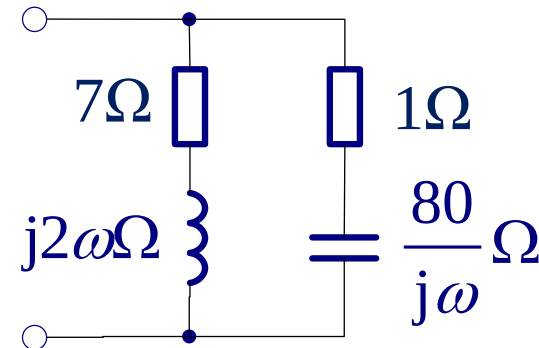


$$\dot{U} = \dot{U}_{oc} - Z_o \dot{I}$$

例：求 $\omega=4\text{rad/s}$ 时的等效模型。

$$Z(j\omega) = (7 + j2\omega) // (1 - j\frac{80}{\omega}) = \frac{(7 + j2\omega)(1 - j\frac{80}{\omega})}{(7 + j2\omega) + (1 - j\frac{80}{\omega})}$$

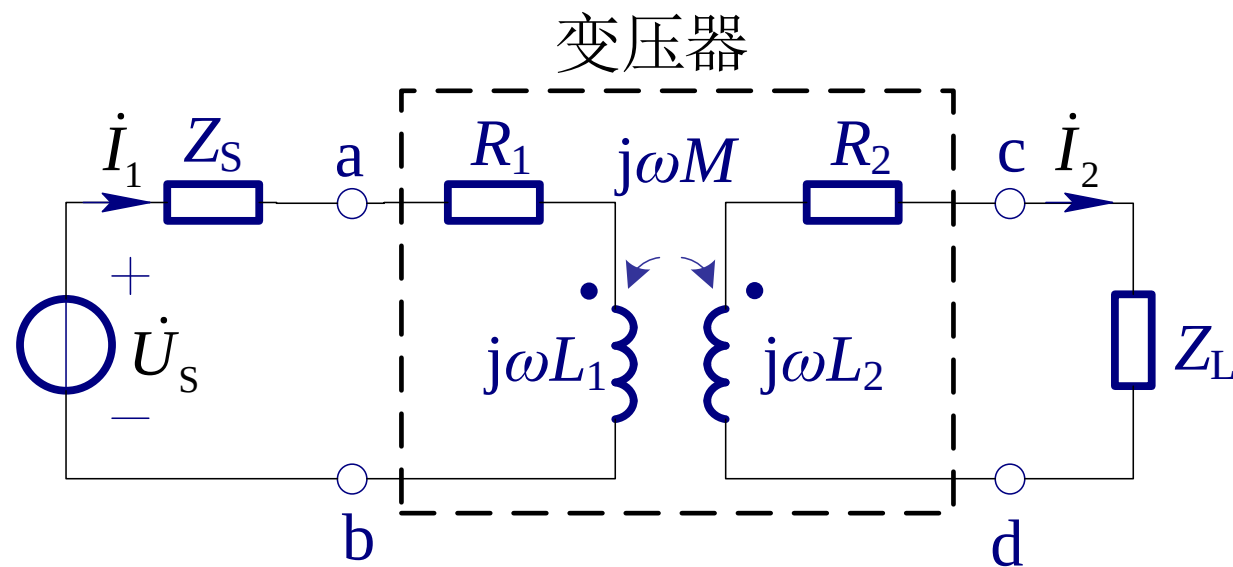
$$Z(j4) = (14.04 + j4.56)\Omega \quad Y(j4) = (0.0644 - j0.0209)\text{S}$$



练习：下述不正确的是

- A. 一个阻抗，可以等效为电阻和电抗的串联，也可等效为电导和电纳的并联。
- B. 一个二端电路的电压、电流分别为 $u = -10\cos t$, $i = -2\sin t$, 参考方向关联，该电路可等效为一个电感元件。
- C. 已知 $Z = (-2 - j2)\Omega$, 则 $\varphi_Z = 45^\circ$ 。
- D. 电阻电路的分析方法可类推应用于正弦稳态电路的分析。

8.4.3 含耦合电感电路的正弦稳态分析 (应用)



初级线圈：接信号源或电源

次级线圈：接负载

铁心变压器：两个线圈的耦合用铁心， $k \sim 1$

空心变压器：不用铁心， $k \ll 1$

$$\begin{cases} (Z_S + R_1 + j\omega L_1)\dot{I}_1 - j\omega M\dot{I}_2 = \dot{U}_S \\ -j\omega M\dot{I}_1 + (R_2 + j\omega L_2 + Z_L)\dot{I}_2 = 0 \end{cases}$$

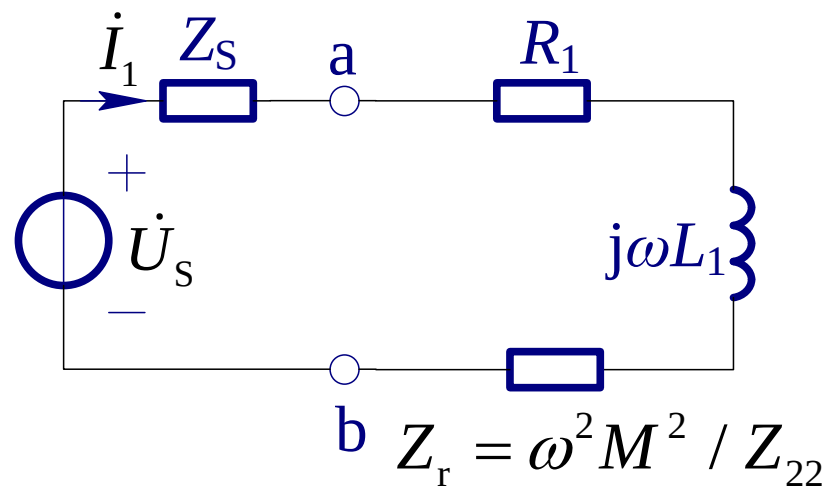
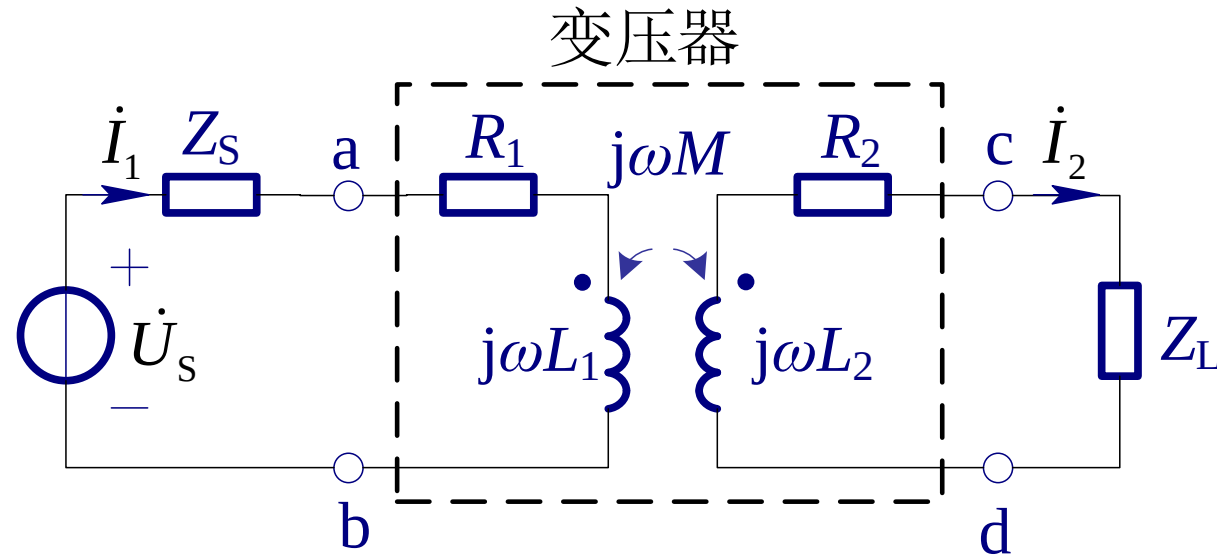
令

$$\begin{cases} Z_{11} = Z_S + R_1 + j\omega L_1 \\ Z_{22} = R_2 + j\omega L_2 + Z_L \end{cases}$$

$$\Rightarrow \begin{cases} \dot{I}_1 = \frac{\dot{U}_S}{Z_{11} + \omega^2 M^2 / Z_{22}} \\ \dot{I}_2 = \frac{j\omega M}{Z_{11}Z_{22} + \omega^2 M^2} \dot{U}_S \end{cases}$$

反映阻抗

$$\Rightarrow \begin{cases} \dot{I}_1 = \frac{\dot{U}_S}{Z_{11} + Z_r} \\ \dot{I}_2 = \frac{j\omega M}{Z_{22}} \dot{I}_1 \end{cases}$$

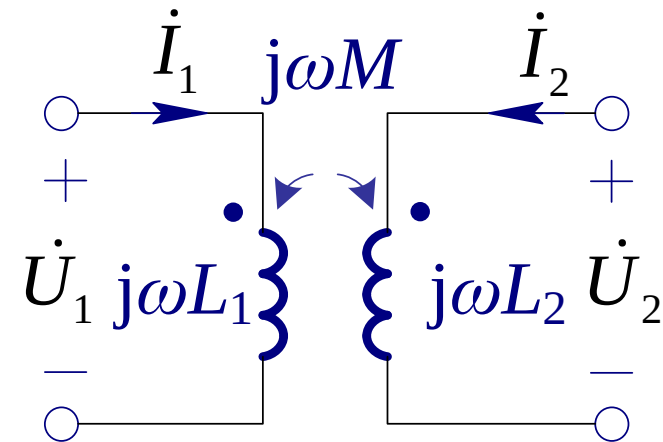


初级回路等效电路

变压器的理想化模型：铁心变压器——理想变压器模型

满足条件： $k=1$ ； L_1 、 $L_2 \rightarrow \infty$ ，且 $\sqrt{L_1/L_2} = n$ ；
电压、电流为时变的。

$$\begin{cases} \dot{U}_1 = j\omega L_1 \dot{I}_1 + j\omega \sqrt{L_1 L_2} \dot{I}_2 \\ \dot{U}_2 = j\omega \sqrt{L_1 L_2} \dot{I}_1 + j\omega L_2 \dot{I}_2 \end{cases}$$



$$\frac{\dot{U}_1}{\dot{U}_2} = \frac{j\omega L_1 \dot{I}_1 + j\omega \sqrt{L_1 L_2} \dot{I}_2}{j\omega \sqrt{L_1 L_2} \dot{I}_1 + j\omega L_2 \dot{I}_2} = \frac{\sqrt{L_1} (j\omega \sqrt{L_1} \dot{I}_1 + j\omega \sqrt{L_2} \dot{I}_2)}{\sqrt{L_2} (j\omega \sqrt{L_1} \dot{I}_1 + j\omega \sqrt{L_2} \dot{I}_2)} = \sqrt{\frac{L_1}{L_2}} = n$$

$$\dot{I}_1 = \frac{\dot{U}_1}{j\omega L_1} - \sqrt{\frac{L_2}{L_1}} \dot{I}_2 = -\frac{1}{n} \dot{I}_2$$

$$\begin{aligned} \dot{U}_1 &= n \dot{U}_2 \\ \dot{I}_1 &= -\dot{I}_2 / n \end{aligned}$$

变压器的理想化模型：铁心变压器——全耦合变压器模型

满足条件： $k=1$ ； L_1 、 L_2 有限

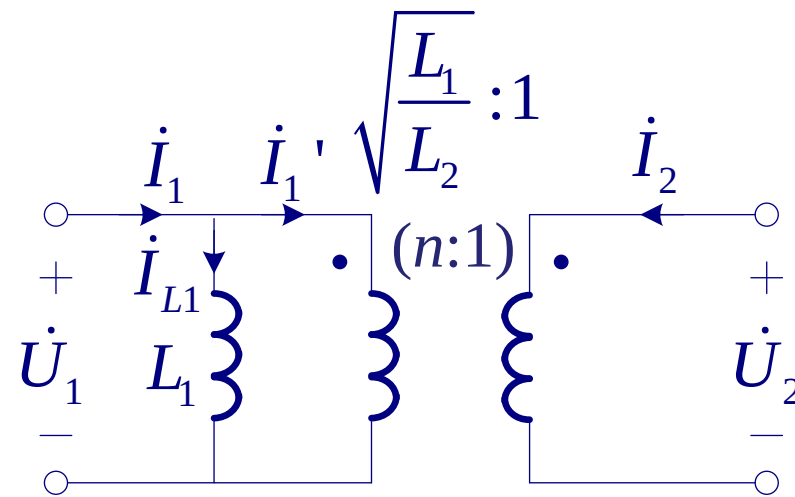
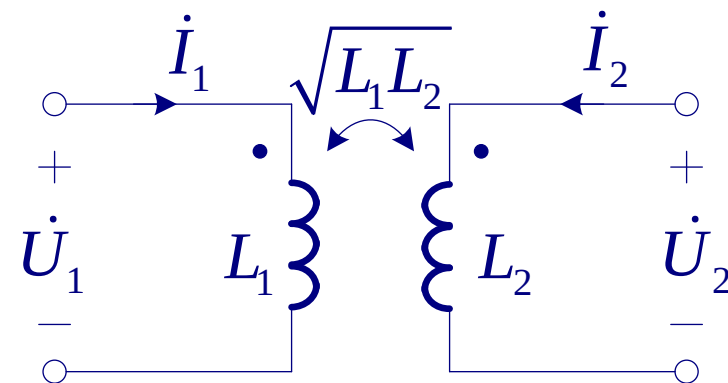
$$\dot{U}_1 = j\omega L_1 \dot{I}_{L1} = j\omega L_1 (\dot{I}_1 - \dot{I}_1')$$

$$\dot{I}_1' = -\dot{I}_2 / n = -\sqrt{\frac{L_2}{L_1}} \dot{I}_2$$

$$\dot{U}_1 = j\omega L_1 (\dot{I}_1 + \dot{I}_2 / n) = j\omega L_1 \dot{I}_1 + j\omega \sqrt{L_1 L_2} \dot{I}_2$$

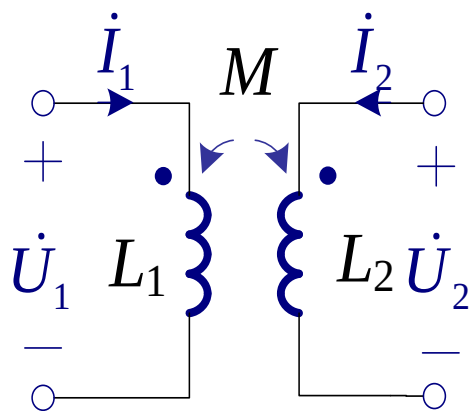
$$\dot{U}_2 = \frac{1}{n} \dot{U}_1 = j\omega \sqrt{L_1 L_2} \dot{I}_1 + j\omega L_2 \dot{I}_2$$

全耦合电感VCR

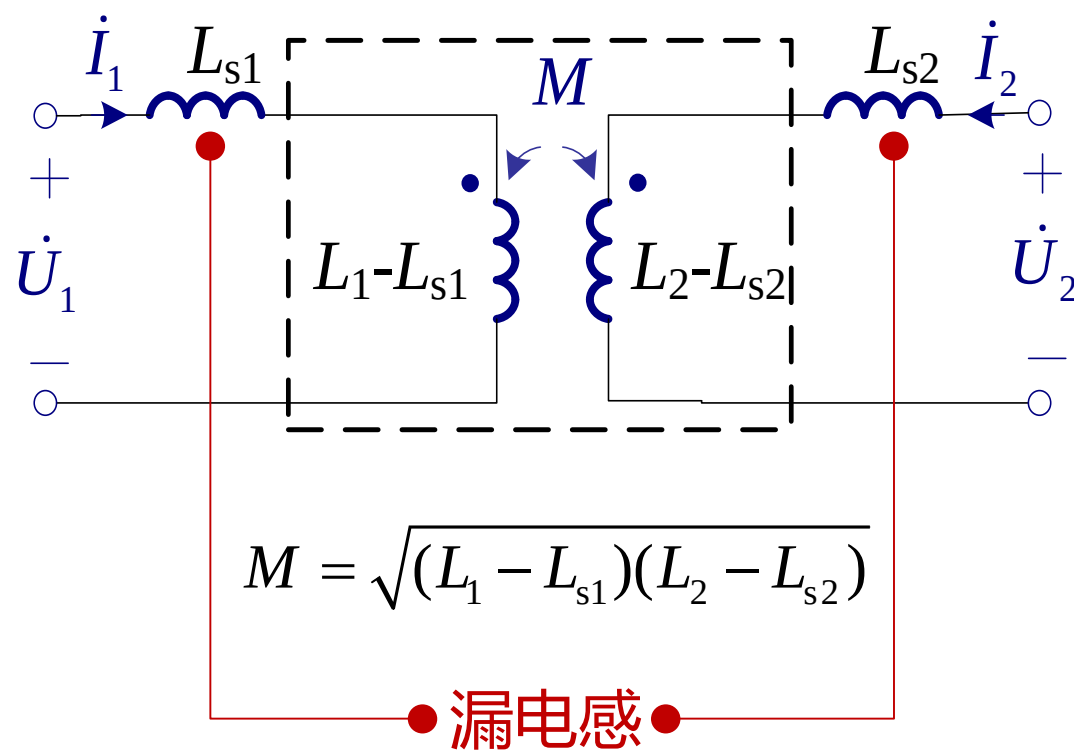


变压器的理想化模型：铁心变压器——一般变压器模型

满足条件： $k < 1$ ； L_1 、 L_2 有限



除互磁通外，还有漏磁通



例：全耦合变压器如图，试求各电压和电流。

解1：回路法（一般方法）

$$\begin{cases} j2\dot{I}_1 - j4\dot{I}_2 = 1\angle 0^\circ \\ -j4\dot{I}_1 + (8 + j8)\dot{I}_2 = 0 \end{cases} \quad \begin{cases} \dot{I}_1 = (0.5 - j0.5)\text{A} \\ \dot{I}_2 = 0.25\angle 0^\circ \text{A} \end{cases}$$

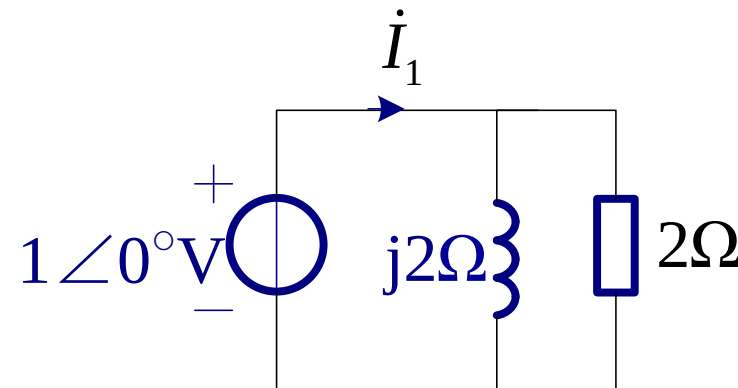
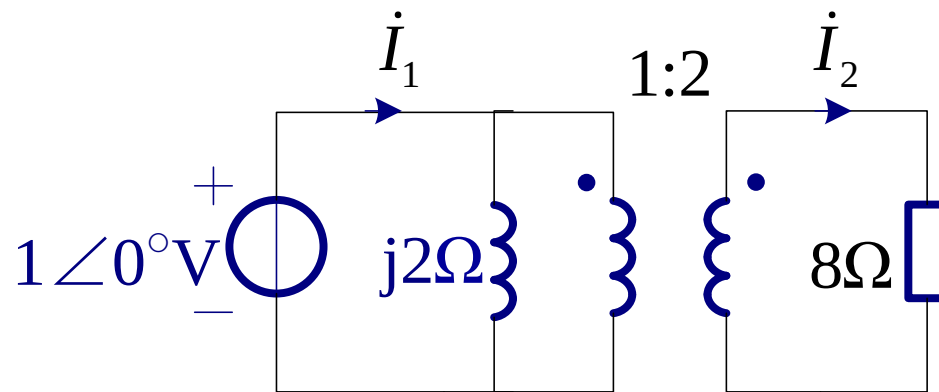
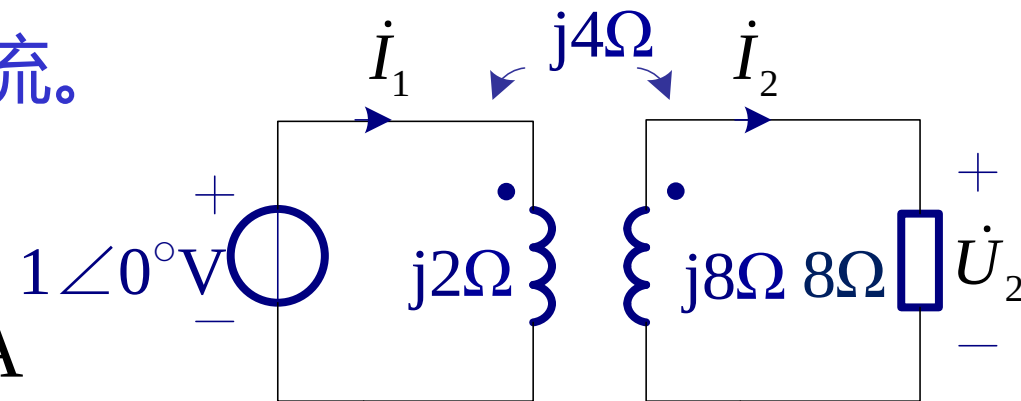
$$\dot{U}_2 = 8 \times \dot{I}_2 = 2\angle 0^\circ \text{V}$$

解2：全耦合变压器模型法

$$\dot{I}_1 = 1\angle 0^\circ / (j2 \parallel 2) = (0.5 - j0.5)\text{A}$$

$$\dot{U}_2 = \dot{U}_1 / n = 2 \times 1\angle 0^\circ = 2\angle 0^\circ \text{V}$$

$$\dot{I}_2 = \dot{U}_2 / 8 = 0.25\angle 0^\circ \text{A}$$



8.4.4 相量图法

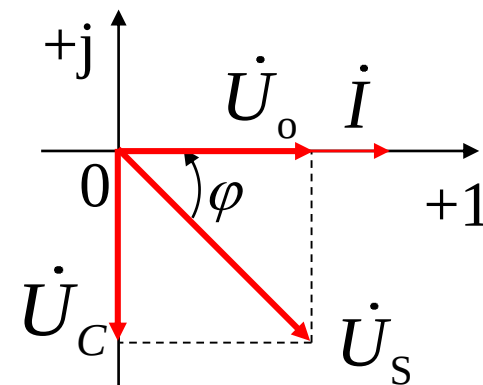
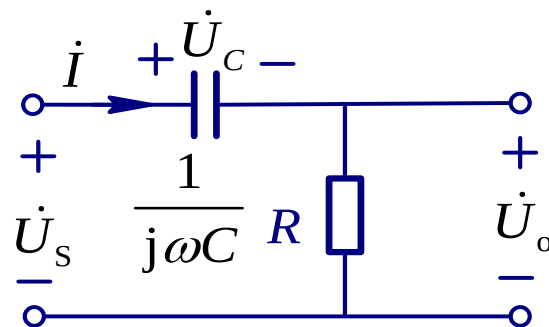
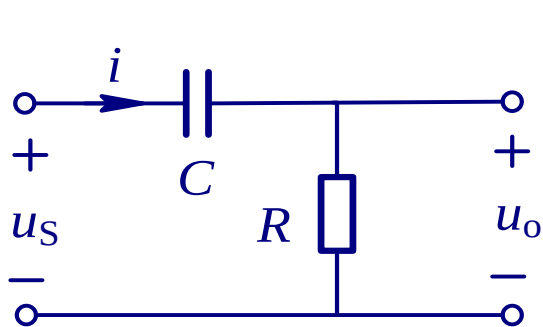
相量法求解正弦稳态电路的两种方法：

- ✓ 相量解析法 准确
- ✓ 相量图法：通过作电路中各相量的相量图来分析电路 直观

运用相量图法（定性）求解相位差问题的步骤：

- (1) 首先在复数坐标中画出参考相量，一般绘在实轴上。参考相量可选电压相量，也可选电流相量。
- (2) 利用元件VCR、KVL、KCL求出其他相量，并绘在复数坐标中。
- (3) 比较有关相量的相位差。

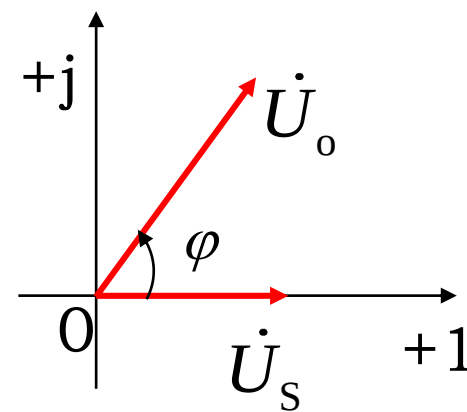
例：求输出电压对输入电压的相位关系。



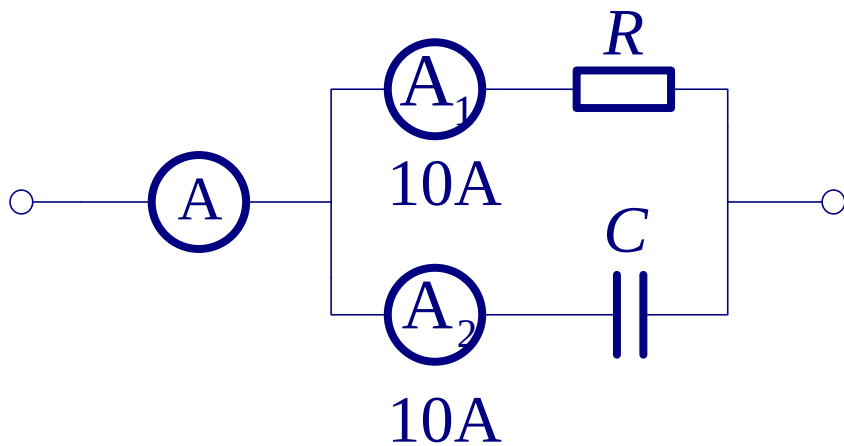
相量解析法：

$$\dot{U}_o = \dot{U}_s \frac{R}{R - j \frac{1}{\omega C}}$$

$$\Rightarrow \varphi_o - \varphi_s = \angle \tan^{-1} \left(\frac{1}{\omega C R} \right) > 0$$



练习：已知正弦稳态电路中的电流表 A_1 、 A_2 的读数如图，则电流表A的读数为



- A. $10A$
- B. $20A$
- C. $10\sqrt{2}A$
- D. $20\sqrt{2}A$
- E. 0

电路实例：电感线圈测试电路

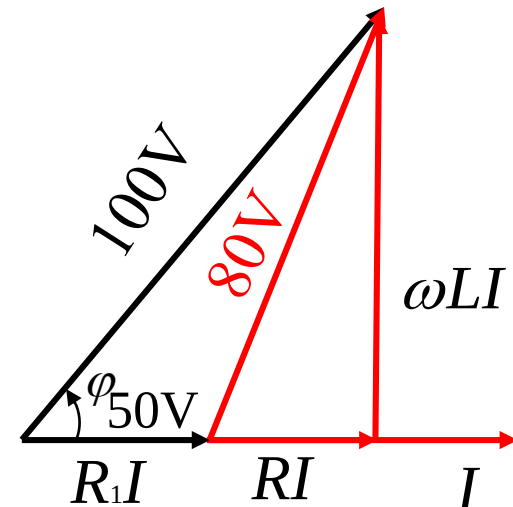
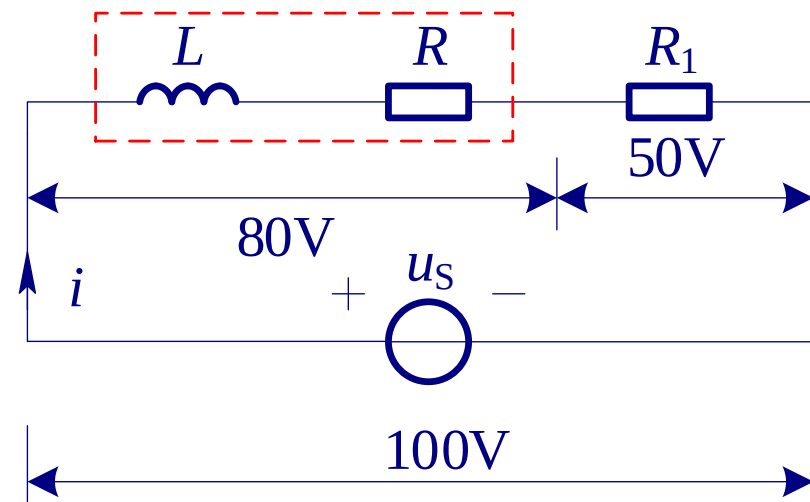
已知电压表的读数， $\omega=314\text{rad/s}$ ， $R_1=25\Omega$ ，求 L 、 R 。

$$R_1 I = 50 \Rightarrow I = \frac{50}{25} = 2\text{A}$$

$$\cos \varphi = \frac{100^2 + 50^2 - 80^2}{2 \times 100 \times 50} = 0.61 \Rightarrow \varphi = 52.4^\circ$$

$$100 \times \sin \varphi = \omega L I \Rightarrow L = \frac{100 \times \sin 52.4^\circ}{314 \times 2} = 0.126\text{H}$$

$$100 \times \cos \varphi - 50 = R I \Rightarrow R = \frac{100 \times \cos 52.4^\circ - 50}{2} = 5.5\Omega$$



已知电压表的读数, $\omega=314\text{rad/s}$, $R_1=25\Omega$, 求 L 、 R 。

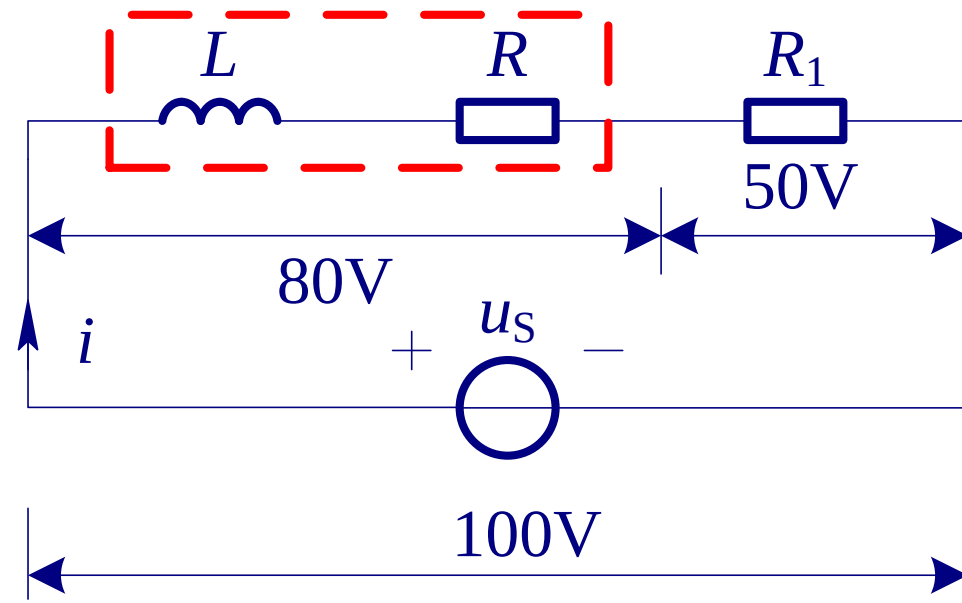
解析法: 假设 $\dot{U}_s = 100\angle 0^\circ\text{V}$

$$\left| \frac{R + j\omega L}{R + j\omega L + R_1} \dot{U}_s \right| = 80$$

$$\left| \frac{R_1}{R + j\omega L + R_1} \dot{U}_s \right| = 50$$

$$\begin{cases} \frac{\sqrt{R^2 + 314^2 L^2}}{\sqrt{(R + 25)^2 + 314^2 L^2}} \times 100 = 80 \\ \frac{25}{\sqrt{(R + 25)^2 + 314^2 L^2}} \times 100 = 50 \end{cases}$$

$$\begin{cases} \sqrt{R^2 + 314^2 L^2} = 40 \\ \sqrt{(R + 25)^2 + 314^2 L^2} = 50 \end{cases} \Rightarrow \begin{cases} R = 5.5\Omega \\ L = 0.126\text{H} \end{cases}$$



8.5 正弦稳态电路的功率

对于正弦稳态电路，功率和能量仍然满足定义：

$$\begin{cases} p(t) = u(t)i(t) \\ w(t_0, t_1) = \int_{t_0}^{t_1} u(t)i(t)dt = w(t_1) - w(t_0) \end{cases}$$

例如，对于电容：

$$\begin{cases} i(t) = C \frac{du(t)}{dt} \\ p(t) = u(t)i(t) = Cu(t) \frac{du(t)}{dt} = \frac{1}{2} C \frac{du^2(t)}{dt} \\ w(t_0, t_1) = \int_{t_0}^{t_1} u(t)i(t)dt = \frac{1}{2} C [u^2(t_1) - u^2(t_0)] \end{cases}$$

8.5.1 正弦稳态一端口电路的功率

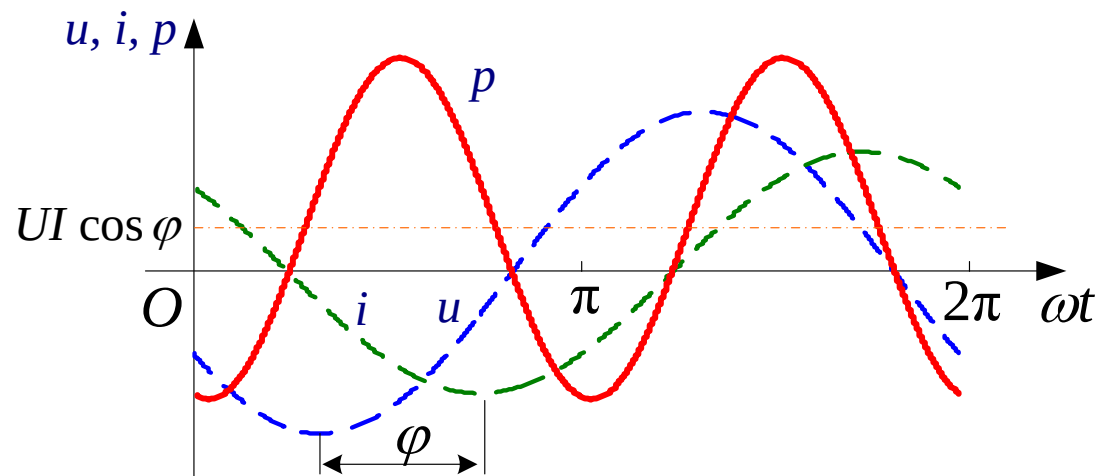
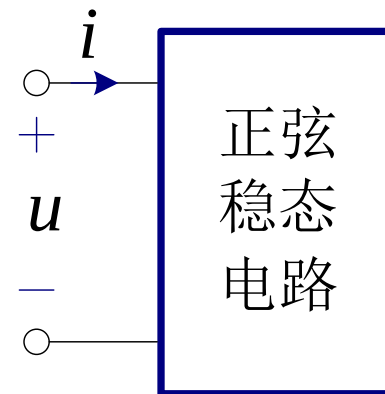
1. 瞬时功率

设端口电压和端口电流分别为

$$u = \sqrt{2}U \cos(\omega t + \varphi_u), \quad i = \sqrt{2}I \cos(\omega t + \varphi_i)$$

$$\begin{aligned} p &= ui = 2UI \cos(\omega t + \varphi_u) \cos(\omega t + \varphi_i) \\ &= UI \cos(\varphi_u - \varphi_i) + UI \cos(2\omega t + \varphi_u + \varphi_i) \end{aligned}$$

可见： p 随时间是变化的，不适合用来表示正弦稳态功率的大小。

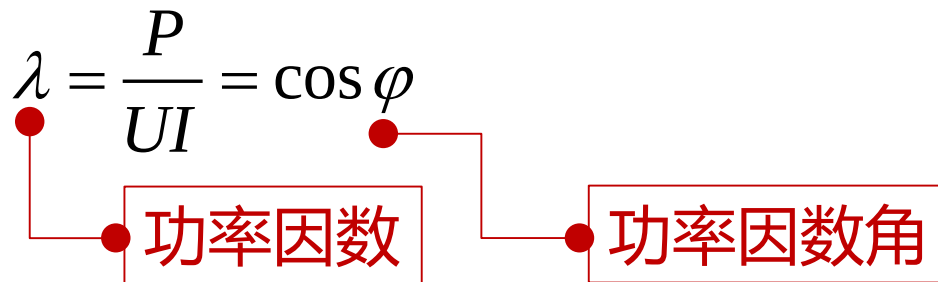


2. 平均功率和无功功率

定义（平均功率）： 瞬时功率在一个周期内的平均值。

平均功率是电路中实际消耗的功率，因此也称为有功功率。

$$\begin{aligned} P &= \frac{1}{T} \int_0^T p dt = \frac{1}{T} \int_0^T [UI \cos(\varphi_u - \varphi_i) + UI \cos(2\omega t + \varphi_u + \varphi_i)] dt \\ &= UI \cos(\varphi_u - \varphi_i) = UI \cos \varphi = UI \lambda \end{aligned}$$

$$\lambda = \frac{P}{UI} = \cos \varphi$$


功率因数

功率因数角

电阻的平均功率

$$P = UI \cos(\varphi_u - \varphi_i) = UI \cos \varphi = UI \lambda$$

$$\Rightarrow P_R = UI = RI^2 = GU^2$$

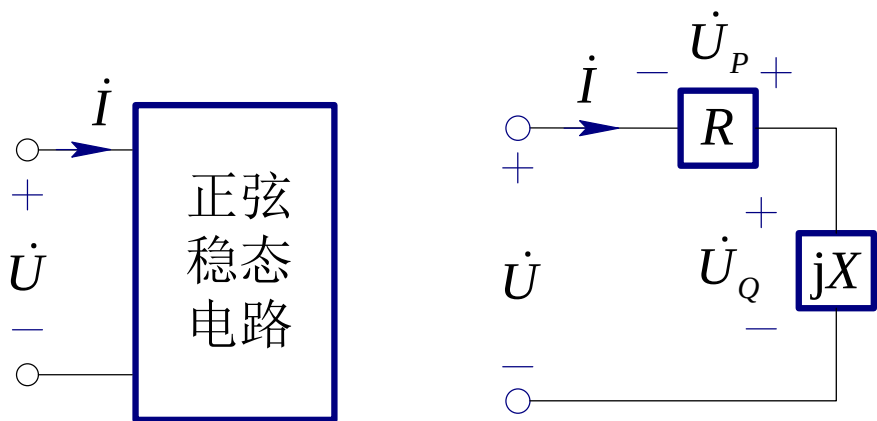
$p_R \geq 0$, 电路中的正电阻总是吸收功率。如果用端口电压、电流有效值来计算电阻的平均功率, 则其计算公式与电阻电路中的对应公式一致。

问题: 平均功率能够用来表征动态元件的功率吗?

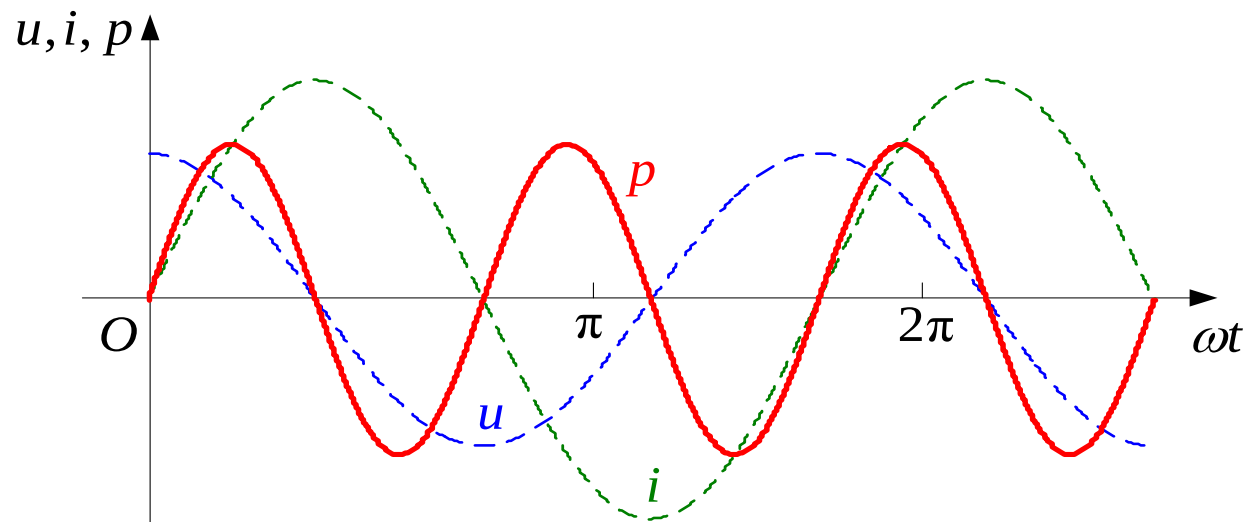
电感的无功功率

$$P_L = UI \lambda = 0$$

如何刻画动态元件的“功率”？



思路与方法： R 上消耗的功率为有功功率， jX 上“消耗”的功率($=I^2X$)为动态元件的功率。



$$p_L = UI \cos 90^\circ + UI \cos(2\omega t + \varphi_u + \varphi_u - 90^\circ) = UI \sin 2(\omega t + \varphi_u)$$

定义： 电感电压、电流有效值之积为无功功率(reactive power)。

$$Q_L = UI \quad \text{单位为乏(var)}$$

电感的平均储能

电感元件的瞬时能量为 $w_L = \frac{1}{2} Li^2 = LI^2 \cos^2(\omega t + \varphi_i) = \frac{1}{2} LI^2 [1 + \cos(2\omega t + 2\varphi_i)]$

电感储存的能量以 2ω 的频率在其平均值上下波动，且 $w_L \geq 0$ 。电感储能平均值为

$$W_L = \frac{1}{2} LI^2$$

$$\because \dot{U} = Z_L \dot{I} = j\omega L \dot{I} \quad U = \omega LI \quad \Rightarrow Q_L = UI = \omega LI^2 = 2\omega W_L$$

如果电感的平均储能越多，与外电路能量往返的频率越高，那么其无功功率就越大。

电容的无功功率和平均储能

电容吸收的瞬时功率和平均功率分别为

$$p_C = UI \cos(-90^\circ) + UI \cos(2\omega t + \varphi_u + \varphi_u + 90^\circ) = -UI \sin 2(\omega t + \varphi_u)$$

$$P_C = UI \lambda = 0$$

与电感元件类似，在电路理论中，把上式中的 $-UI$ 定义为电容元件的无功功率，记为 Q_C

$$Q_C = -UI$$

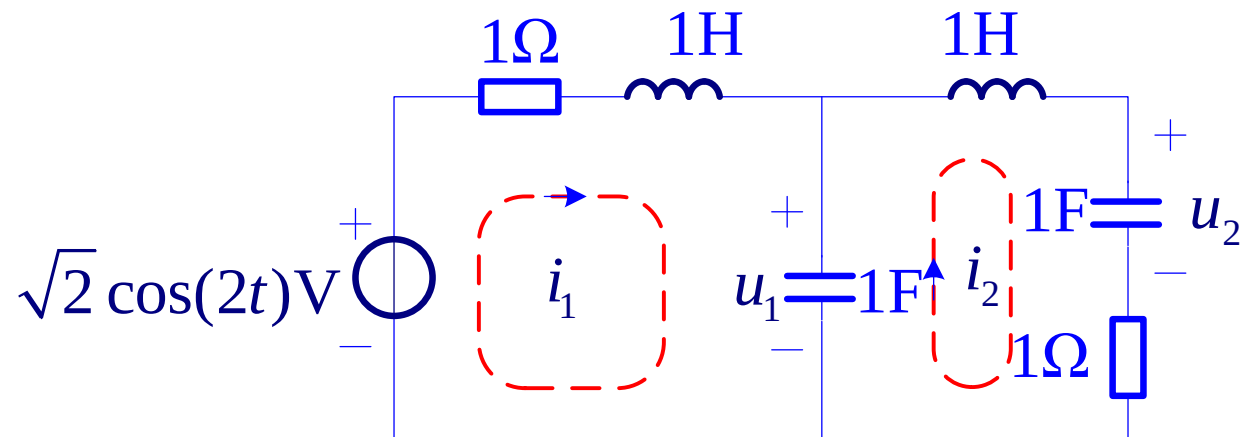
$$w_C = \frac{1}{2} Cu^2 = CU^2 \cos^2(\omega t + \varphi_u) = \frac{1}{2} CU^2 [1 + \cos(2\omega t + 2\varphi_u)]$$

$$W_C = \frac{1}{2} CU^2 \quad Q_C = -UI = -\omega CU^2 = -2\omega W_C$$

例：求各电阻的平均功率总和，各电感、电容的平均储能总和。

$$\begin{cases} (1 + j2 - j0.5)\dot{I}_1 - (-j0.5)\dot{I}_2 = 1\angle 0^\circ \\ -(-j0.5)\dot{I}_1 + (1 + j2 - j)\dot{I}_2 = 0 \end{cases}$$

$$\Rightarrow \begin{cases} \dot{I}_1 = 0.563\angle -50.7^\circ \text{ A} \\ \dot{I}_2 = 0.2\angle 174.3^\circ \text{ A} \end{cases}$$



$$P = I_1^2 R + I_2^2 R = 0.36 \text{ W}$$

$$W_L = LI_1^2 / 2 + LI_2^2 / 2 = 0.18 \text{ J}$$

$$U_1 = |(\dot{I}_1 - \dot{I}_2) \times (-j0.5)| = 0.36 \text{ V}$$

$$U_2 = |\dot{I}_2 \times (-j0.5)| = 0.1 \text{ V}$$

$$W_C = CU_1^2 / 2 + CU_2^2 / 2 = 0.0698 \text{ J}$$

平均功率和无功功率的物理含义

一端口吸收的平均功率为

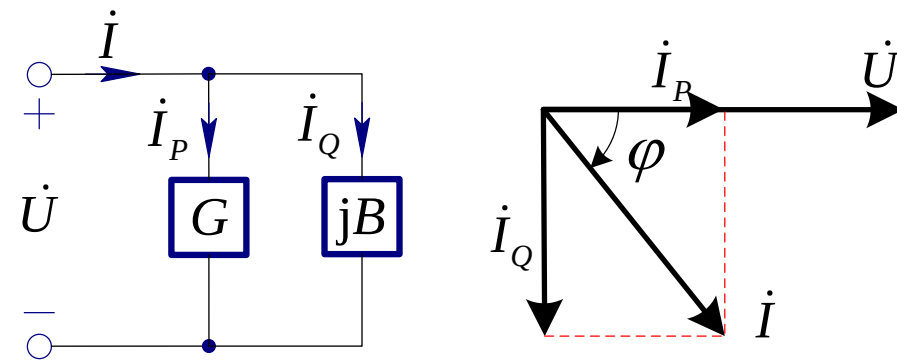
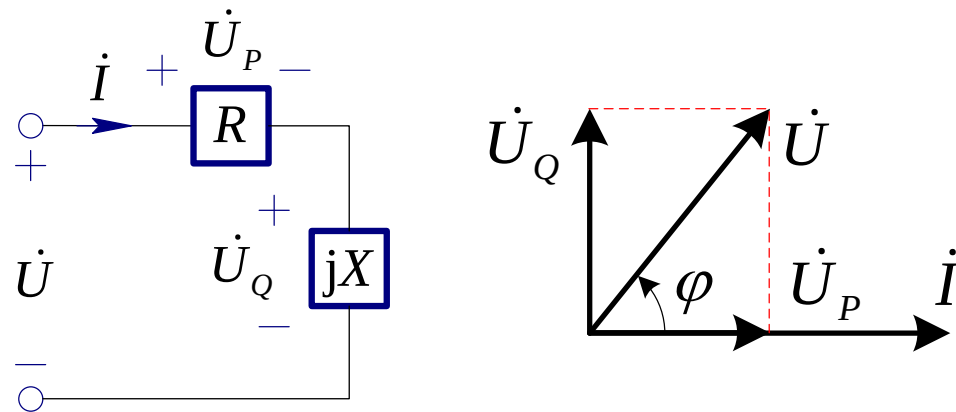
$$P = UI \cos \varphi = U_P I$$

称 $U_P = U \cos \varphi$ 为电压的有功分量或有功电压

相量图中的 $\dot{U}_Q$ 是与电流 i 正交的电压分量，
两者有效值之积得到无功功率

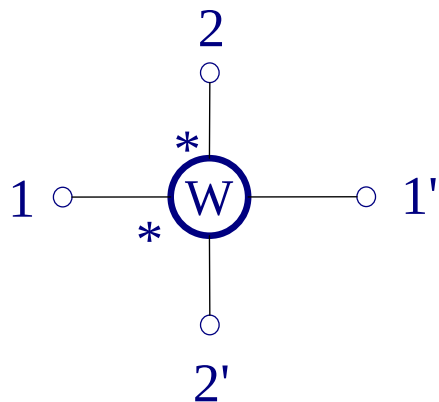
$$Q = UI \sin \varphi = U_Q I$$

称 $U_Q = U \sin \varphi$ 为电压的无功分量或无功电压

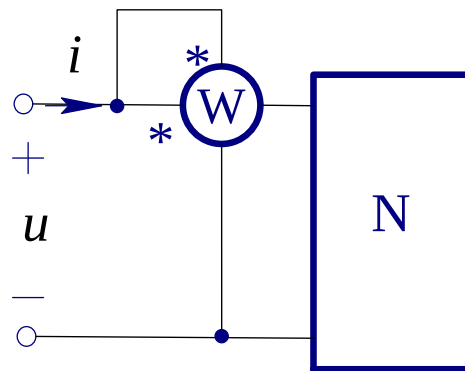


可作类似分析

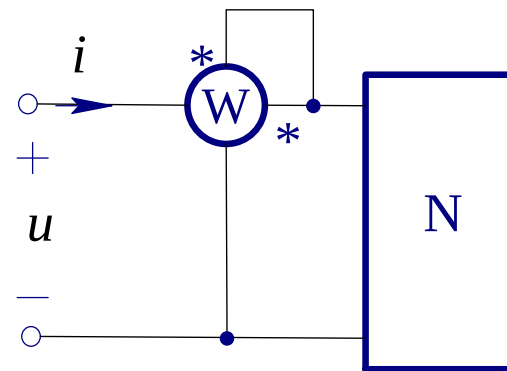
平均功率的测量



(a)



(b)



(c)

$$P = \frac{1}{T} \int_0^T u i dt = UI \cos(\varphi_u - \varphi_i) \quad P = -UI \cos(\varphi_u - \varphi_i)$$

练习：已知单口 $u = 300\cos(100t + 10^\circ)\text{V}$,
 $i = 50\sin(100t + 70^\circ)\text{A}$, 电压、电流
为关联参考方向, 则网络吸收功率为

A. $-\frac{1}{2} \times 300 \times 50 \times \frac{\sqrt{3}}{2} \text{W}$

B. $300 \times 50 \times \frac{\sqrt{3}}{2} \text{W}$

C. $-300 \times 50 \times \frac{\sqrt{3}}{2} \text{W}$

D. $\frac{1}{2} \times 300 \times 50 \times \frac{\sqrt{3}}{2} \text{W}$

例：如图，已知 $\dot{i} = 12.65\angle 18.5^\circ \text{A}$, $\dot{i}_1 = 20\angle -53.1^\circ \text{A}$, $\dot{i}_2 = 20\angle 90^\circ \text{A}$, $\dot{U} = 100\angle 0^\circ \text{V}$ ，求单口网络的平均功率。

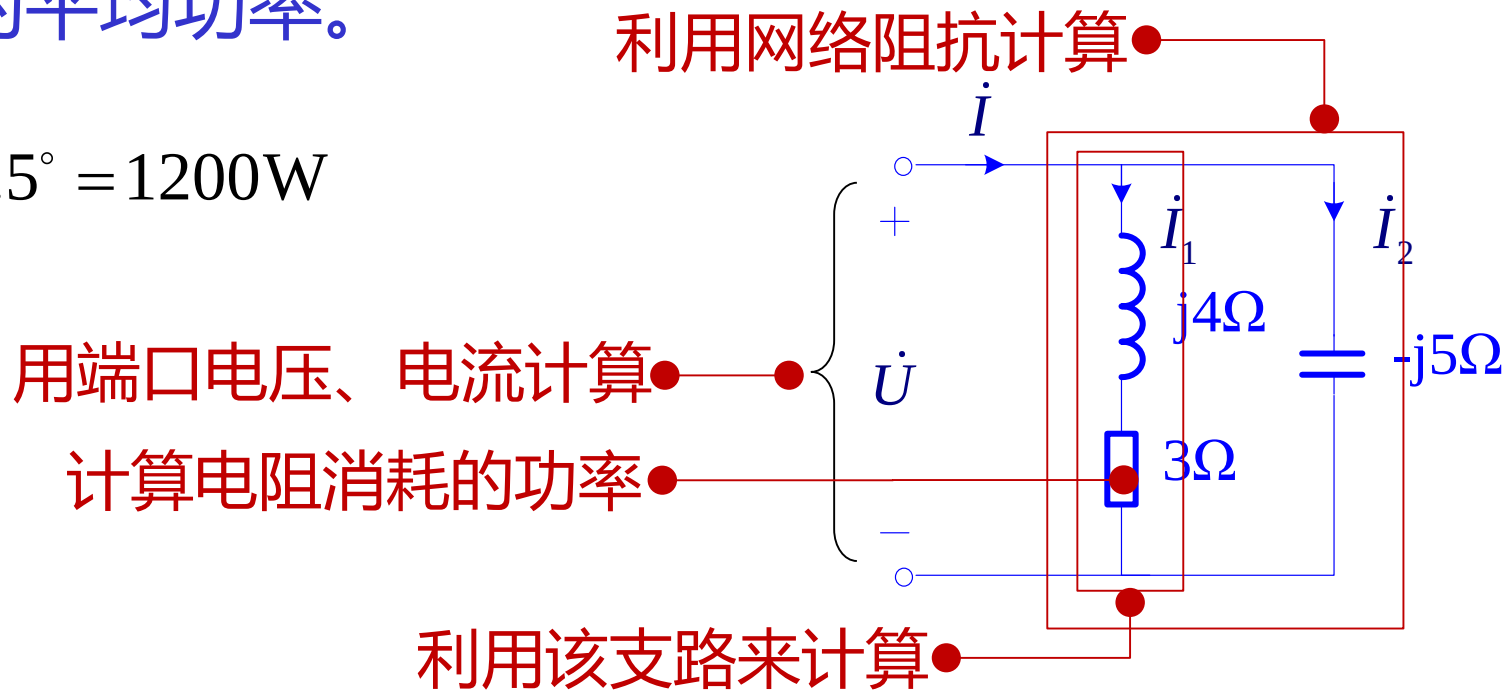
$$P = UI \cos \varphi = 100 \times 12.65 \cos 18.5^\circ = 1200 \text{W}$$

$$P = I_1^2 R = 20^2 \times 3 = 1200 \text{W}$$

$$\begin{aligned} P &= U_1 I_1 \cos(0^\circ + 53.1^\circ) \\ &= 100 \times 20 \cos(53.1^\circ) = 1200 \text{W} \end{aligned}$$

$$Z = \frac{(3 + j4)(-j5)}{3 + j4 - j5} = (7.5 - j2.5)\Omega$$

$$P = UI \cos \varphi = I^2 |Z| \cos \varphi = I^2 |Z| \frac{\text{Re}(Z)}{|Z|} = I^2 \text{Re}(Z) = 12.65^2 \times 7.5 = 1200 \text{W}$$



例：求发电机端电压、功率因数、提供的功率。

$$\lambda = 0.8 \Rightarrow \varphi = \cos^{-1} 0.8 = 36.87^\circ$$

$$P = UI_L \cos \varphi = 220 \times I_L \times 0.8 = 20 \times 10^3$$

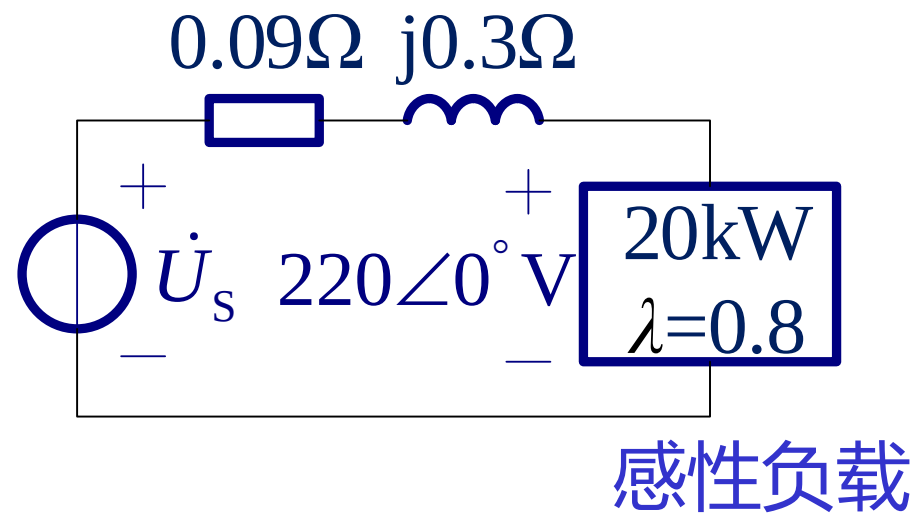
$$\Rightarrow I_L = 113.64 \text{ A}$$

$$\dot{I}_L = 113.64 \angle -36.87^\circ \text{ A}$$

$$\dot{U}_S = 220 \angle 0^\circ + (0.09 + j0.3)(113.64 \angle -36.87^\circ) = 249.53 \angle 4.86^\circ \text{ V}$$

$$\lambda_{\text{发}} = \cos[4.86^\circ - (-36.87^\circ)] = \cos 41.73^\circ = 0.746$$

$$P_S = U_S I_L \lambda_{\text{发}} = 21162 \text{ W}$$

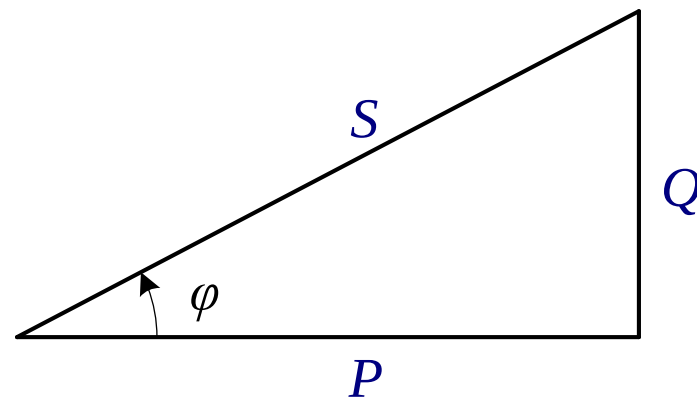


3. 视在功率

有功功率和无功功率的计算都涉及电压、电流有效值之积 UI 。在电路理论中，把这一乘积**定义**为**视在功率**。

$$S = UI$$

$$\begin{cases} S = \sqrt{P^2 + Q^2} \\ P = S \cos \varphi \\ Q = S \sin \varphi \end{cases}$$



功率三角形

视在功率的SI单位是伏安（VA）。电气设备往往用视在功率来表示其本身的容量。

例：发电厂向三个地方输送功率，仪表读数见右。试求发电厂端的功率因数。

$$\varphi = \cos^{-1} \frac{P}{S} = \sin^{-1} \frac{Q}{S} = \tan^{-1} \frac{Q}{P}$$

$$\Leftrightarrow \begin{cases} P = \sum P_k \\ Q = \sum Q_k \end{cases} \Leftrightarrow \begin{cases} P_k = S_k \lambda_k = S_k \cos \varphi_k \\ Q_k = \sqrt{S_k^2 - P_k^2} = S_k \sin \varphi_k \end{cases}$$

1号线：5MW 0.95 (感性)

2号线：2MV·A 0.9 (容性)

3号线：0.1Mvar 和1MW (容性)

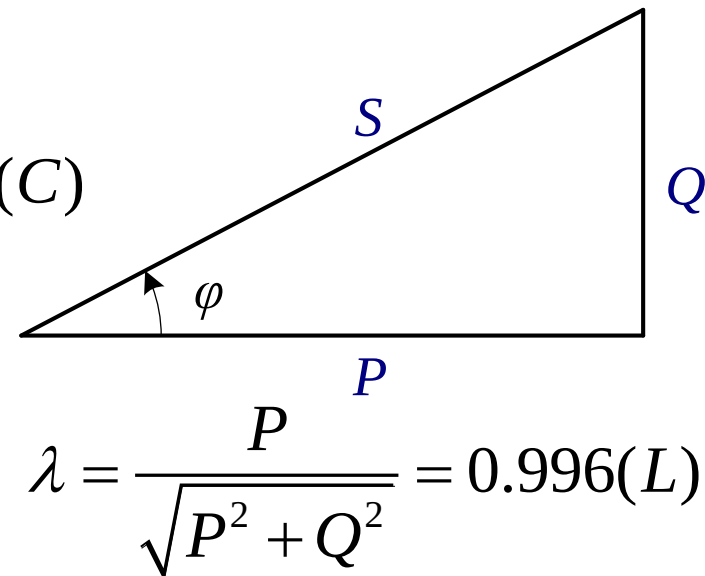
$$P_1 = 5\text{MW}, Q_1 = P_1 \tan \varphi_1 = 1.644\text{M var}(L)$$

$$P_2 = 2 \times 0.9\text{MW} = 1.8\text{MW}, Q_2 = P_2 \tan(\arccos 0.9) = 0.872\text{M var}(C)$$

$$P_3 = 1\text{MW}, Q_3 = 0.1\text{M var}(C)$$

$$P = P_1 + P_2 + P_3 = (5 + 1.8 + 1)\text{MW} = 7.8\text{MW}$$

$$Q = Q_1 + Q_2 + Q_3 = (1.644 - 0.872 - 0.1)\text{M var} = 0.672\text{M var}(L)$$



8.5.2 平均功率的叠加

不同激励单独作用产生的平均功率可否叠加？

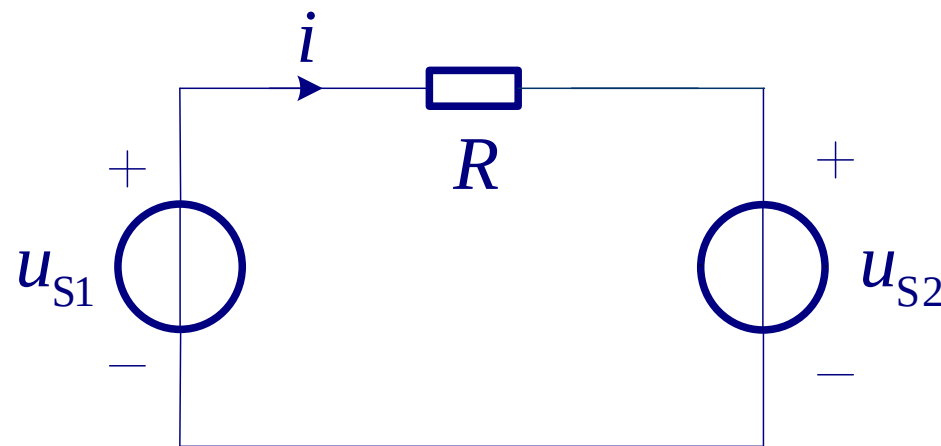
$$i(t) = i_1(t) + i_2(t)$$

$$p = R(i_1 + i_2)^2 = p_1 + p_2 + 2Ri_1i_2 \neq p_1 + p_2$$

$$P = \frac{1}{T} \int_0^T p dt = \frac{1}{T} \int_0^T (p_1 + p_2 + 2Ri_1i_2) dt$$

$$= P_1 + P_2 + \frac{2R}{T} \int_0^T i_1i_2 dt$$

● 如积分项等于零，则平均功率可叠加



设

$$i_1(t) = I_{1m} \cos(\omega_1 t + \varphi_1)$$
$$i_2(t) = I_{2m} \cos(\omega_2 t + \varphi_2)$$

$$\frac{\omega_1}{\omega_2} = \frac{r_1}{r_2} \Rightarrow \begin{cases} T = r_2 T_2 = r_1 T_1 \\ \omega = 2\pi / T \end{cases}$$

$$\begin{aligned} \int_0^T i_1 i_2 dt &= I_{1m} I_{2m} \int_0^{2\pi/\omega} \cos(r_1 \omega t + \varphi_1) \cos(r_2 \omega t + \varphi_2) dt \\ &= \frac{1}{2} I_{1m} I_{2m} \int_0^{2\pi/\omega} \{ \cos[(r_1 + r_2) \omega t + \varphi_1 + \varphi_2] + \cos[(r_1 - r_2) \omega t + \varphi_1 - \varphi_2] \} dt \\ &= \begin{cases} \frac{I_{1m} I_{2m} \pi \cos(\varphi_1 - \varphi_2)}{\omega}, & r_1 = r_2 \\ 0, & r_1 \neq r_2 \end{cases} \end{aligned}$$

结论：（1）频率相同的正弦量，其平均功率不满足叠加原理；（2）不同频率的正弦电流（电压）产生的平均功率等于每一正弦电流（电压）单独作用时所产生的平均功率之和。

非正弦周期稳态平均功率的计算

设流过电阻 R 的周期电流为

$$i(t) = I_0 + \sqrt{2}I_1\cos(\omega_1 t + \varphi_1) + \cdots + \sqrt{2}I_N\cos(\omega_N t + \varphi_N)$$

$$\Rightarrow P = I_0^2 R + I_1^2 R + \cdots + I_N^2 R = P_0 + P_1 + \cdots + P_N$$

利用有效值的概念，可得周期电流的有效值 I

$$I^2 R = I_0^2 R + I_1^2 R + \cdots + I_N^2 R$$

$$\Rightarrow I = \sqrt{I_0^2 + I_1^2 + \cdots + I_N^2}$$

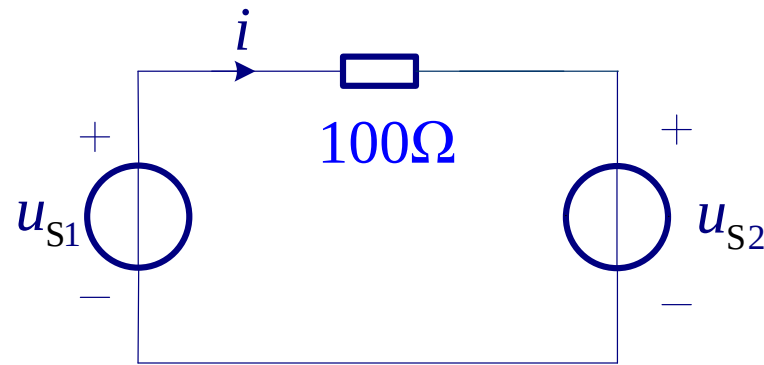
$$\Rightarrow U = \sqrt{U_0^2 + U_1^2 + \cdots + U_N^2}$$

例：已知电压源值如下，试求 R 的平均功率。

$$(1) u_{S1}(t) = 100 \cos(314t + 60^\circ) \text{V}, u_{S2}(t) = 50 \cos(314t) \text{V}$$

$$(2) u_{S1}(t) = 100 \cos(314t + 60^\circ) \text{V}, u_{S2}(t) = 50 \text{V}$$

$$(3) u_{S1}(t) = 100 \cos(314t + 60^\circ) \text{V}, u_{S2}(t) = 50 \cos(471t) \text{V}$$



$$\dot{I}' = \frac{1}{\sqrt{2}} \angle 60^\circ \text{ A}$$

$$P_1 = \frac{U_{S1}^2}{R} = \frac{(100/\sqrt{2})^2}{100} = 50 \text{ W}$$

$$P_1 = \frac{(100/\sqrt{2})^2}{100} = 50 \text{ W}$$

$$\dot{I}'' = -\frac{1}{2\sqrt{2}} \angle 0^\circ \text{ A}$$

$$P_2 = \frac{U_{S2}^2}{R} = \frac{(50)^2}{100} = 25 \text{ W}$$

$$P_2 = \frac{(50/\sqrt{2})^2}{100} = 12.5 \text{ W}$$

$$\dot{I} = \dot{I}' + \dot{I}'' = j \frac{0.866}{\sqrt{2}} \text{ A}$$

$$P = P_1 + P_2 = 75 \text{ W}$$

$$P = P_1 + P_2 = 62.5 \text{ W}$$

$$P = \left(\frac{0.866}{\sqrt{2}} \right)^2 (100) = 37.5 \text{ W}$$

单口网络吸收的功率

假设电压、电流为关联参考方向，端口电压、电流为

$$\begin{cases} i(t) = I_0 + I_{1m} \cos(\omega_1 t + \varphi_{i1}) + \cdots + I_{Nm} \cos(\omega_N t + \varphi_{iN}) \\ u(t) = U_0 + U_{1m} \cos(\omega_1 t + \varphi_{u1}) + \cdots + U_{Nm} \cos(\omega_N t + \varphi_{uN}) \end{cases}$$

$$\begin{aligned} P &= \frac{1}{T} \int_0^T [U_0 + U_{1m} \cos(\omega_1 t + \varphi_{u1}) + \cdots][I_0 + I_{1m} \cos(\omega_1 t + \varphi_{i1}) + \cdots] dt \\ &= \frac{1}{T} \int_0^T (U_0 I_0 + U_{1m} I_{1m} \cos(\omega_1 t + \varphi_{u1}) \cos(\omega_1 t + \varphi_{i1}) + \cdots) dt \\ &= U_0 I_0 + U_1 I_1 \cos(\varphi_{u1} - \varphi_{i1}) + \cdots + U_N I_N \cos(\varphi_{uN} - \varphi_{iN}) \\ &= P_0 + P_1 + \cdots + P_N \end{aligned}$$

练习：已知端口电压、电流为

$$u(t) = [100 + 100 \cos t + 50 \cos 2t + 30 \cos 3t] \text{V}$$

$$i(t) = [10 \cos(t - 60^\circ) + 2 \cos(3t + 120^\circ)] \text{A}$$

取关联参考方向，则端口吸收的功率为

A. 265W

B. 235W

C. 530W

D. 470W

E. 0

8.5.3 复功率 复功率守恒

平均功率、无功功率能否通过电压、电流相量直接进行计算？

$$\tilde{S} = P + jQ = UI \cos \varphi + jUI \sin \varphi = UIe^{j(\varphi_u - \varphi_i)} = Ue^{j\varphi_u} \times Ie^{-j\varphi_i} = \dot{U}\dot{I}^*$$

称 $\dot{U}\dot{I}^*$ 为复功率(VA)，记为 $\tilde{S}$ 。

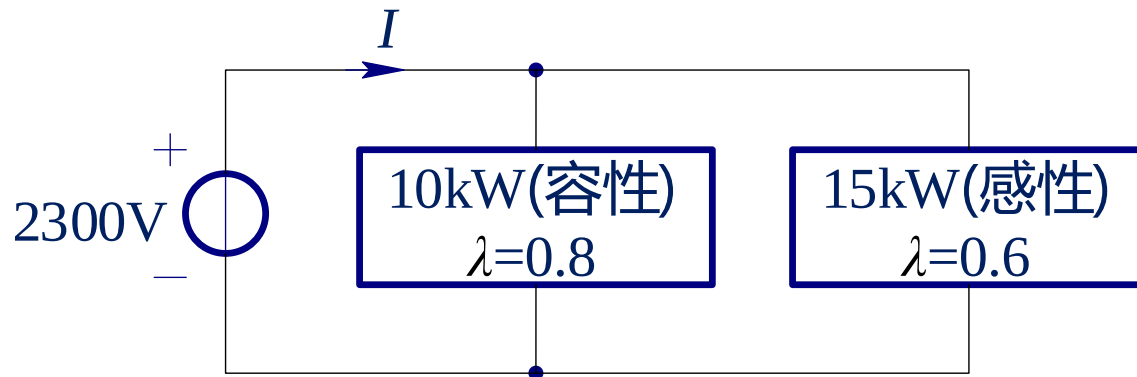
$$\tilde{S} = P + jQ = \dot{U}\dot{I}^* = Z\dot{I}\dot{I}^* = ZI^2 = RI^2 + jXI^2$$

复功率守恒： 在任意复杂的电路网络中，某些支路发出的复功率之和等于其余各支路吸收的复功率之和。

证明：

$$\sum_{k=1}^K \dot{I}_k = 0 \quad \sum_{k=1}^K \dot{U}_k = 0 \Rightarrow \sum_{k=1}^K \dot{I}_k^* = 0 \Rightarrow \sum_{k=1}^K \dot{U}_k \dot{I}_k^* = 0$$

例：试求输入电流和总功率因数。



$$\tilde{S} = \tilde{S}_1 + \tilde{S}_2 \quad I = S/U = |\tilde{S}|/U$$

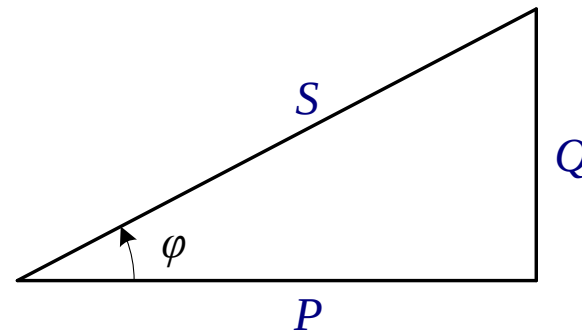
$$\lambda = \text{Re}(\tilde{S})/|\tilde{S}|$$

$$\tilde{S}_1 = P_1 + jP_1 \frac{-\sqrt{1-\lambda_1^2}}{\lambda_1} = (10000 - j7500) \text{V} \cdot \text{A}$$

$$\tilde{S}_2 = P_2 + jP_2 \frac{\sqrt{1-\lambda_2^2}}{\lambda_2} = (15000 + j20000) \text{V} \cdot \text{A}$$

$$\tilde{S} = \tilde{S}_1 + \tilde{S}_2 = (25000 + j12500) \text{V} \cdot \text{A} = 27951 \angle 26.56^\circ \text{V} \cdot \text{A}$$

$$I = |\tilde{S}|/U = \frac{27951}{2300} = 12.2 \text{A} \quad \lambda = \text{Re}(\tilde{S})/|\tilde{S}| = \frac{25000}{27951} = 0.8957$$

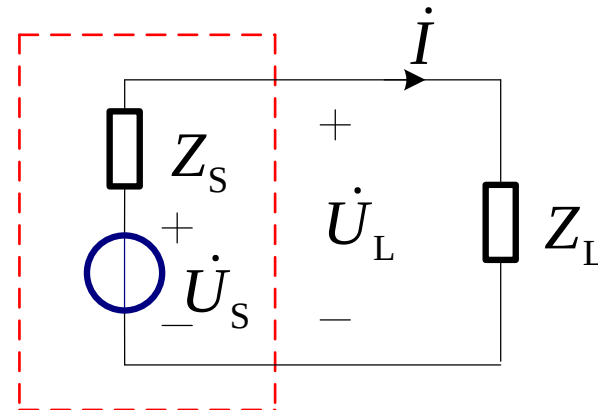
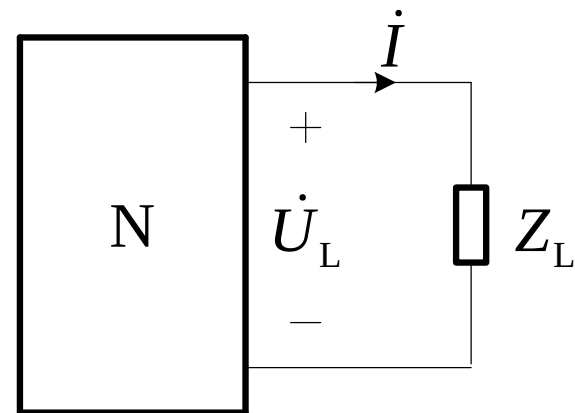


8.5.4 正弦稳态最大功率传递定理

问题： 给定电源及其阻抗，求负载阻抗为何值时，负载阻抗获得的有功功率（负载电阻获得的功率）为最大？

条件：

- (1) 负载电阻及电抗均可变化
- (2) 负载阻抗角固定而模可改变
- (3) 负载阻抗角可变而模固定



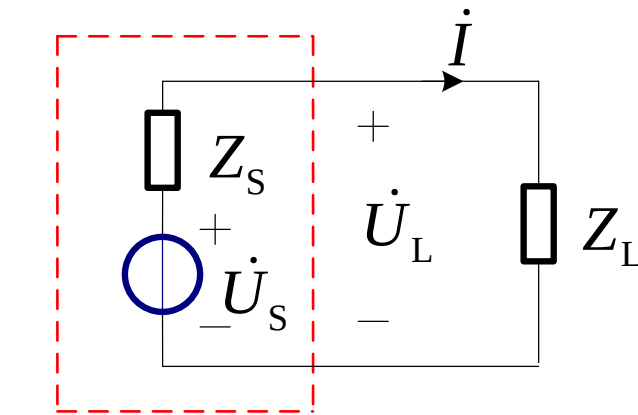
(1) 已知 $\dot{U}_S$, $Z_S = R_S + jX_S$, 求 $Z_L = R_L + jX_L$ 。

$$\dot{I} = \frac{\dot{U}_S}{(R_S + R_L) + j(X_S + X_L)} \quad I = \frac{U_S}{\sqrt{(R_S + R_L)^2 + (X_S + X_L)^2}}$$

$$\Rightarrow P_L = I^2 R_L = \frac{U_S^2}{(R_S + R_L)^2 + (X_S + X_L)^2} R_L$$

令 $\frac{\partial P_L}{\partial R_L} = 0, \frac{\partial P_L}{\partial X_L} = 0$

$$\begin{cases} \frac{U_S^2}{[(R_S + R_L)^2 + (X_S + X_L)^2]^2} [(R_S + R_L)^2 - 2(R_S + R_L)R_L] = 0 \\ \frac{2U_S^2}{[(R_S + R_L)^2 + (X_S + X_L)^2]^2} (X_S + X_L) = 0 \end{cases} \Rightarrow \begin{cases} R_L = R_S \\ X_L = -X_S \end{cases}$$



负载阻抗与电源内阻抗为最大功率匹配或共轭匹配

$$Z_L = Z_S^*$$

$$P_{L\max} = \frac{U_S^2}{4R_S}$$

(2) 可设 $Z_L = |Z| \angle \varphi = |Z| \cos \varphi + j|Z| \sin \varphi$

$$\dot{I} = \frac{\dot{U}_S}{(R_S + |Z| \cos \varphi) + j(X_S + |Z| \sin \varphi)}$$

$$I = \frac{U_S}{\sqrt{(R_S + |Z| \cos \varphi)^2 + (X_S + |Z| \sin \varphi)^2}}$$

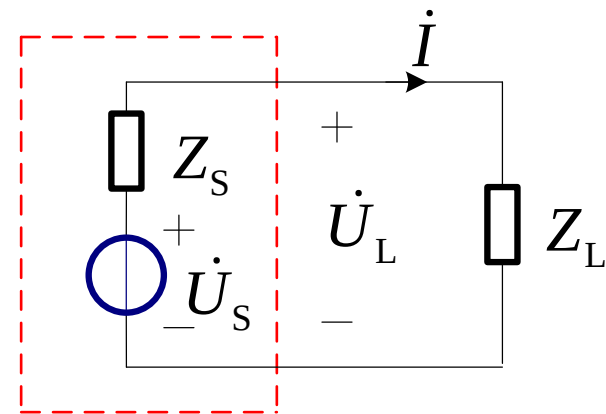
$$\Rightarrow P_L = I^2 R_L = \frac{U_S^2 |Z| \cos \varphi}{(R_S + |Z| \cos \varphi)^2 + (X_S + |Z| \sin \varphi)^2}$$

令 $dP_L / d|Z| = 0$

$$\frac{U_S^2 \cos \varphi}{[(R_S + |Z| \cos \varphi)^2 + (X_S + |Z| \sin \varphi)^2]^2} \times \{(R_S + |Z| \cos \varphi)^2 + (X_S + |Z| \sin \varphi)^2 - 2|Z|[(R_S + |Z| \cos \varphi) \cos \varphi + (X_S + |Z| \sin \varphi) \sin \varphi]\} = 0$$

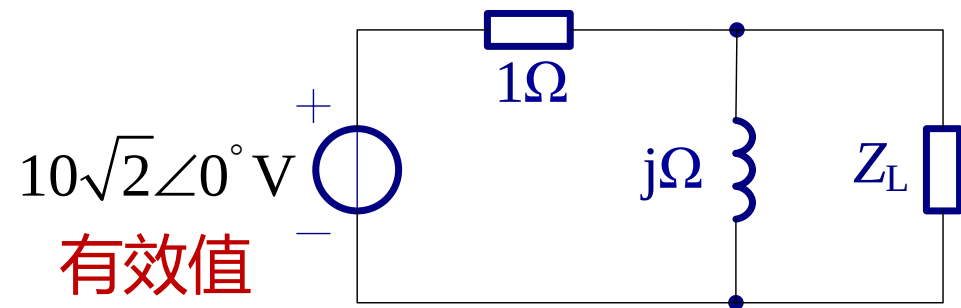
$$\Rightarrow |Z|^2 = R_S^2 + X_S^2$$

$$\Rightarrow |Z| = \sqrt{R_S^2 + X_S^2}$$



负载阻抗与电源内阻抗的模相等，称为模匹配

练习：负载为何值时，获得最大功率？



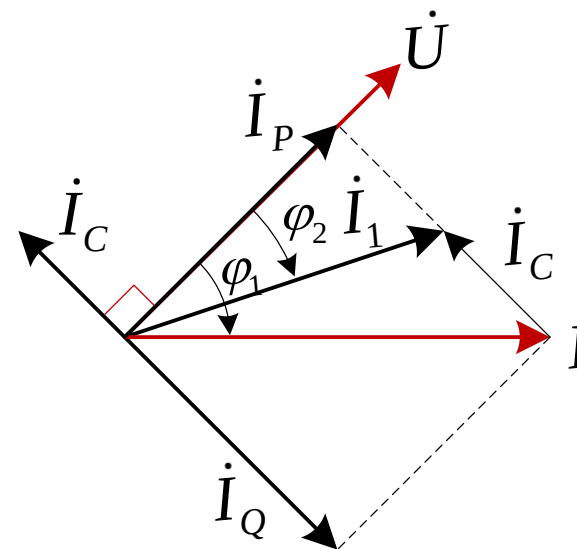
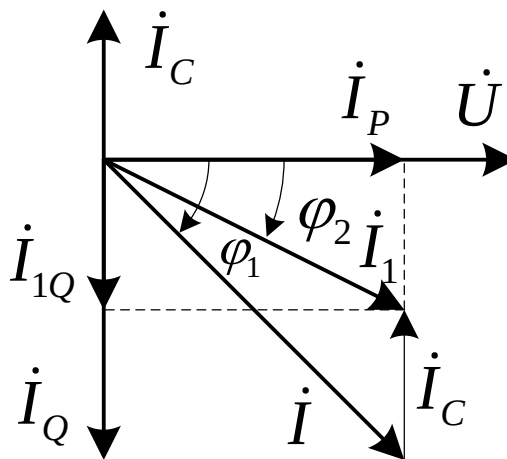
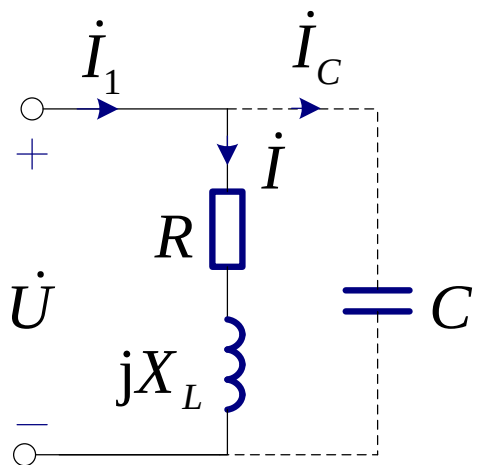
- A. $Z_L = (0.5 - j0.5)\Omega$, $P_{L\max} = 25W$
- B. $Z_L = (0.5 + j0.5)\Omega$, $P_{L\max} = 50W$
- C. $Z_L = (0.5 + j0.5)\Omega$, $P_{L\max} = 25W$
- D. $Z_L = (0.5 - j0.5)\Omega$, $P_{L\max} = 50W$
- E. $Z_L = (0.5 - j0.5)\Omega$, $P_{L\max} = 100W$

8.5.5 功率因数的提高

问题 (1) 为什么要提高功率因数?

由 $P = UI\lambda$ 知, 负载端的有功功率小于供电端的视在功率为 UI 。

(2) 如何提高功率因数: 将功率因数从 λ_1 提高到 λ_2 , 需并联多大电容?



已知: P 、 U 、 λ_1 、 λ_2 。 **求:** C 。 [无功功率补偿]

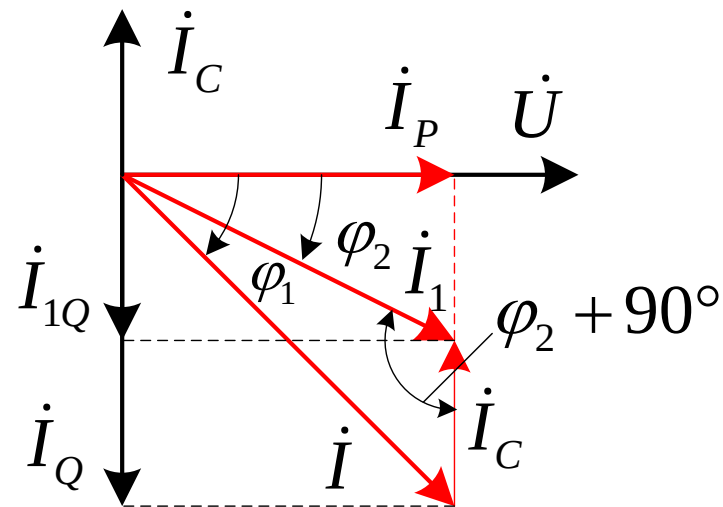
解1: 解直角三角形

$$\tan \varphi_2 = \frac{I_{1Q}}{I_P} = \frac{I \sin \varphi_1 - \omega CU}{I \cos \varphi_1}$$

$$\Rightarrow C = \frac{P}{\omega U^2} (\tan \varphi_1 - \tan \varphi_2)$$

$$I = \frac{P}{U \cos \varphi_1}$$

已知: P 、 U 、 λ_1 、 λ_2 。求: C 。



解2: 解直角三角形

$$\tan \varphi_1 = \frac{I_C + I_P \tan \varphi_2}{I_P} = \frac{\omega CU}{P/U} + \tan \varphi_2$$

$$\Rightarrow C = \frac{P}{\omega U^2} (\tan \varphi_1 - \tan \varphi_2)$$

解3: 利用正弦定理

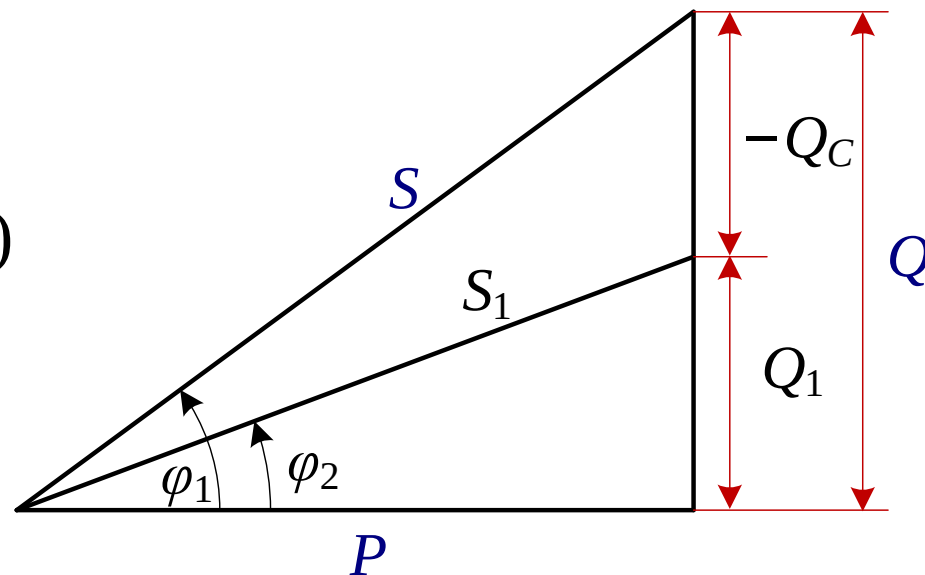
$$\frac{I}{\sin(\varphi_2 + \pi/2)} = \frac{I_C}{\sin(\varphi_1 - \varphi_2)}$$

$$\Rightarrow \frac{P}{U \cos \varphi_1 \cos \varphi_2} = \frac{\omega CU}{\sin(\varphi_1 - \varphi_2)}$$

解4：利用功率三角形

$$-Q_C = \omega C U^2 = Q - Q_1 = P(\tan \varphi_1 - \tan \varphi_2)$$

$$\Rightarrow C = \frac{P}{\omega U^2} (\tan \varphi_1 - \tan \varphi_2)$$



解5：利用阻抗/导纳的概念

$$Y = G + jB \quad \leftarrow \quad \varphi_1 \quad \text{阻抗角}$$

$$\tan(-\varphi_1) = \frac{B}{G}$$

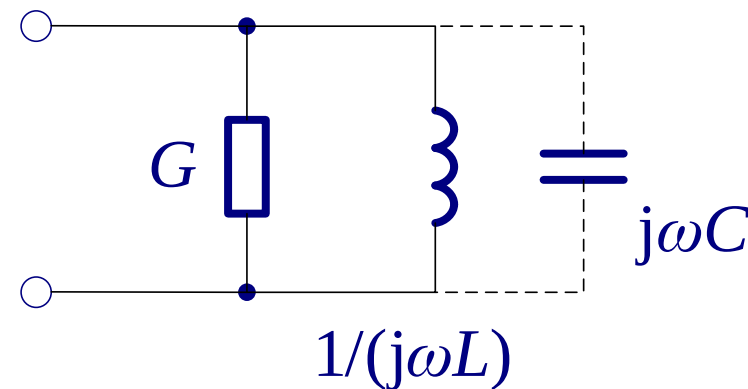
$$Y' = G + jB + j\omega C \quad \leftarrow \quad \varphi_2$$

$$\tan(-\varphi_2) = \frac{B + \omega C}{G}$$

$$\tan \varphi_1 - \tan \varphi_2 = \frac{\omega C}{G}$$

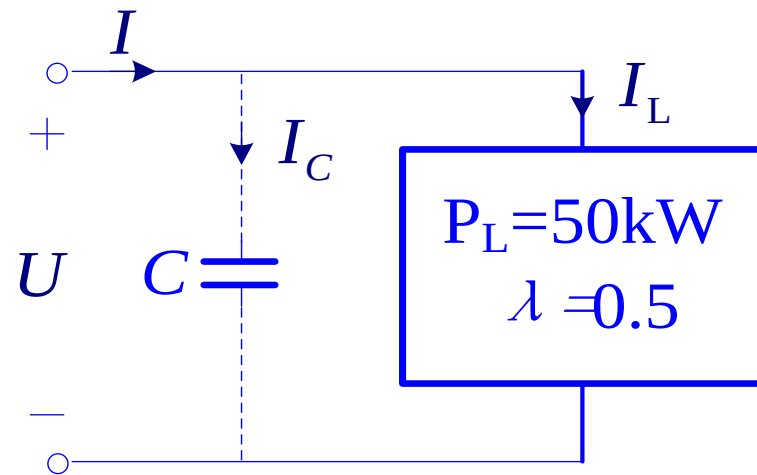
$$P = G U^2$$

$$\Rightarrow C = \frac{P}{\omega U^2} (\tan \varphi_1 - \tan \varphi_2)$$



例：如图，负载为220V、50Hz的感应电动机。试求电源供应的电流及无功功率。若并联电容使功率因数达到1，问电容的大小及电源提供的电流。

解1:
$$C = \frac{P}{\omega U^2} (\tan \varphi_1 - \tan \varphi_2)$$
$$= \frac{50000}{314 \times 220^2} \times \frac{\sqrt{1-0.5^2}}{0.5} = 5.698 \times 10^{-3} \text{ F}$$



并联电容前
$$I = I_L = \frac{P_L}{U \cos \varphi_L} = \frac{50000}{220 \times 0.5} = 454.5 \text{ A}$$

并联电容后
$$I = \frac{P}{U \cos \varphi} = \frac{P}{U} = \frac{50000}{220} = 227.3 \text{ A}$$

解2：并联电容后，网络阻抗为电阻性。

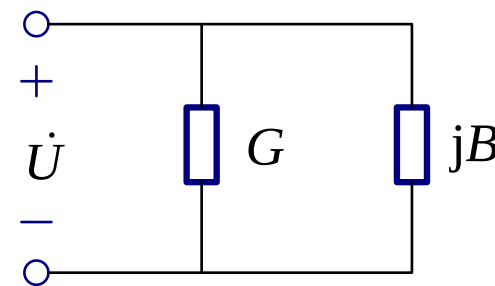
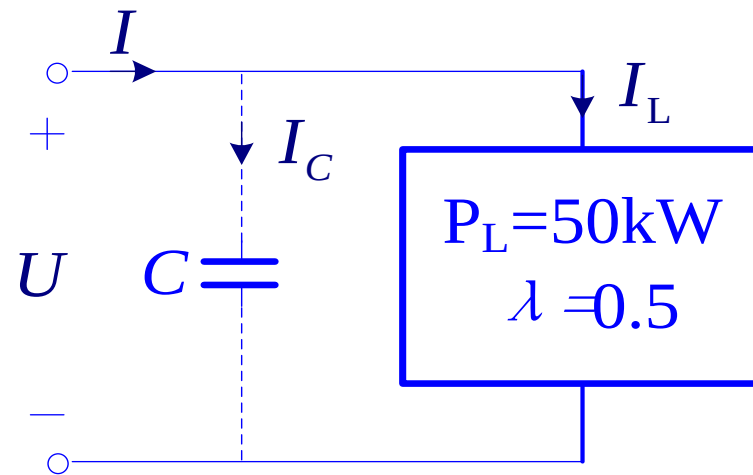
$$Y' = G + jB + j\omega C = G \Rightarrow B + \omega C = 0$$

$$P = UI \cos \varphi_L = U^2 |Y| \cos \varphi_L$$

$$\Rightarrow |Y| = \frac{P}{U^2 \cos \varphi_L}$$

$$\Rightarrow B = -|Y| \sin \varphi_L = -|Y| \sqrt{1 - \cos^2 \varphi_L}$$

$$\Rightarrow C = -\frac{B}{\omega} = \frac{|Y| \sqrt{1 - \cos^2 \varphi_L}}{\omega} = \frac{P \sqrt{1 - \cos^2 \varphi_L}}{\omega U^2 \cos \varphi_L}$$



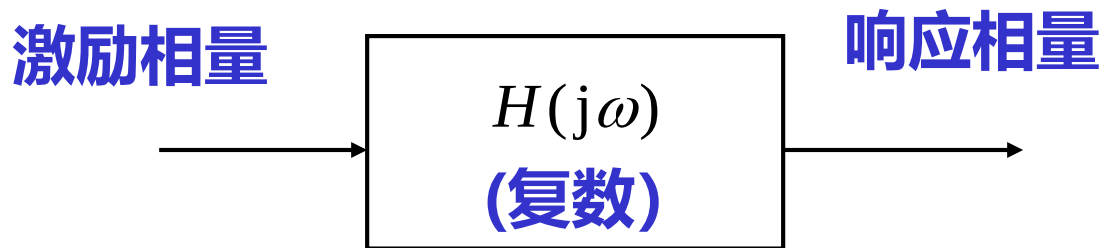
$$\begin{cases} P = GU^2 \\ \tan(-\varphi_L) = \frac{B}{G} \end{cases}$$

8.6 正弦稳态网络函数和频率响应

8.6.1 正弦稳态网络函数的定义

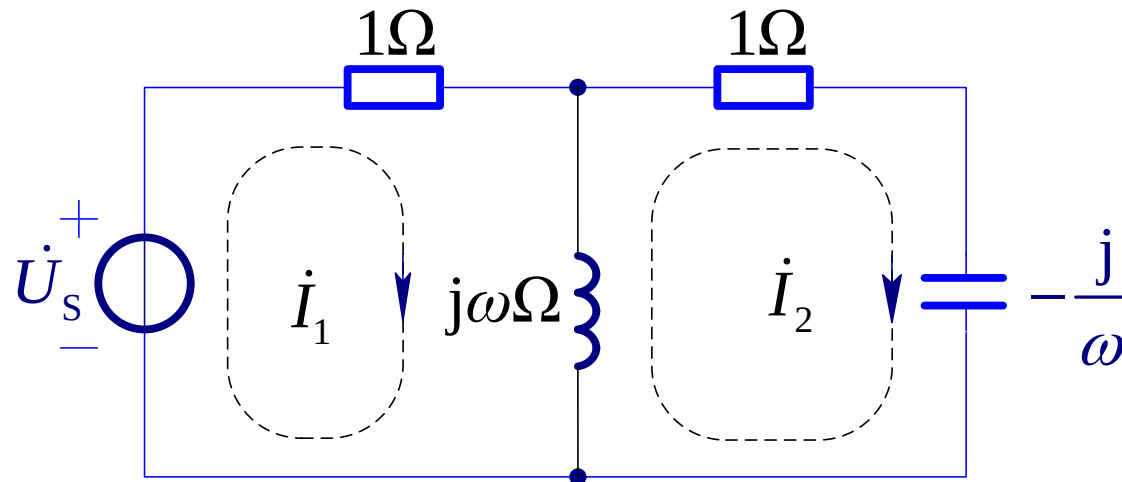
对相量模型，在单一激励下的网络函数定义为

$$H(j\omega) = \frac{\text{响应相量}}{\text{激励相量}} = |H(j\omega)| \angle \varphi(\omega)$$



网络函数(或响应)随频率变化的规律称为频率响应。

例：求转移导纳 $i_2/\dot{U}_s$ 。



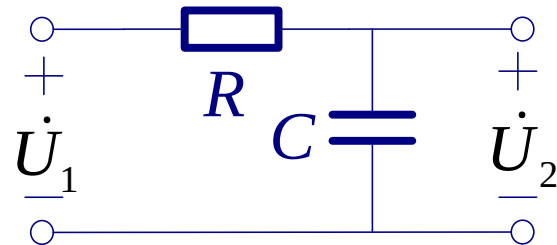
$$\begin{cases} (1 + j\omega)\dot{I}_1 - j\omega\dot{I}_2 = \dot{U}_s \\ -j\omega\dot{I}_1 + [1 + j(\omega - 1/\omega)]\dot{I}_2 = 0 \end{cases}$$

$$\Rightarrow \dot{I}_1 = \frac{[1 + j(\omega - 1/\omega)]\dot{I}_2}{j\omega} \Rightarrow \frac{(1 + j\omega)[1 + j(\omega - 1/\omega)]\dot{I}_2}{j\omega} - j\omega\dot{I}_2 = \dot{U}_s$$

$$\frac{\dot{I}_2}{\dot{U}_s} = \frac{j\omega}{(1 + j\omega)[1 + j(\omega - 1/\omega)] + \omega^2} = \frac{j\omega^2}{2\omega + j(2\omega^2 - 1)}$$

$$= \frac{\omega^2}{\sqrt{(2\omega)^2 + (2\omega^2 - 1)^2}} \angle \left[\frac{\pi}{2} - \tan^{-1} \left(\frac{2\omega^2 - 1}{2\omega} \right) \right]$$

例：RC低通电路。



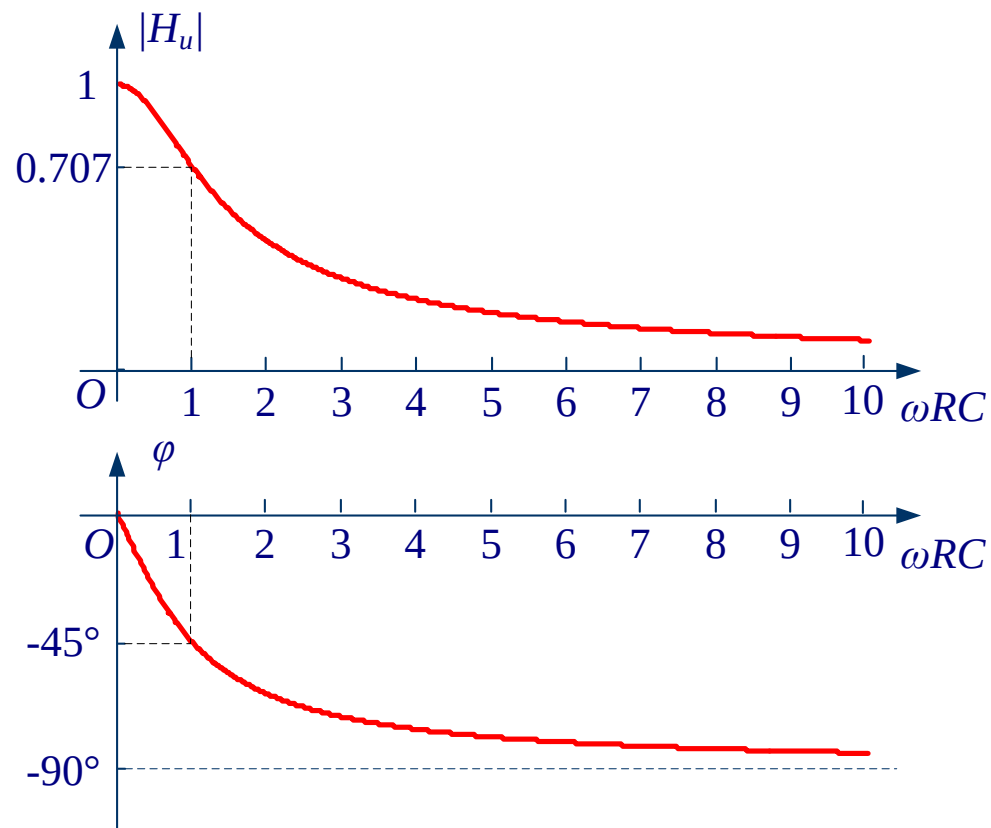
$$H_u = \frac{\dot{U}_2}{\dot{U}_1} = \frac{1/(j\omega C)}{R + 1/(j\omega C)} = \frac{1}{1 + j\omega RC}$$

$$= \frac{1}{\sqrt{1 + (\omega RC)^2}} \angle [-\tan^{-1}(\omega RC)]$$

$$|H_u| = \frac{U_2}{U_1} = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\varphi = \varphi_2 - \varphi_1 = -\tan^{-1}(\omega RC)$$

上述电路也称为滞后网络；截止频率 $\omega_c = 1/RC$



8.6.2 网络函数与正弦稳态网络函数的关系

$$H(s) \Leftrightarrow H(j\omega)$$

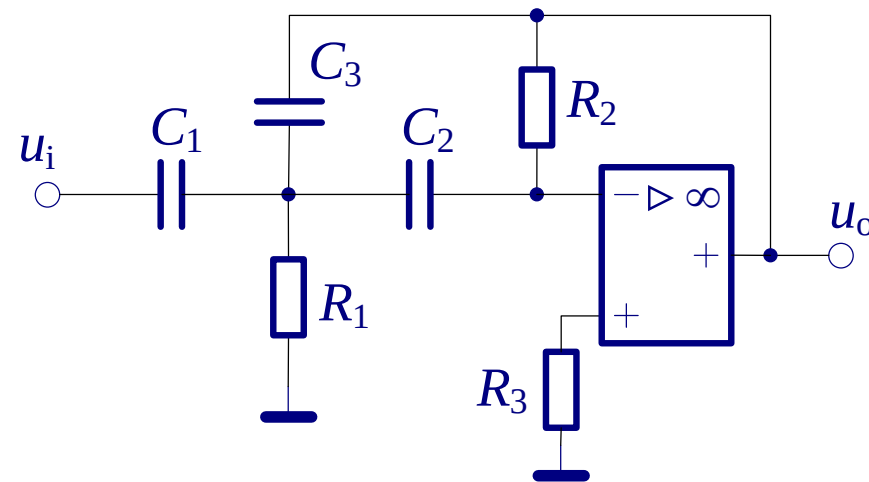
$$H(j\omega) = H(s)\big|_{s=j\omega} \quad \text{或} \quad H(s) = H(j\omega)\big|_{j\omega=s}$$

例：试求图示电路的频率特性。

$$\begin{cases} (sC_1 + sC_2 + sC_3 + 1/R_1)U_1(s) - sC_2U_-(s) \\ \quad - sC_3U_o(s) - sC_1U_i(s) = 0 \\ -sC_2U_1(s) + (sC_2 + 1/R_2)U_-(s) - (1/R_2)U_o(s) = 0 \\ U_-(s) = U_+(s) = 0 \end{cases}$$

$$H(s) = \frac{U_o(s)}{U_i(s)} = -\frac{\frac{C_1}{C_3}s^2}{s^2 + \frac{C_1 + C_2 + C_3}{R_2C_2C_3}s + \frac{1}{R_1R_2C_2C_3}} = -\frac{As^2}{s^2 + \frac{\omega_0}{Q}s + \omega_0^2}$$

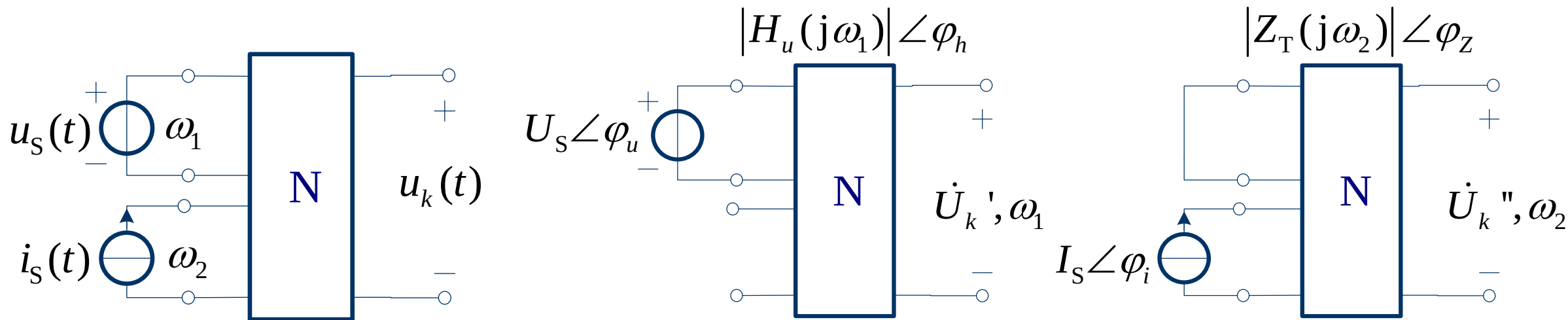
$$H(j\omega) = \frac{U_o(j\omega)}{U_i(j\omega)} = -\frac{A\omega^2}{\omega^2 - j\frac{\omega_0}{Q}\omega - \omega_0^2}$$



$$\begin{cases} \omega_0 = 1/\sqrt{R_1R_2C_2C_3} \\ Q = \frac{\sqrt{R_2C_2C_3/R_1}}{C_1 + C_2 + C_3} \\ A = C_1/C_3 \end{cases}$$

特征频率/转折频率
品质因数
通带增益

8.6.3 正弦稳态的叠加



$$\begin{cases} \dot{U}_k' = H_u(j\omega_1)\dot{U}_S \\ \dot{U}_k'' = Z_T(j\omega_2)\dot{I}_S \end{cases} \Rightarrow \begin{cases} u_k'(t) = \sqrt{2}|H_u(j\omega_1)|U_S \cos(\omega_1 t + \varphi_h + \varphi_u) \\ u_k''(t) = \sqrt{2}|Z_T(j\omega_2)|I_S \cos(\omega_2 t + \varphi_Z + \varphi_i) \end{cases}$$

1) 如果 $\omega_1 \neq \omega_2 \Rightarrow u_k(t) = u_k'(t) + u_k''(t)$

2) 如果 $\omega_1 = \omega_2 = \omega \Rightarrow \dot{U}_k = \dot{U}_k' + \dot{U}_k'' = U_k \angle \varphi \Rightarrow u_k(t) = \sqrt{2}U_k \cos(\omega t + \varphi)$

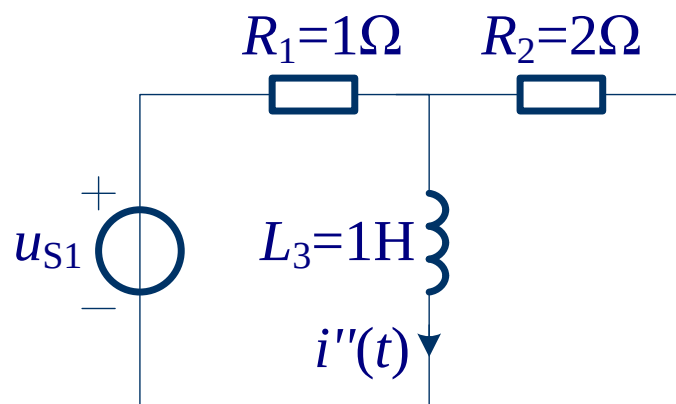
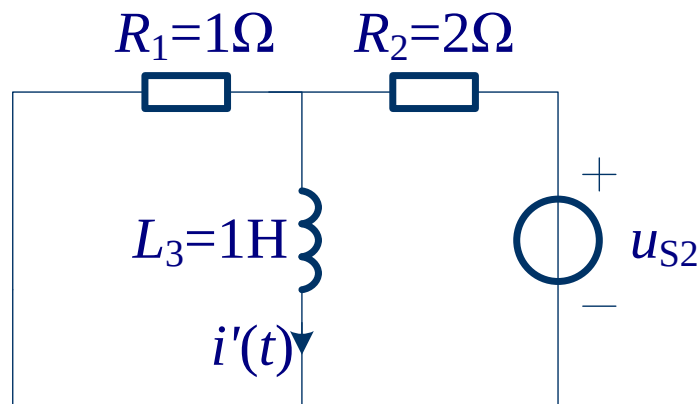
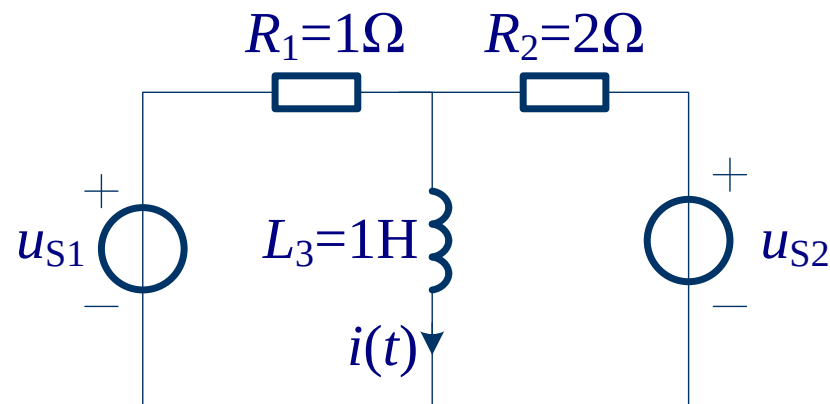
例：求 $i(t)$ 。已知 $u_{S1}(t) = 5\sqrt{2}\cos(2t)V$, $u_{S2}(t) = 10\sqrt{2}\cos(2t + 90^\circ)V$ 。

$$\dot{I}' = \frac{\dot{U}_{S2}}{Z_2 + Z_1 Z_3 / (Z_1 + Z_3)} \times \frac{Z_1}{(Z_1 + Z_3)} = 1.58 \angle 18.4^\circ \text{ A}$$

$$\dot{I}'' = \frac{\dot{U}_{S1}}{Z_1 + Z_2 Z_3 / (Z_2 + Z_3)} \times \frac{Z_2}{(Z_2 + Z_3)} = 1.58 \angle -71.6^\circ \text{ A}$$

$$\dot{I} = \dot{I}' + \dot{I}'' = 1.58 \angle 18.4^\circ + 1.58 \angle -71.6^\circ = 2.24 \angle -26.6^\circ \text{ A}$$

$$i(t) = 2.24\sqrt{2} \cos(2t - 26.6^\circ) \text{ A}$$



例：求 $i_o(t)$ 。

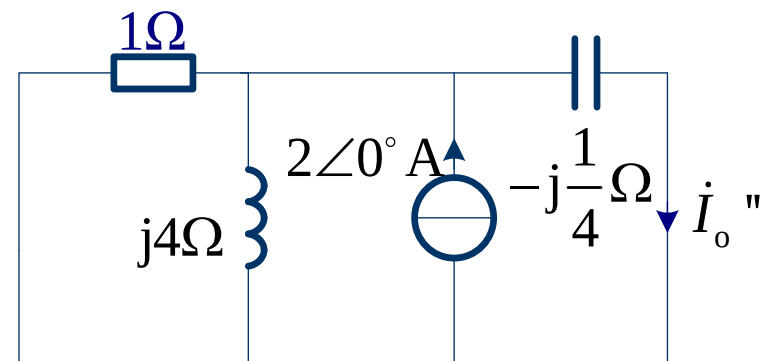
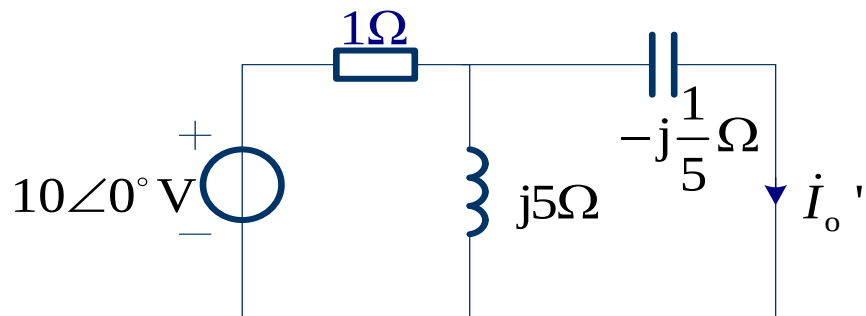
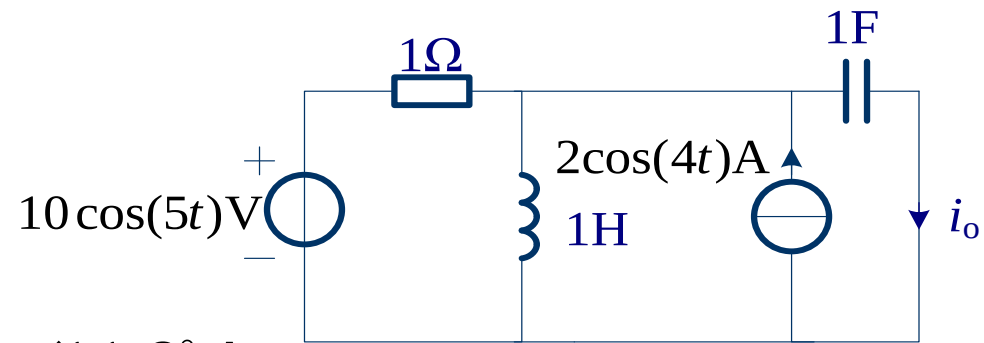
$$\dot{I}_o' = \frac{10\angle 0^\circ}{1 + (j5)(-j/5)/(j5 - j/5)} \times \frac{j5}{(j5 - j/5)} = 10.2\angle 11.8^\circ \text{ A}$$

$$i_o'(t) = 10.2 \cos(5t + 11.8^\circ) \text{ A}$$

$$\dot{I}_o'' = 2\angle 0^\circ \times \frac{j4}{1 + 1/(j4) + j4} = 2.06\angle 14.9^\circ \text{ A}$$

$$i_o''(t) = 2.06 \cos(4t + 14.9^\circ) \text{ A}$$

$$i_o(t) = [2.06 \cos(4t + 14.9^\circ) + 10.2 \cos(5t + 11.8^\circ)] \text{ A}$$



练习：某电路含三个不同频率的正弦激励，激励频率分别为 1rad/s 、 2rad/s 、 $2\pi\text{rad/s}$ ，下述正确的是

- A. 经过一段时间后，该电路进入非正弦周期稳态。
- B. 分析该电路的响应，可以在时域内应用叠加定理。
- C. 分析该电路的响应，可以在相量域内应用叠加定理。

例：求 R 的电流有效值。已知

$$u_s = [10 + \sqrt{2} \cos t + \sqrt{2} \sin 2t] \text{V}$$

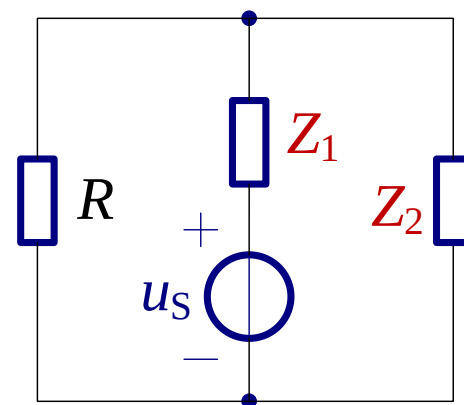
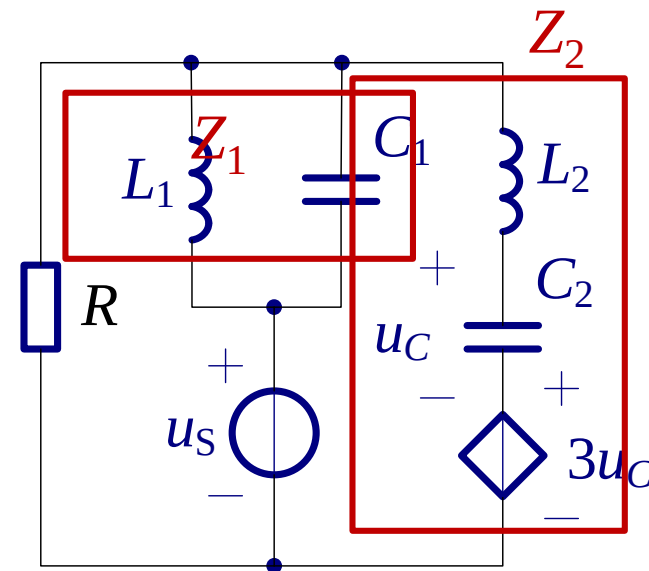
$$L_1 = L_2 = 1\text{H}, C_1 = C_2 = 1\text{F}, R = 10\Omega$$

直流单独作用 $Z_1 = 0, Z_2 = \infty \quad I_{R0} = 1\text{A}$

基频单独作用 $Z_1 = \infty \quad I_{R1} = 0$

倍频单独作用 $Z_2 = 0 \quad I_{R2} = 0$

$$\Rightarrow I_R = 1\text{A}$$

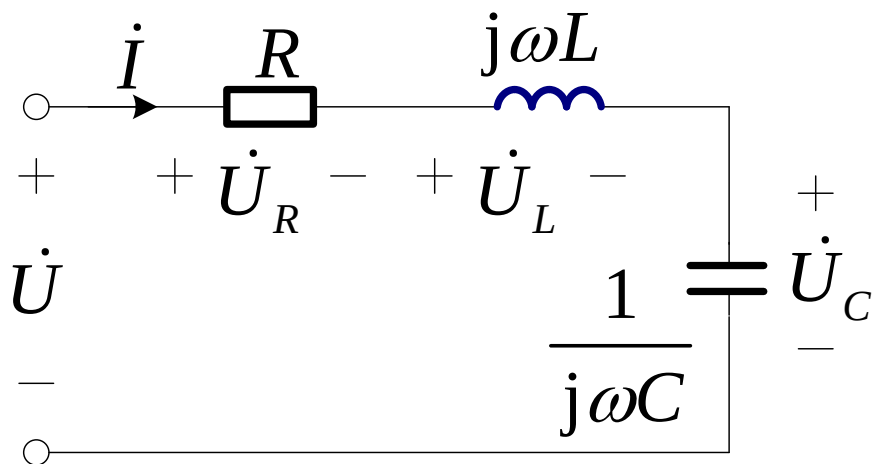


$$Z_1 = j\omega L_1 \parallel \frac{1}{j\omega C_1} \quad Z_2 = j\omega L_2 + \frac{1}{j\omega C_2 / 4}$$

8.7 谐振电路

8.7.1 RLC电路的谐振

定义：对于任何含有电感和（或）电容的单口网络，在一定的条件下呈现电阻性，其端口电压与电流同相，则称此单口网络发生谐振。

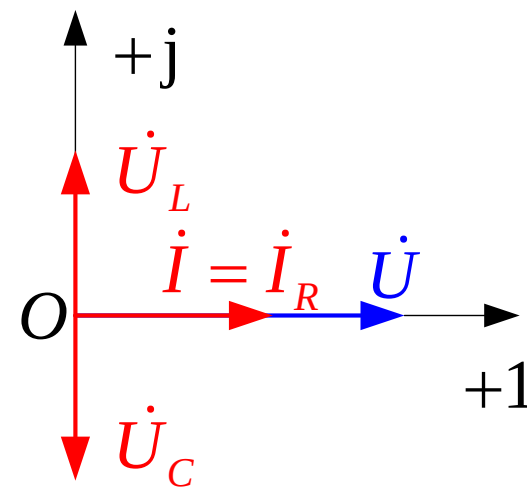


$$Z(j\omega) = R + j(\omega L - \frac{1}{\omega C})$$

$$\Rightarrow \omega L = \frac{1}{\omega C}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

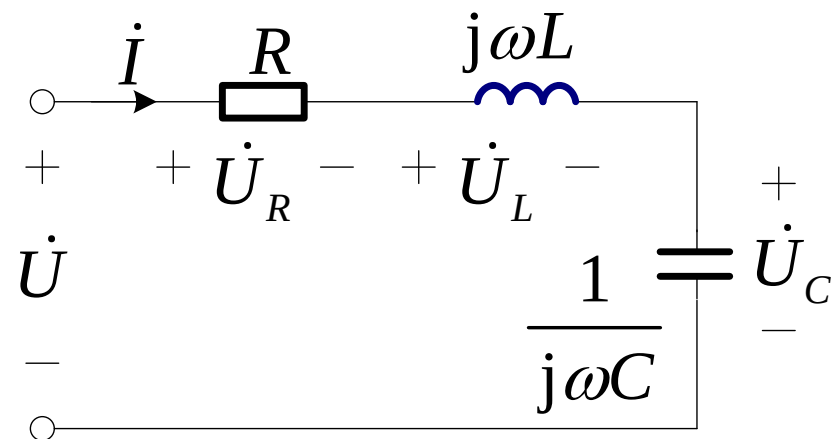
$$Z(j\omega_0) = R$$



定义：串联 RLC 电路谐振时动态元件的电压与激励电压之比为 RLC 电路的**品质因数**。用 Q 表示。

$$\begin{cases} \frac{U_L}{U} = \frac{I\omega_0 L}{IR} = \frac{\omega_0 L}{R} = Q \\ \frac{U_C}{U} = \frac{I / (\omega_0 C)}{IR} = \frac{1}{\omega_0 RC} = \frac{\omega_0 L}{R} = Q \end{cases}$$

在谐振时，电容和电感两端的电压相等，且为电源电压的 Q 倍。



讨论： Q 的能量解释。

$$i = \sqrt{2}I \cos \omega_0 t$$

$$w = \frac{1}{2} L i_L^2 + \frac{1}{2} C u_C^2 = \frac{1}{2} L I^2 (1 + \cos 2\omega_0 t) + \frac{1}{2} C \left[\frac{\sqrt{2}I}{\omega_0 C} \sin \omega_0 t \right]^2$$

$$Q = \omega_0 \frac{L I^2}{R I^2} = 2\pi \frac{L I^2}{R I^2 T}$$

$$= \frac{1}{2} L I^2 (1 + \cos 2\omega_0 t) + \frac{1}{2} L I^2 (1 - \cos 2\omega_0 t) = L I^2$$

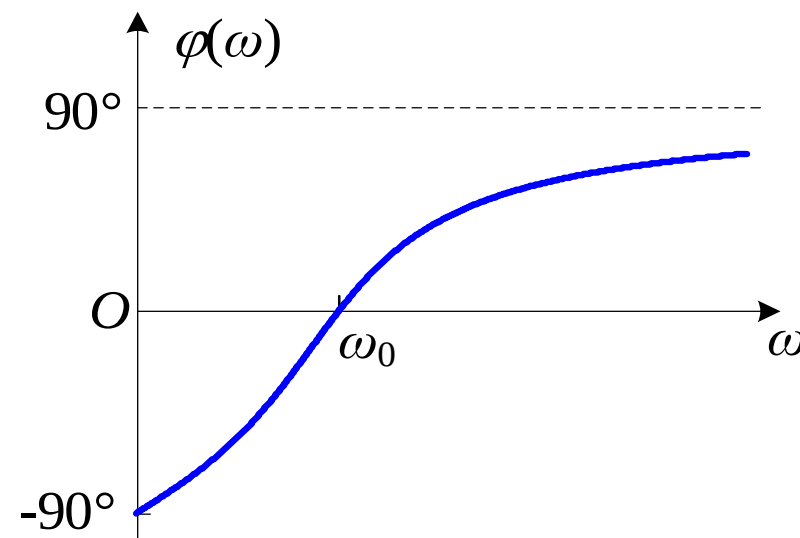
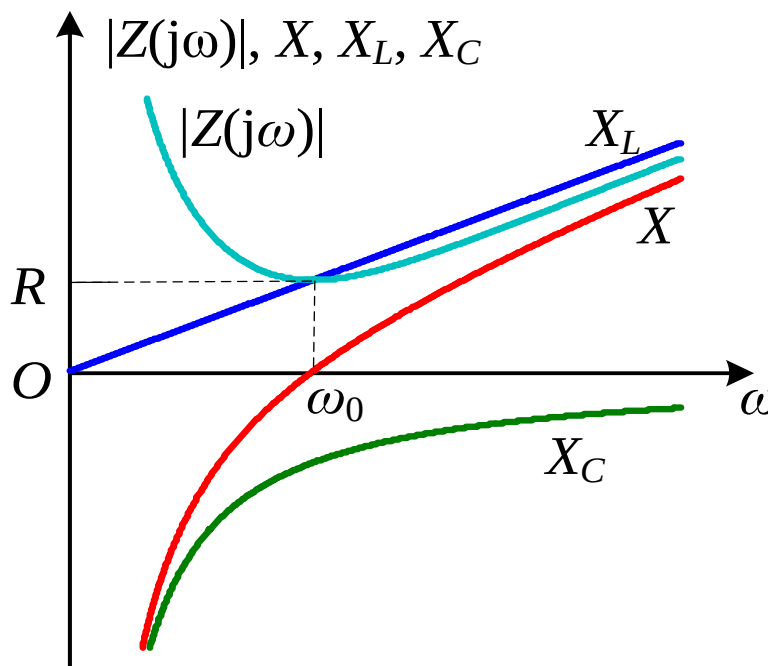
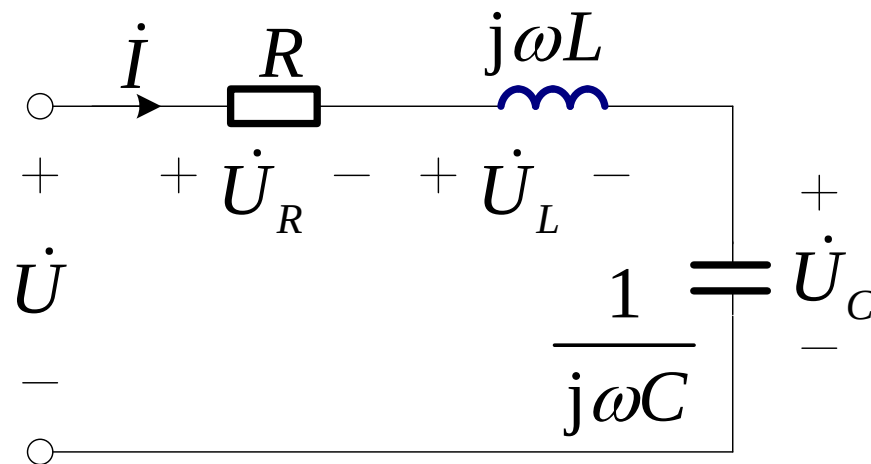
$$Q = 2\pi \frac{\text{谐振时电路储存的电磁能量}}{\text{谐振时电路一个周期内消耗的能量}}$$

8.7.2 RLC谐振电路的频率特性

一、输入阻抗的频率特性

$$Z(j\omega) = R + j(\omega L - \frac{1}{\omega C})$$

$$\begin{cases} |Z(j\omega)| = \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2} \\ \varphi_Z = \tan^{-1} \frac{\omega L - \frac{1}{\omega C}}{R} \end{cases}$$

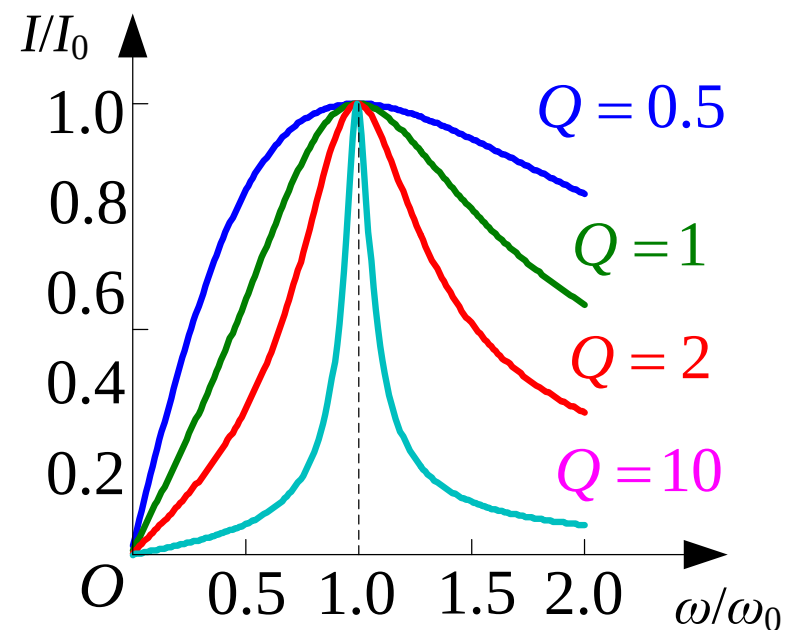
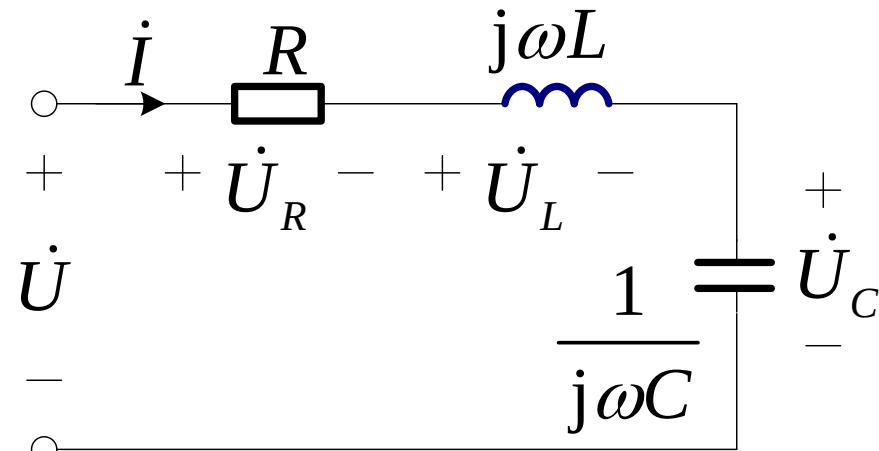


二、电流谐振曲线[电流随元件参数、频率变化的关系]

$$I = \frac{U}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} = \frac{U}{R \sqrt{1 + \frac{1}{R^2} (\omega L - \frac{1}{\omega C})^2}}$$

$$= \frac{I_0}{\sqrt{1 + \frac{1}{R^2} (\omega L - \frac{1}{\omega C})^2}} = \frac{I_0}{\sqrt{1 + Q^2 (\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})^2}}$$

$$\frac{I}{I_0} = \frac{1}{\sqrt{1 + Q^2 (\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})^2}}$$



截止频率、带宽

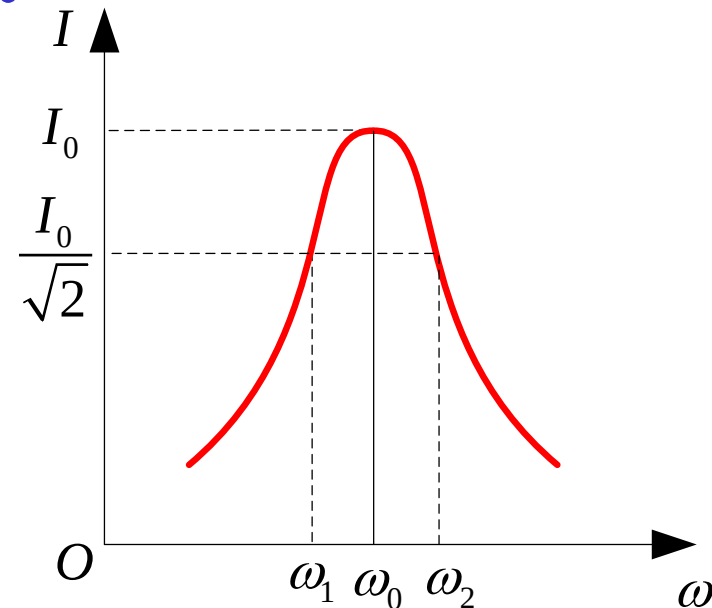
*RLC*串联谐振电路对电流来说是一个带通滤波电路。

$$BW = \Delta\omega = \omega_2 - \omega_1$$

$$\frac{I}{I_0} = \frac{1}{\sqrt{2}} = \frac{1}{\sqrt{1 + Q^2 \left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega} \right)^2}}$$

$$\begin{cases} \omega_1 = \omega_0 \left(-\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right) \stackrel{Q \gg 1}{\approx} \omega_0 \left(1 - \frac{1}{2Q} \right) \\ \omega_2 = \omega_0 \left(\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right) \stackrel{Q \gg 1}{\approx} \omega_0 \left(1 + \frac{1}{2Q} \right) \end{cases}$$

$\Rightarrow \Delta\omega = \omega_2 - \omega_1 = \omega_0 / Q$ 对一定谐振频率而言，品质因数越高，通带越窄。



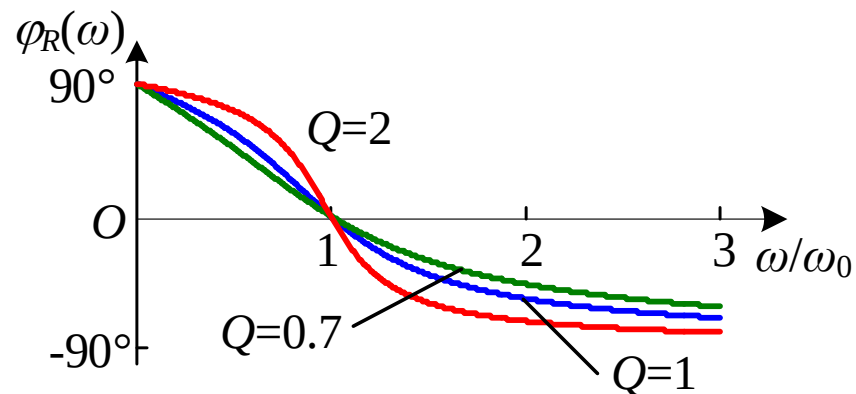
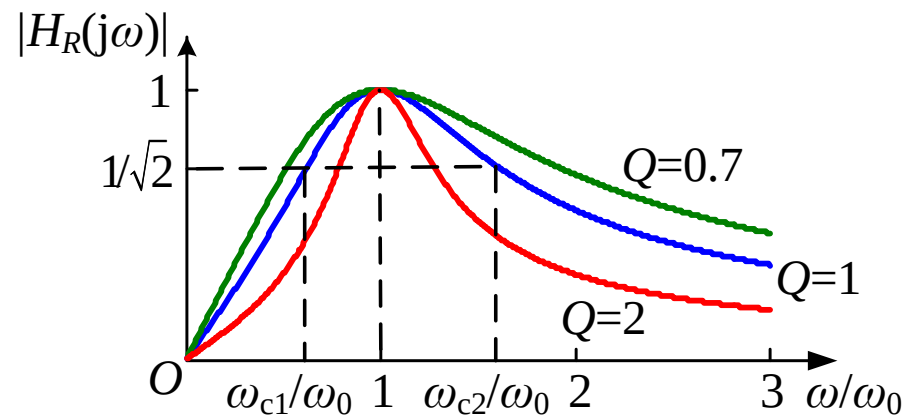
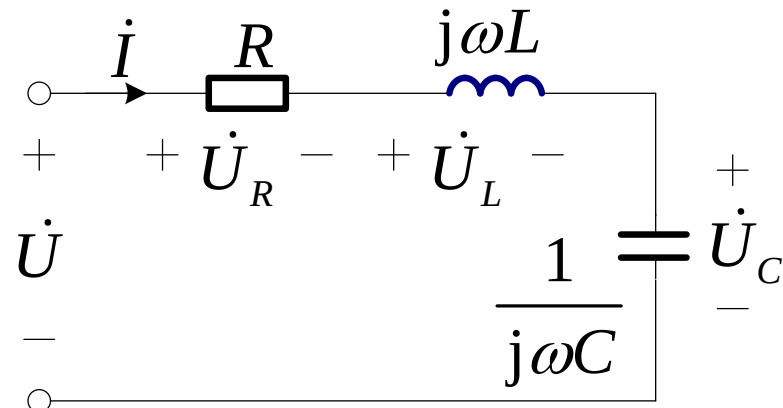
应用：带通电路——以电阻电压为响应

$$H_R(j\omega) = \frac{\dot{U}_R}{\dot{U}} = \frac{R}{R + j(\omega L - \frac{1}{\omega C})} = \frac{1}{1 + jQ(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})}$$

$$\begin{cases} |H_R(j\omega)| = \frac{1}{\sqrt{1 + Q^2(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})^2}} \\ \varphi_R(\omega) = -\tan^{-1}[Q(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})] \end{cases}$$

$$\begin{cases} \omega_{c1} = \omega_0 \left(-1/(2Q) + \sqrt{1/(4Q^2) + 1} \right) \\ \omega_{c2} = \omega_0 \left(1/(2Q) + \sqrt{1/(4Q^2) + 1} \right) \end{cases}$$

$$BW = \frac{\omega_0}{Q} = \frac{R}{L}$$



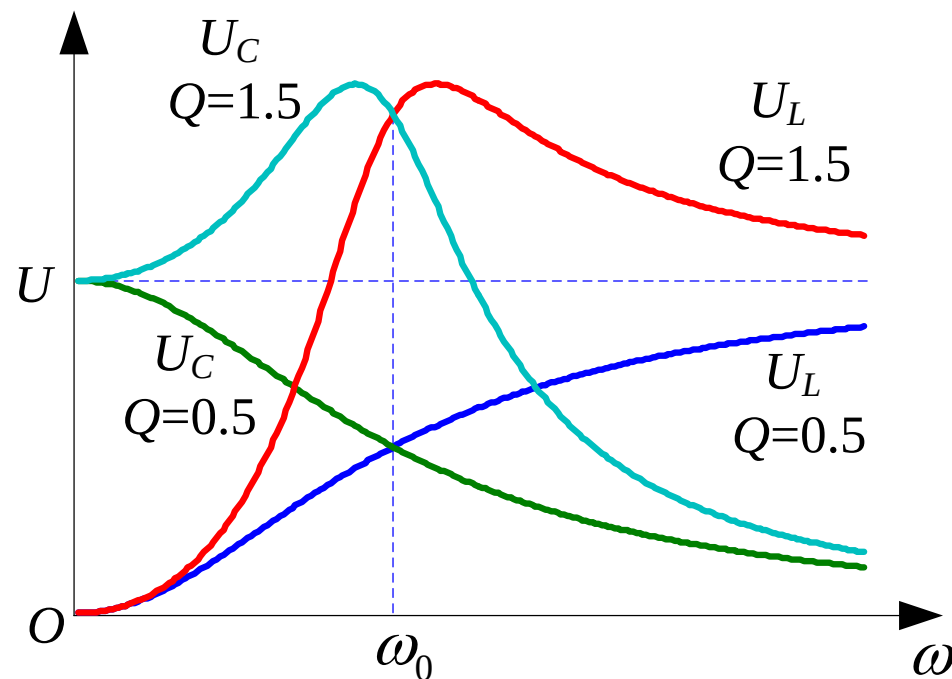
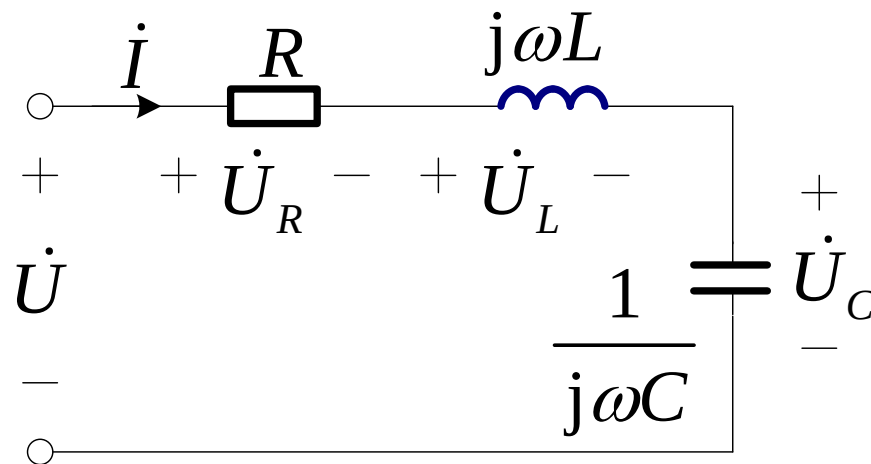
三、电压谐振曲线

$$U_L = I\omega L$$

$$= \frac{U\omega L}{\sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} = \frac{\omega QU}{\omega_0 \sqrt{1 + Q^2 (\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})^2}}$$

$$U_C = I \frac{1}{\omega C}$$

$$= \frac{U}{\omega C \sqrt{R^2 + (\omega L - \frac{1}{\omega C})^2}} = \frac{\omega_0 QU}{\omega \sqrt{1 + Q^2 (\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega})^2}}$$

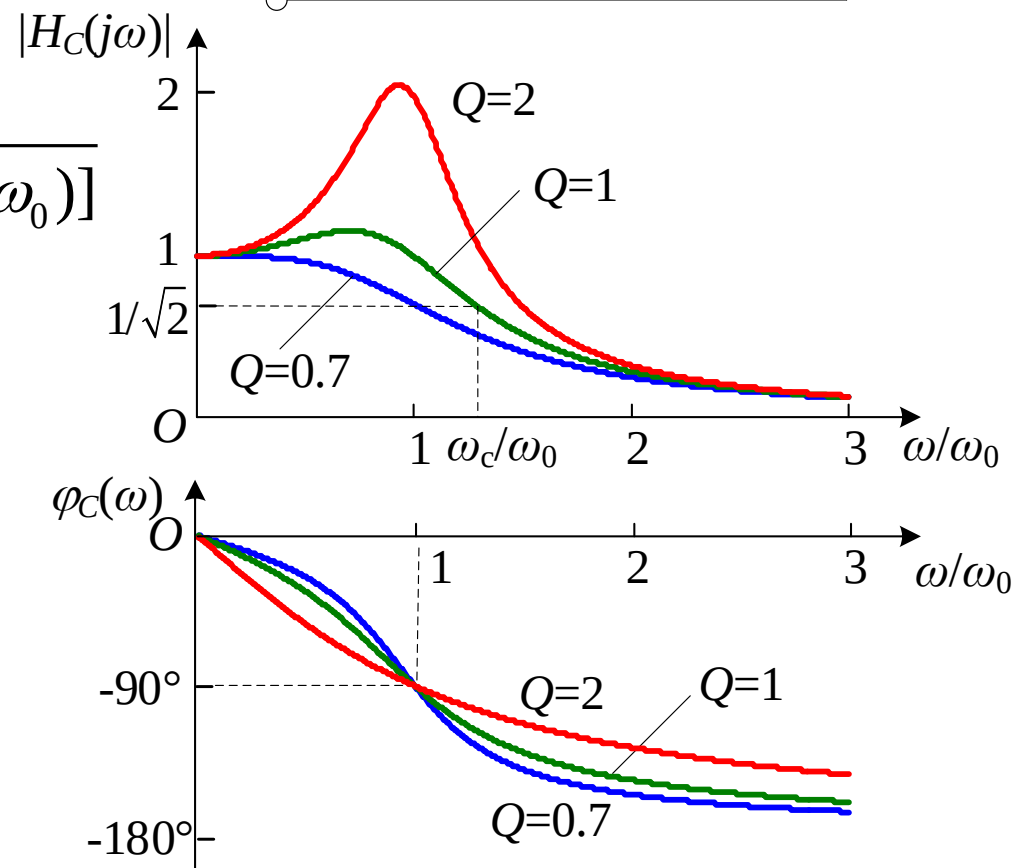
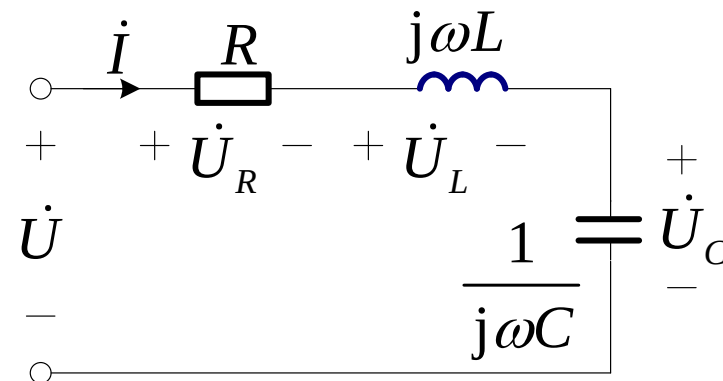


应用：低通电路——以电容电压为响应

$$H_C(j\omega) = \frac{\dot{U}_C}{\dot{U}} = \frac{1/(j\omega C)}{R + j[\omega L - 1/(\omega C)]}$$

$$= \frac{1}{(1 - \omega^2 LC) + j\omega CR} = \frac{1}{[1 - (\omega/\omega_0)^2] + j[\omega/(Q\omega_0)]}$$

$$\left\{ \begin{aligned} |H_C(j\omega)| &= \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_0}\right)^2\right]^2 + \frac{1}{Q^2} \left(\frac{\omega}{\omega_0}\right)^2}} \\ \varphi_C(\omega) &= \tan^{-1} \frac{1}{Q\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)} \end{aligned} \right.$$



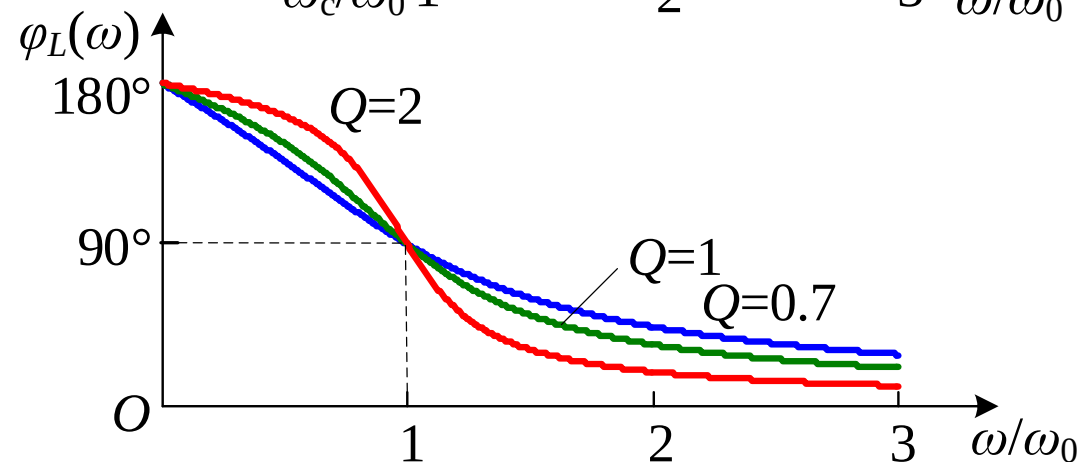
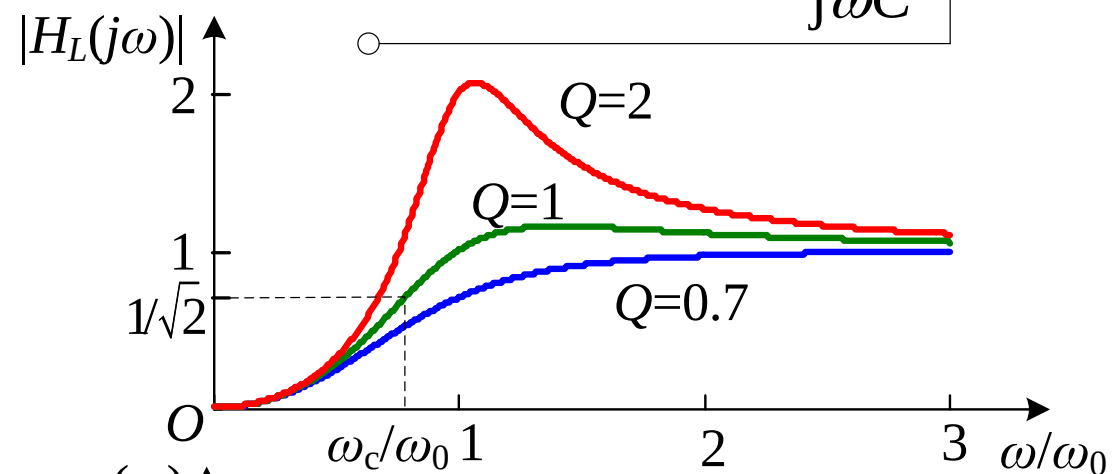
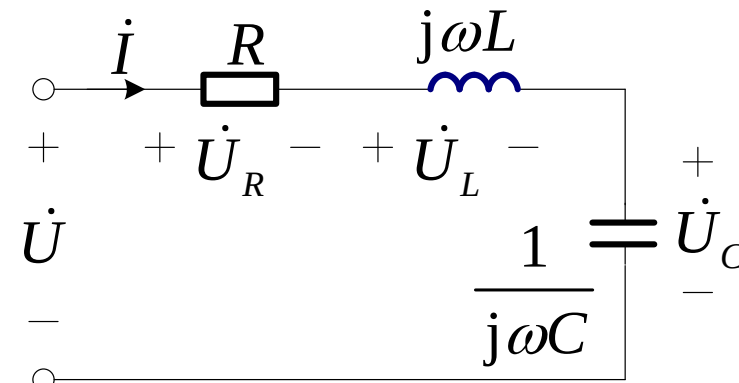
应用：高通电路——以电感电压为响应

$$H_L(j\omega) = \frac{\dot{U}_L}{\dot{U}} = \frac{j\omega L}{R + j[\omega L - 1/(\omega C)]}$$

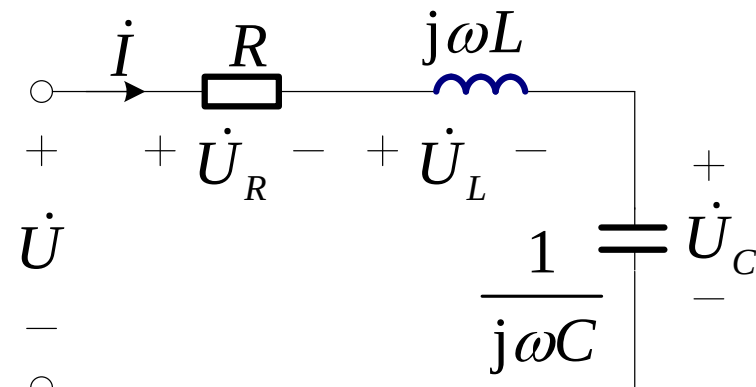
$$= \frac{1}{[1 - 1/(\omega^2 LC) + jR/(\omega L)]}$$

$$= \frac{1}{[1 - (\omega_0/\omega)^2] + j\omega_0/(Q\omega)}$$

$$\left\{ \begin{array}{l} |H_L(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega_0}{\omega}\right)^2\right]^2 + \frac{1}{Q^2} \left(\frac{\omega_0}{\omega}\right)^2}} \\ \varphi_L(\omega) = -\tan^{-1} \frac{1}{Q\left(\frac{\omega}{\omega_0} - \frac{\omega_0}{\omega}\right)} \end{array} \right.$$



应用：带阻电路——以电感、电容电压之和为响应

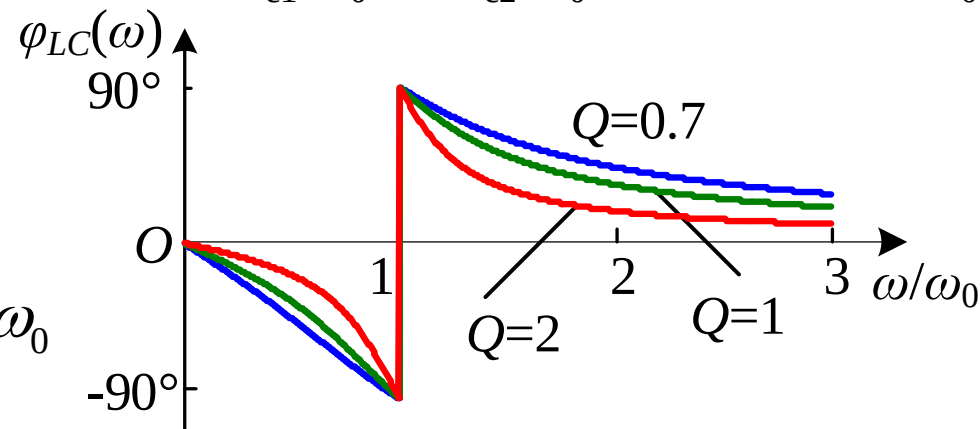
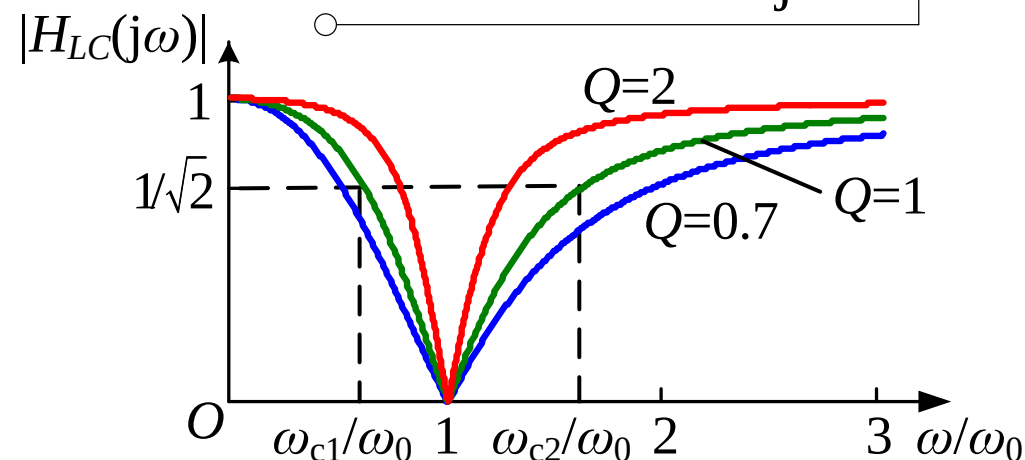


$$H_{LC}(j\omega) = \frac{\dot{U}_L + \dot{U}_C}{\dot{U}}$$

$$= \frac{j\omega L + 1/(j\omega C)}{R + j(\omega L - 1/\omega C)} = \frac{\omega_0^2 - \omega^2}{(\omega_0^2 - \omega^2) + j\omega_0\omega/Q}$$

$$\begin{cases} |H_{LC}(j\omega)| = \frac{|\omega_0^2 - \omega^2|}{\sqrt{(\omega_0^2 - \omega^2)^2 + \omega_0^2\omega^2/Q^2}} \\ \varphi_{LC}(\omega) = -\arctan \frac{\omega_0\omega}{Q(\omega_0^2 - \omega^2)} \end{cases}$$

$$\omega_{c1} = \left(-\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right) \omega_0, \omega_{c2} = \left(\frac{1}{2Q} + \sqrt{1 + \frac{1}{4Q^2}} \right) \omega_0$$



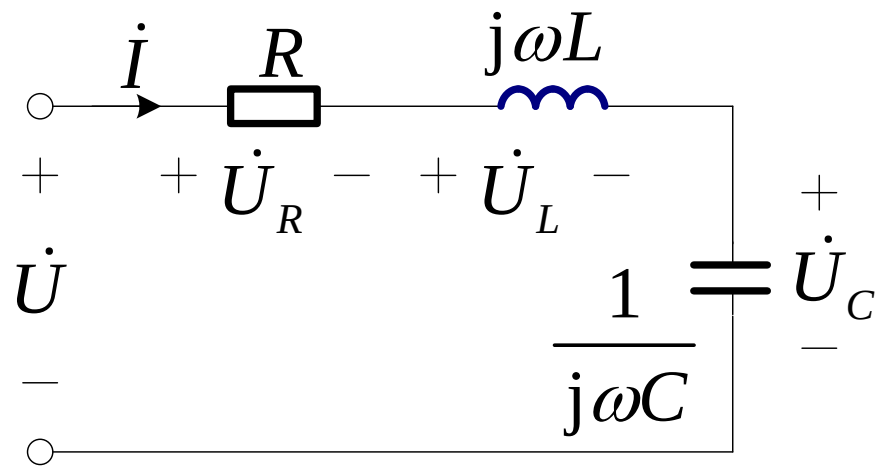
例：求电容电压为最大时的频率。

$$|H_C(j\omega)| = \frac{1}{\sqrt{\left[1 - \left(\frac{\omega}{\omega_0}\right)^2\right]^2 + \frac{1}{Q^2} \left(\frac{\omega}{\omega_0}\right)^2}}$$

$$F = \left[1 - \left(\frac{\omega}{\omega_0}\right)^2\right]^2 + \frac{1}{Q^2} \left(\frac{\omega}{\omega_0}\right)^2 \rightarrow \min$$

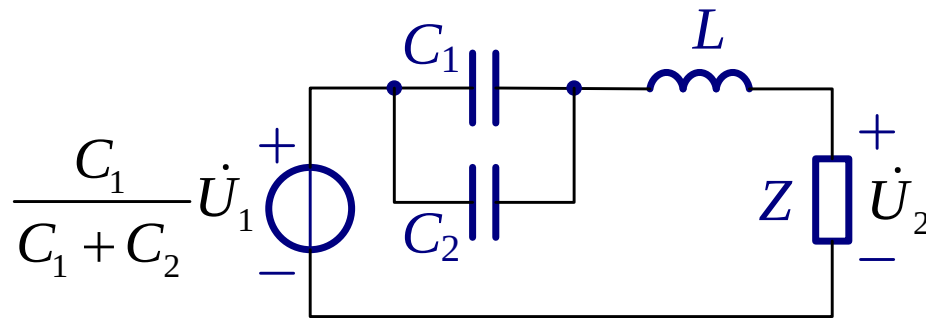
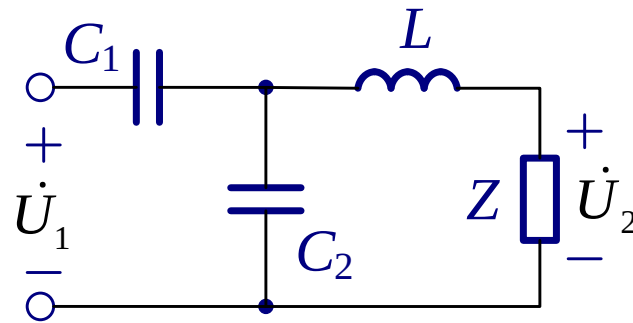
$$\frac{dF}{d\omega} = 4 \left[\left(\frac{\omega}{\omega_0}\right)^2 - 1 \right] \frac{\omega}{\omega_0} \frac{1}{\omega_0} + \frac{2}{Q^2} \frac{\omega}{\omega_0} \frac{1}{\omega_0} = 0$$

$$\Rightarrow \omega = \omega_0 \sqrt{1 - \frac{1}{2Q^2}}$$



例：如图，要求 $\dot{U}_2$ 与 Z 无关，则激励 $\dot{U}_1$ 的频率 ω 应为多少？

解：只有当从 Z 两端看出去的等效电路为一个电压源时， $\dot{U}_2$ 与 Z 无关。



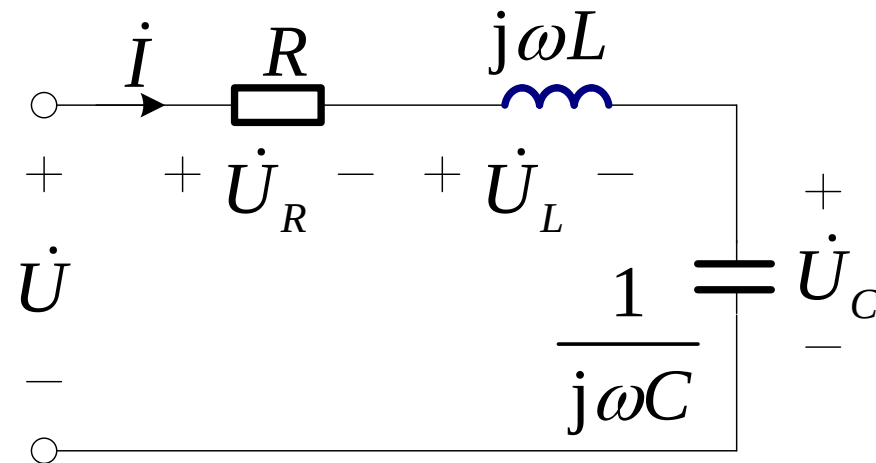
$$\omega = \frac{1}{\sqrt{L(C_1 + C_2)}}$$

例：设计 RLC 串联电路，谐振频率为 10^4Hz ，通频带为 100Hz 。串联电阻及负载电阻分别为 10Ω 、 15Ω 。截止频率为多少？

$$R = 10 + 15 = 25\Omega$$

$$Q = \frac{\omega_0}{\omega_2 - \omega_1} = \frac{f_0}{f_2 - f_1} = 100$$

$$\left. \begin{aligned} \omega_0 &= \frac{1}{\sqrt{LC}} \\ BW &= \frac{\omega_0}{Q} = \frac{R}{L} \end{aligned} \right\} \Rightarrow \begin{cases} L = \frac{QR}{\omega_0} = 39.8\text{mH} \\ C = \frac{1}{\omega_0 QR} = 6360\text{pF} \end{cases}$$



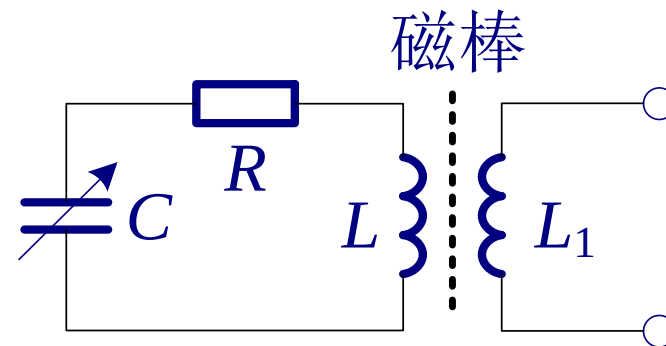
$$\begin{cases} f_1 = f_0 \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right) = 9950\text{Hz} \\ f_2 = f_0 \left(\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right) = 10050\text{Hz} \end{cases}$$

电路实例：收音机输入调谐电路(内置天线)

已知 $R=0.6\Omega$, $L=0.1\mu\text{H}$, 现将收音机调谐在调频波段106.1MHz。试求：(1) 电容 C 和品质因数 Q ；(2) 106.1MHz与107MHz信号电压在电感两端产生的电压之比，设两者的信号电压幅值相同。

$$(1) \quad C = \frac{1}{\omega_0^2 L} = \frac{1}{(2\pi \times 106.1 \times 10^6)^2 \times 0.1 \times 10^{-6}} = 22.5 \text{pF}$$

$$Q = \frac{\omega_0 L}{R} = \frac{2\pi \times 106.1 \times 10^6 \times 0.1 \times 10^{-6}}{0.6} = 111.1$$



(2) 设信号电压幅值为 U , 则谐振时信号电压在电感两端产生的电压幅值为 $U_0=QU=111.1U$; 在107MHz, 信号电压在电感两端产生的电压幅值为

$$U_1 = \frac{U \omega L}{\sqrt{R^2 + [\omega L - 1/(\omega C)]^2}} = \dots = 52.84U$$

$$\frac{U_1}{U_0} = \frac{52.84U}{111.1U} = 0.4756$$

8.7.3 GLC并联电路的谐振

练习：下述公式**不**正确的是 [运用对偶原理]

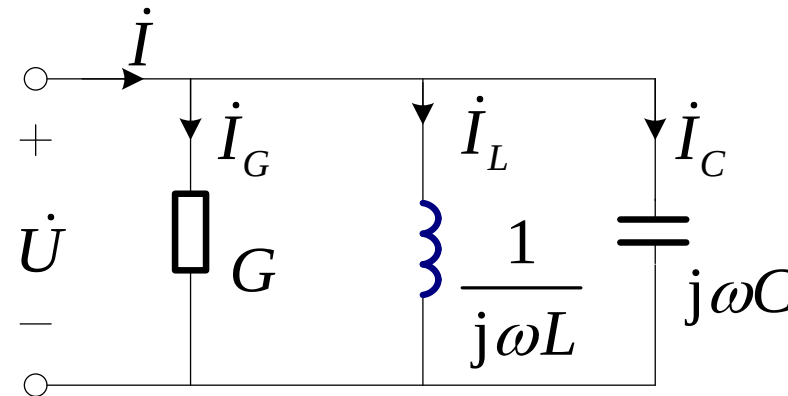
A. $\omega_0 = \frac{1}{\sqrt{LC}}$

B. $Q = \omega_0 GC = \frac{1}{\omega_0 LG}$

C. $BW = \frac{G}{C} = \frac{\omega_0}{Q}$

D. $\omega_1 = \omega_0 \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$

E. $\omega_2 = \omega_0 \left(\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$



电路实例：收音机输入调谐电路(外置天线)

GLC电路常用于收音机电路中。设立体声收音机调谐在FM98MHz, 已知 $L = 0.1\mu\text{H}$, $Q = 120$ 。(1) 求 G 和 C ; (2) 设信号电流幅度不变, 试求100MHz信号电流产生的电压与谐振时电压之比。

$$(1) \quad C = \frac{1}{\omega_0^2 L} = 26.4\text{pF}; \quad G = \frac{1}{\omega_0 L Q} = 135\mu\text{S}$$

$$(2) \quad \dot{U}_0 = \frac{\dot{I}}{Y(j\omega)} = \frac{\dot{I}}{G} = \frac{\dot{I}}{135 \times 10^{-6}} = 7407\dot{I}$$

$$\Rightarrow U_0 = 7407I$$

100MHz时

$$U = \frac{I}{\sqrt{G^2 + (\omega C - 1/\omega L)^2}} = \dots = 1459I \quad \frac{U}{U_0} = 0.197$$

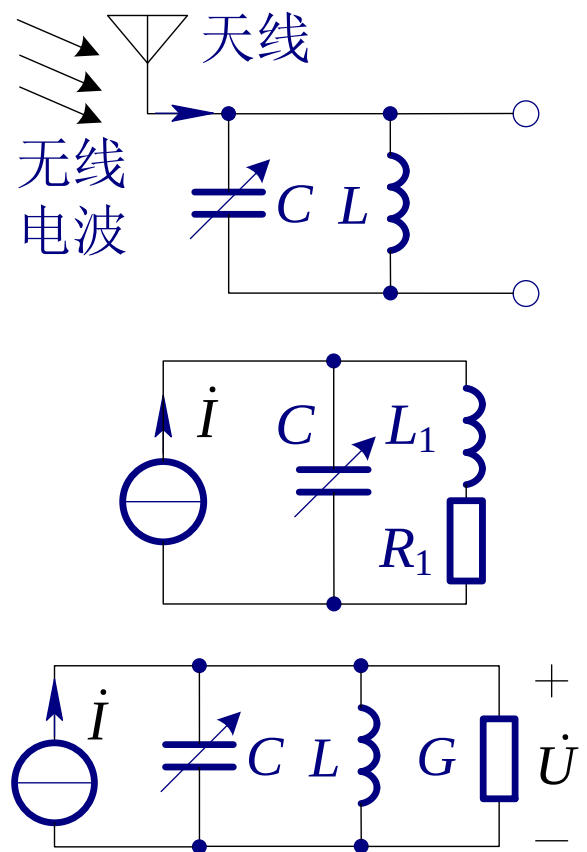
$$\omega_0 = 1/\sqrt{LC}$$

$$Q = \frac{\omega_0 C}{G} = \frac{1}{\omega_0 L G}$$

$$BW = G/C = \omega_0/Q$$

$$\omega_1 = \omega_0 \left(-\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$

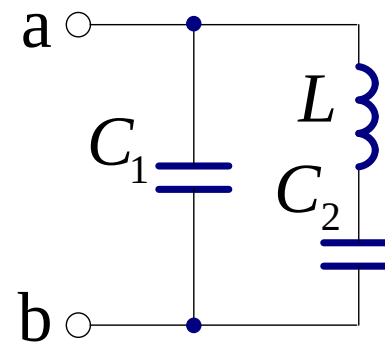
$$\omega_2 = \omega_0 \left(\frac{1}{2Q} + \sqrt{\frac{1}{4Q^2} + 1} \right)$$



电路实例：石英晶体谐振电路

石英晶体具有非常稳定的机械和压电特性，常用来作为基本的时钟器件。在电路中石英晶体主要起选频、鉴频和稳频的作用，应用非常广泛。如图所示电路为石英晶体的电路模型，试求出其谐振频率。

$$Z_{ab} = \frac{(j\omega L + \frac{1}{j\omega C_2}) \frac{1}{j\omega C_1}}{j\omega L + \frac{1}{j\omega C_2} + \frac{1}{j\omega C_1}} = \frac{1 - \omega^2 LC_2}{j[\omega(C_1 + C_2) - \omega^3 LC_1 C_2]}$$



$$1 - \omega^2 LC_2 = 0 \Rightarrow \omega_s = \sqrt{\frac{1}{LC_2}}$$

$$C_1 + C_2 = \omega^2 LC_1 C_2 \Rightarrow \omega_p = \sqrt{\frac{C_1 + C_2}{LC_1 C_2}}$$

练习：请选择正确的表述。

- A. 某一端口电路，其端口电压比电流滞后 25° ，则该电路对外呈感性。
- B. 电阻性电路的功率因数 λ 小于1。
- C. 正弦稳态 RLC 串联电路，其端口电压有效值 U 与 R 、 L 、 C 各自电压有效值满足 $U < U_R + U_L + U_C$ 。
- D. 不同频率的电压、电流形成瞬时功率，也产生平均功率。
- E. 任意复数都可表达一个相量。